

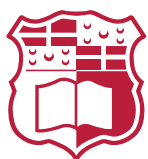
Evaluating the Quality of the LAION-2B Dataset: Insights into Large-Scale Image Dataset Assessment

Jerome Agius

Supervisor: Dr Dylan Seychell

May 2026

*Submitted in partial fulfilment of the requirements
for the degree of Master of Science in Artificial Intelligence [Taught and
Research (Mainly by Research)] (M.Sc.(Melit.)).*



L-Università ta' Malta

**Faculty of Information &
Communication Technology**

Abstract

Generative Artificial Intelligence (AI) systems have enabled large-scale image creation, bringing advanced image generation capabilities within reach of non-specialist users. However, the development of these systems depends heavily on massive web-scraped training datasets such as LAION-5B, with this research focusing on its re-released version, Re-LAION-5B, and the related Stable Diffusion v1.5 model. Due to the nature of web-scraped training data, concerns have increased regarding the ethical implications of these datasets, particularly in relation to bias and unintended correlations embedded within the training corpus. Specifically, non-semantic or spurious features present within the training dataset may serve as shortcuts in downstream tasks, negatively impacting model behaviour and fairness outcomes. While these shortcuts can emerge across many tasks, demographic attributes represent a particularly concerning case, especially when predictions rely on features unrelated to the inherent semantic content of the image.

This thesis investigates demographic bias within the Re-LAION-5B dataset and in generative images produced by Stable Diffusion v1.5, with a direct focus on identifying spurious features in both sources and determining whether the generative model amplifies, alters, or mitigates the patterns observed in the dataset. The study adopts a classifier-based approach derived from prior research, training discriminative models to distinguish between Re-LAION-5B images and Stable Diffusion v1.5 generated images. A set of transformations is introduced to systematically reduce the semantic information available to the classifier and force reliance on specific low-level image features, thereby exposing dataset-specific spurious patterns that can be exploited for image source identification.

Building upon this classifier-based analysis, the images used in the discrimination task are further annotated according to gender and skin-tone attributes. This secondary analysis investigates whether the previously identified dataset-specific spurious features correlate with these sensitive demographic attributes, thereby identifying potential unintended shortcuts for demographic classification. To further contextualise these findings and assess the real-world implications of the observed biases, the demographic distributions of retrieved and generated images across 200 occupation categories are compared against real-world demographic statistics, revealing patterns of representation bias and potential amplification by the generative model.

Finally, based on the spurious features identified in the preceding analyses, this thesis implements and evaluates a combined set of inference-time interventions to debias Stable Diffusion v1.5 without requiring full model retraining. These

interventions reduce spurious demographic correlations across low-level, geometric, and semantic representations, showing that inference-time debiasing can produce measurable and substantive departures from the demographic and spurious feature profile of standard generation. By explicitly linking dataset characteristics, spurious feature learning mechanisms, and generative bias outcomes, this work contributes methodological insights for evaluating large-scale multimodal datasets and practical interventions for developing fairer text-to-image generation systems.

Acknowledgements

I wish to express my sincere appreciation to my supervisor Dr. Dylan Seychell, for his guidance and support throughout the completion of this Master's thesis. His expertise and mentorship played a crucial role in navigating the challenges encountered during this endeavour. Furthermore, I am deeply grateful to my parents, Reno and Graziella, and my brother Julian, for their encouragement and support throughout this academic pursuit.

Contents

Abstract	i
Acknowledgements	iii
1 Introduction	1
1.1 Problem Definition	1
1.2 Motivation	2
1.3 Aims and Objectives	2
1.4 Document Structure	3
2 Background & Literature Review	4
2.1 Re-LAION-5B Dataset	4
2.2 Stable Diffusion: Architecture, Training, and Dataset Lineage	6
2.2.1 Low-Rank Adaptation of Diffusion Models	7
2.2.2 Aesthetic Quality Assessment in Large-Scale Image Datasets	8
2.2.3 Face Detection Models and Evaluation Benchmarks	8
2.2.4 Facial Landmark Estimation: MediaPipe FaceMesh	9
2.3 Similarity Search in High-Dimensional Embedding Spaces	9
2.4 Spurious Features in Machine Learning	11
2.5 Demographic Annotation in Large-Scale Image Datasets	19
2.5.1 Human and Automated Annotation Practices	20
2.5.2 Measurement Bias and Label Uncertainty	21
2.5.3 Skin Tone Annotation Frameworks	21
2.5.4 Limitations of Visual Demographic Inference	23
2.6 Inference-Time Debiasing of Text-to-Image Diffusion Models	24
2.7 Occupations as a Probe for Demographic Bias in Text-to-Image Generation	28
2.7.1 Occupational Stereotyping in Language and Vision Systems	28
2.7.2 Occupational Bias in Text-to-Image Models	29
2.7.3 Occupation Classification: The ISCO-08 Framework	29
3 Methodology	30
3.1 Image Retrieval	30

3.1.1	Data Acquisition and Subsetting	30
3.1.2	Image Source Distribution	30
3.1.3	Image Validity Criteria	32
3.1.4	Decay Rate	32
3.1.5	Image Preprocessing and Embedding Computation	32
3.2	Profession-Conditioned Image Retrieval	33
3.2.1	Profession-Conditioned Subset Construction	33
3.2.2	FAISS Index Construction	34
3.2.3	Prompt Design	34
3.2.4	Negative Prompt Dominance Filtering	34
3.2.5	Embedding-Level Deduplication	35
3.2.6	Retrieval Parameters	35
3.2.7	Face Detection Model Selection	35
3.2.8	Human-Centric Structural Filtering	37
3.2.9	Filtering Outcomes and Acceptance Rate Distribution	37
3.2.10	Subset Design and Statistical Justification	38
3.3	Image Generation Pipeline	39
3.3.1	Model Selection	39
3.3.2	Prompt Design and Justification	40
3.3.3	Guidance Scale and Its Methodological Implications	40
3.3.4	Structural Validation Criteria and Unfiltered Control Subset	42
3.3.5	Structural Acceptance Rates for Generated Images	42
3.4	Demographic Annotation	43
3.4.1	Pre-processing	44
3.4.2	Gender Annotation	44
3.4.3	Skin Tone Annotation	46
3.4.4	Real-World Occupational Demographic Reference Data	49
3.4.5	Race to Skin Tone Distribution Mapping	50
3.5	Spurious Feature Probing Pipeline	51
3.6	Inference-Time Debiasing of Stable Diffusion v1.5	66
3.6.1	Experimental Design	66
3.6.2	Demographic Conditioning	67
3.6.3	Pose Conditioning via ControlNet	67
3.6.4	Colour Alignment	68
3.6.5	Generation Configuration	68
4	Evaluation	70
4.1	Demographic Distribution Analysis	70
4.1.1	Aggregate Gender and Skin Tone Distributions	70

4.1.2	Per-Profession and Per-ISCO-Group Analysis	71
4.2	Dataset Classification	78
4.3	Demographic Attribute Separability	83
4.3.1	Gender Artefact Analysis	83
4.3.2	Skin Tone Artefact Analysis	87
4.3.3	Connections to Dataset Classification Results	92
4.4	Debiasing Evaluation	93
4.4.1	Demographic Distribution Shift	94
4.4.2	Probing Accuracy on the Debaised Corpus	94
4.5	Limitations	98
4.6	Synthesis of Evaluation Findings	100
5	Conclusions and Future Work	102
5.1	Conclusions	102
5.2	Future Work	104
A	Occupation List	119
B	Code Repository	122

List of Figures

Figure 3.1	Overview of the methodology pipeline.	31
Figure 3.2	Precision-recall curves at IoU= 0.5 on the WIDER Face validation set for all four evaluated face detection models. RetinaFace and MediaPipe both exhibit a recall ceiling below 0.35 at near-perfect precision, indicating systematic detection suppression. SCRFD maintains high precision to approximately 0.6 recall before degrading. YOLOv12l-Face achieves the widest high-precision operating range, sustaining near-perfect precision to approximately 0.8 recall, consistent with its superior AP@0.5 and small-face detection performance.	36
Figure 3.3	Distribution of structural acceptance rates across 200 professions following CLIP retrieval and YOLO-based structural filtering of Re-LAION-5B images. The distribution is approximately unimodal with a mean of 33.51% and median of 33.67%, with a visible left tail reflecting professions whose online visual representation is dominated by non-person content.	38
Figure 3.4	Sample outputs from the ten model configurations screened during preliminary evaluation, generated using a fixed Doctor prompt at 512×512 resolution. Four samples are shown per configuration. Of these, only SD v1.5, SD v1.5 LCM-LoRA, and SD v1.2 satisfy the LAION training lineage criterion and are formally evaluated for model selection.	41
Figure 3.5	Distribution of structural acceptance rates across 200 professions for SD v1.5 generated images. The distribution is left-skewed with a mean of 66.38% and median of 71.00%, substantially higher than the Re-LAION-5B retrieval acceptance rate of 33.51%.	43
Figure 3.6	Structural transformations: depth map (Depth-Anything-V2), Canny edge map, and YOLOv8l-Pose skeleton overlay. The classifier input for pose is the 34-dimensional normalised keypoint vector, not the rendered overlay. .	63
Figure 3.7	Frequency domain transformations: low-pass reconstruction retaining smooth luminance gradients (cutoff $r = 40$) and high-pass residual isolating fine textures and compression artefacts. Both derived from a single FFT pass per image.	64

Figure 3.8	Spatial decomposition: pixel shuffle (global colour preserved, spatial structure destroyed), patch shuffle 16×16 (local texture preserved, global layout disrupted), and mean RGB (uniform per-channel colour fill).	64
Figure 3.9	Semantic and latent-space transformations: YOLOv8x object detection bounding boxes on white canvas (OpenImages V7), BLIP Large captions with gender-neutral remapping, and SD v1.5 VAE reconstructions.	65
Figure 3.10	Segmentation and segmentation-derived occlusion: ADE20K pixel-label map (Mask2Former), full person isolated (no background), person masked with white fill (background retained), and white silhouette on black background.	65
Figure 3.11	Rectangular occlusion: white rectangle replacing the person region (background retained) and inverse mask retaining only the occluded boundary region on a black background.	66
Figure 4.1	Aggregate gender distribution across all 200 professions for Re-LAION-5B and Stable Diffusion v1.5. The dashed line indicates demographic parity at 50%.	71
Figure 4.2	Aggregate skin tone distribution using the Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) three-bin grouping for Re-LAION-5B and Stable Diffusion v1.5.	72
Figure 4.3	Aggregate MST 1–10 skin tone distribution for Re-LAION-5B and Stable Diffusion v1.5. Shaded regions indicate the Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) bin boundaries.	72
Figure 4.4	Female proportion by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5. The dashed line indicates demographic parity at 50%.	76
Figure 4.5	Skin tone distribution (Light, Mid, Dark bins) by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5.	76
Figure 4.6	MST 1–10 score distribution (%) by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5. Cell values indicate the proportion of images in each group assigned to each MST score.	77
Figure 4.7	Representative debiased generation samples at a fixed seed.	95

List of Tables

Table 2.1	LAION-5B subset composition [2, 17, 18].	5
Table 3.1	Face detection model evaluation on the WIDER Face validation set. AP@0.5 is reported overall and stratified by face size. mAP@0.50:0.95 is the COCO-style average across IoU thresholds. [†] Evaluated using the <code>batch_face</code> library, which uses a different model checkpoint and preprocessing pipeline from the original RetinaFace implementation; results reflect this deployment configuration rather than the model's published capability.	36
Table 3.2	Per-image inference times for LAION-derived model configurations at 512×512 resolution. Times are the mean over four generation runs on an NVIDIA RTX 3060 Ti. [†] LCM-LoRA acceleration applied.	39
Table 3.3	Gender classification accuracy on segmented face-only crops from CCv2 and FACET. Overall, male (M), and female (F) accuracies are reported. DeepFace was not evaluated on CCv2 due to the absence of a batch inference interface. The Realistic Gender Classifier achieves the highest overall accuracy on CCv2 and the best balance between male and female subgroup accuracy.	45
Table 3.4	Per-MST-level accuracy of the Realistic Gender Classifier on CCv2 segmented crops. Overall, male (M), and female (F) subgroup accuracies and sample counts (N) are reported. Accuracy is highest for mid-range tones (MST 4–6) and degrades at both extremes, with female accuracy declining more sharply at darker tones.	46
Table 3.5	Global accuracy of baseline skin tone models. E = exact, ± 1 = within-one, B = three-bin. All trainable models were trained on CCv2.	47
Table 3.6	VGG-16 MST10 variant accuracy. E = exact, ± 1 = within-one, B = three-bin. BB = black background, AB = average skin tone background. Bold indicates the best value in each column.	48
Table 3.7	VGG-16 MST3 variant accuracy. Since the label space contains only three classes, exact accuracy and three-bin accuracy are identical. Bold indicates the best value in each column.	48
Table 3.8	Final selected model performance: VGG-16 MST10 Regression RGB Black-Background.	49

Table 3.9	Object detection model inference time on a 10,000-image subset of Re-LAION-5B at 640×640 pixel input resolution.	57
Table 3.10	Detection performance on the OpenImages V7 validation set.	57
Table 3.11	Label remapping configurations applied during object detection.	58
Table 3.12	Image captioning model throughput assessment on a single GPU.	59
Table 3.13	Classical caption quality metrics on the Karpathy test splits of COCO 2014 and Flickr30k. Low 4-gram diversity (underlined) indicates degenerate template-based generation.	59
Table 3.14	CLIP-based caption quality metrics on the Karpathy test splits of COCO 2014 and Flickr30k. CLIP-S measures image-caption alignment scaled to [0, 2.5]; RefCLIP-S is the harmonic mean of CLIP-S and maximum reference similarity.	60
Table 3.15	Pose estimation model evaluation on COCO 2017 validation set. Standard mode uses the model’s integrated person detector. Oracle mode provides ground-truth bounding boxes. FPS is measured end-to-end on a single GPU.	61
Table 3.16	Segmentation model evaluation on single-person images from the COCO 2017 validation set, used for model selection only.	62
Table 4.1	Ten professions with the strongest gender skew in Re-LAION-5B and Stable Diffusion v1.5. Gender skew is the absolute deviation of the female proportion from 0.5; higher values indicate stronger skew regardless of direction. F% = female percentage; M% = male percentage.	77
Table 4.2	Ten professions with the highest Light-bin (MST 1–3) skin tone proportion in Re-LAION-5B and Stable Diffusion v1.5.	78
Table 4.3	Dataset classification results per image transformation (three-class: Re-LAION-5B, Stable Diffusion v1.5, COCO). All models trained with ConvNeXt-Tiny, early stopping (patience = 2), seed = 0.	79
Table 4.4	Gender separability results per image transformation and dataset. Reported values correspond to the selected primary run, seed 0 unless otherwise noted. AUC is the primary metric; chance level is 0.500 for binary classification.	84
Table 4.5	Skin tone separability results per image transformation and dataset (MST 1–8). All probes trained with seed 0 unless otherwise noted. Macro-averaged AUC is the primary metric; chance level is 0.500.	88
Table 4.6	Per-class skin tone AUC per image transformation and dataset (MST 1–8). MST 9 and MST 10 are absent from both datasets and omitted. All probes trained with seed 0 unless otherwise noted.	88

Table 4.7 Gender probing AUC on the standard Stable Diffusion v1.5 corpus and the debiased corpus. Chance level is 0.500 for binary classification. Labels for the debiased corpus are derived from the generation configuration rather than automated annotation. † Standard SD v1.5 value reflects near-complete majority-class collapse; see Section 4.3.1 for detail. 96

Table 4.8 Skin tone probing macro-averaged AUC on the standard Stable Diffusion v1.5 corpus and the debiased corpus. The SD v1.5 column reports 8-class macro-averaged AUC (MST 1–8, as MST 9 and MST 10 are absent from the standard corpus); the debiased column reports 10-class macro-averaged AUC (MST 1–10, from the generation configuration). Chance level is 0.500 in both cases. Labels for the debiased corpus are derived from the generation configuration rather than automated annotation. 96

List of Abbreviations

AI Artificial Intelligence.

AP Average Precision.

AUC Area Under the Curve.

BLEU Bilingual Evaluation Understudy.

BLS Bureau of Labour Statistics.

CCv2 Casual Conversations v2.

CFG Classifier-Free Guidance.

CIDeR Consensus-based Image Description Evaluation.

CLIP Contrastive Language-Image Pre-training.

CNN Convolutional Neural Network.

COCO Common Objects in Context.

CSAM Child Sexual Abuse Material.

DDPM Denoising Diffusion Probabilistic Model.

FACET Fairness in Computer Vision Evaluation Benchmark.

FAISS Facebook AI Similarity Search.

FFT Fast Fourier Transform.

FPS Frames Per Second.

FST Fitzpatrick Skin Type.

GAN Generative Adversarial Network.

GPU Graphics Processing Unit.

Grad-CAM Gradient-weighted Class Activation Mapping.

ILOSTAT International Labour Organization Statistics.

IoU Intersection over Union.

ISCO International Standard Classification of Occupations.

ITI-GEN Inclusive Text-to-Image Generation.

LAB CIE L*a*b* Colour Space.

LAION Large-scale Artificial Intelligence Open Network.

LCM Latent Consistency Model.

LDA Latent Dirichlet Allocation.

LDM Latent Diffusion Model.

LoRA Low-Rank Adaptation.

mAP Mean Average Precision.

METEOR Metric for Evaluation of Translation with Explicit ORdering.

MIAP More Inclusive Annotations for People.

ML Machine Learning.

MLP Multilayer Perceptron.

MST Monk Skin Tone.

MSTE Monk Skin Tone Examples.

NSFW Not Safe For Work.

RGB Red, Green, Blue.

ROI Region of Interest.

SAM Segment Anything Model.

SCRFD Sample and Computation Redistribution for Efficient Face Detection.

SDv1.5 Stable Diffusion version 1.5.

UCE Unified Concept Editing.

URL Uniform Resource Locator.

VAE Variational Autoencoder.

WISCO WageIndicator Standard Classification of Occupations.

1 Introduction

1.1 Problem Definition

Text-to-image generative models have transformed visual content creation, with diffusion-based architectures achieving high image quality and prompt controllability. Among these, Stable Diffusion [1] has emerged as particularly influential due to its open-source availability, widespread deployment, and the extensive ecosystem of tools and fine-tuned variants built upon it.

These systems rely on massive web-scraped datasets to learn the mapping between text and visual content. Re-LAION-5B, containing over 5 billion image-text pairs harvested from the internet, represents one of the largest publicly available training corpora for such models [2]. However, the scale and automated nature of web scraping introduce systematic biases that reflect and potentially amplify societal inequities present in online content [3].

Demographic bias in AI is well-documented across natural language processing [4], machine learning [5], and computer vision [6], yet a gap remains between detecting bias in generative outputs and understanding the spurious features that cause and perpetuate it. Current research predominantly measures demographic distributions in generated images against desired benchmarks [7, 8], thereby treating models as black boxes and diagnosing symptoms without exposing the underlying mechanisms.

A fundamental yet under explored dimension involves spurious features, which are low-level non-semantic image characteristics such as colour statistics, texture patterns, edge distributions, frequency domain properties, and compression artefacts [9, 10]. When these features correlate with sensitive demographic attributes, models can exploit spurious statistical regularities rather than semantic content, creating a form of bias that operates beneath standard evaluation metrics. It also remains unclear how generative training transforms these spurious features, making it difficult to predict how training data biases manifest in deployed systems.

Existing mitigation approaches typically require expensive interventions: curating balanced training data [11], fairness-constrained training [12], or fine-tuning on carefully constructed demographic distributions [13]. These demand substantial computational resources, creating barriers for researchers working with pretrained models. Moreover, outcome-focused corrections may balance demographic distributions without addressing the spurious features that enable biased shortcuts, yielding superficial fixes that fail to generalise.

This thesis addresses these gaps through a mechanistic investigation of spurious features in Re-LAION-5B and Stable Diffusion v1.5. Rather than solely measuring

outcomes, it applies established methodologies to expose the spurious features that distinguish dataset sources and correlate with sensitive attributes, then evaluates targeted inference-time interventions that address root causes rather than symptoms. Crucially, this investigation reveals that Re-LAION-5B and Stable Diffusion v1.5 encode demographic information through qualitatively different mechanisms, a distinction that has direct consequences for how bias is measured, attributed, and mitigated.

1.2 Motivation

Influence and Impact: Re-LAION-5B serves as training data for multiple widely-deployed text-to-image models, while Stable Diffusion v1.5 has achieved widespread adoption across platforms such as DreamStudio and Hugging Face. Despite this pervasive influence, comprehensive analysis of spurious features in Re-LAION-5B and their propagation through Stable Diffusion v1.5 remains limited, making the study of bias mechanisms in these systems directly relevant to fairness at scale.

Computational Accessibility: Training or fine-tuning large-scale diffusion models is beyond reach for most researchers and practitioners. By focusing on inference-time methods applicable to frozen models, this research demonstrates fairness interventions accessible to smaller groups and organisations with limited computational budgets.

Reproducibility and Transparency: The open-source nature of Stable Diffusion v1.5 and the research availability of Re-LAION-5B enable findings to be independently verified, replicated, and extended, strengthening the credibility and impact of this work.

Beyond Outcome Measurement: Understanding why biases emerge, and specifically which spurious features enable biased shortcuts, supports interventions that address root causes rather than symptoms, enabling solutions that generalise rather than superficially correct specific evaluation metrics.

1.3 Aims and Objectives

The overarching aim is to apply established spurious feature detection methodologies to Re-LAION-5B and Stable Diffusion v1.5, systematically characterise the spurious features present in each, quantify their correlation with sensitive demographic attributes, and evaluate the effectiveness of existing inference-time debiasing

techniques on this dataset-model pair. This aim is addressed through the following set of objectives:

1. **Dataset Acquisition, Preparation, and Annotation.**

Retrieve occupation-conditioned images from Re-LAION-5B and generate a corresponding set using Stable Diffusion v1.5, annotating all images for gender and skin tone to support subsequent spurious feature and demographic analysis.

2. **Spurious Feature Detection.**

Detect and characterise spurious features that distinguish Re-LAION-5B images from Stable Diffusion v1.5 outputs using transformation-based classifier probing.

3. **Demographic Correlation Assessment.**

Probe whether identified spurious features encode demographic information differently across the two sources, and compare occupational demographic distributions against real-world statistics to assess representation bias and generative amplification.

4. **Inference-Time Debiasing.**

Implement and evaluate an inference-time debiasing pipeline for Stable Diffusion v1.5, assessing its effectiveness across demographic representation and spurious feature mitigation without model retraining.

5. **Synthesis of Bias Mechanisms and Mitigation.**

Synthesise findings to characterise how spurious features and demographic biases manifest in Re-LAION-5B and are transformed by the Stable Diffusion v1.5 generative process.

1.4 Document Structure

The remainder of this dissertation is structured as follows. Chapter 2 reviews the Re-LAION-5B dataset, Stable Diffusion v1.5, spurious feature learning, demographic annotation frameworks, and inference-time debiasing approaches for text-to-image models. Chapter 3 describes the image retrieval and generation pipelines, demographic annotation procedure, transformation-based probing framework, and debiased generation configuration. Chapter 4 presents the demographic distribution analysis across occupations, dataset classification results, demographic attribute separability findings, and debiasing evaluation. Chapter 5 summarises the main findings, discusses their implications for dataset and model assessment, and outlines directions for future work.

2 Background & Literature Review

This chapter reviews the technical foundations and prior work underpinning this thesis, organised around the core pillars of the investigation. Sections 2.1 and 2.2 introduce Re-LAION-5B and Stable Diffusion v1.5 as the primary subjects of study, covering their construction, training lineage, and relevant auxiliary components. Section 2.3 establishes the retrieval foundations for the dataset acquisition pipeline. Section 2.4 reviews spurious feature learning and the classifier-based and occlusion-based probing frameworks that form the primary methodological basis of this work. Section 2.5 addresses demographic annotation practices, skin tone frameworks, and their limitations. Finally, Sections 2.6 and 2.7 survey inference-time debiasing approaches and the use of occupation prompts as a probe for demographic bias in text-to-image generation.

2.1 Re-LAION-5B Dataset

Re-LAION-5B is a large-scale, openly released image-text dataset comprising approximately 5.85 billion image-text pairs harvested from the Common Crawl [2, 14]. It is a re-release of the original LAION-5B, produced in response to the discovery of CSAM within the original corpus. Rather than constituting a newly crawled dataset, Re-LAION-5B preserves the broad construction lineage of LAION-5B while applying improved safety filtering, including enhanced perceptual hashing and classifier-based detection, to remove harmful content [15].

The original LAION-5B construction pipeline parsed HTML pages indexed by Common Crawl and extracted image URLs paired with alt text. These candidate pairs were filtered using cosine similarity between image and text embeddings computed with OpenAI’s ViT-B/32 CLIP model for English pairs and a multilingual CLIP variant for all others [16], retaining pairs above 0.28 for English content and 0.26 for non-English content. This filtering step removed approximately 90% of the original 50 billion candidates [2]. Additional stages removed near-duplicate pairs, applied NSFW classifiers, and enforced minimum resolution and caption length requirements. The resulting LAION-5B dataset was stratified into three subsets summarised in Table 2.1. Re-LAION-5B retains this underlying LAION-5B construction lineage, but applies the subsequent safety re-filtering described above to identify and remove CSAM and related harmful content. The LAION-1B-nolang subset frequently contains images of products, places, or content dominated by proper nouns, for which no language could be reliably detected.

LAION-Aesthetics v2 5+ is a quality-filtered offshoot of LAION-2B-en rather

Subset	Approx. Size	Description
LAION-2B-en	2.32B pairs	English-language image-text pairs
LAION-2B-multi	2.26B pairs	Multilingual pairs (>100 languages)
LAION-1B-nolang	1.27B pairs	No detectable language (products, places, proper nouns)
LAION-Aesthetics v2 5+	~600M pairs	Quality-filtered subset of LAION-2B-en

Table 2.1 LAION-5B subset composition [2, 17, 18].

than a separate corpus, constructed using a linear classifier trained on CLIP ViT-L/14 embeddings to predict human aesthetic ratings [17]. It serves as the training corpus for Stable Diffusion v1.5.

Despite its scale, Re-LAION-5B remains subject to significant concerns. Birhane et al. [3] audited LAION-400M, a predecessor, and identified substantial quantities of misogynistic, pornographic, and racially derogatory content, confirming that automated filtering remains insufficient at billion-scale. Beyond explicit harmful content, the web-scraped construction process introduces systematic demographic imbalances, as the statistical composition of online imagery reflects societal inequities in content production and visual representation.

A significant limitation of the Re-LAION-5B release is the absence of pre-computed embeddings. The original LAION-5B included pre-computed CLIP embeddings that enabled efficient nearest-neighbour retrieval. In the re-release, these embeddings were absent because the embedding infrastructure was not reconstructed for the updated dataset composition [15]. Hence, any embedding-based retrieval requires CLIP embeddings to be computed independently from raw image downloads, introducing substantial overhead. For this thesis, this absence requires embeddings to be derived as part of the data acquisition pipeline.

A further consequence of Re-LAION-5B’s web-scraped construction is link attrition: the proportion of URLs that resolve successfully degrades over time as hosting services expire or content is removed [19]. Because Re-LAION-5B re-filters URLs originally crawled by Common Crawl, the gap between crawl date and download date introduces a decay component that cannot be precisely quantified without access to per-URL crawl timestamps. The surviving image pool is therefore not a random sample of the original corpus, but a temporally filtered subset shaped by domain-specific hosting practices and content removal patterns. Additionally, web-scraped datasets exhibit systematic domain concentration, whereby a disproportionate share of images originates from a small number of large hosting platforms. Birhane et al. [3] demonstrate that such structural biases propagate directly into the resulting dataset, and Zeng et al. [10] similarly show that large-scale web-scraped datasets are reliably distinguishable by neural networks based on their dataset-specific content distributions.

2.2 Stable Diffusion: Architecture, Training, and Dataset Lineage

Diffusion models offer several advantages over alternative generative architectures. Unlike GANs [20], they avoid mode collapse and training instability [21], and they bypass the surrogate loss limitation of VAEs [22] that constrains reconstruction fidelity. These properties have established diffusion models as a dominant architecture for high-quality image synthesis.

Stable Diffusion is a family of open-source text-to-image models based on the Latent Diffusion Model (LDM) framework of Rombach et al. [1]. The central innovation is performing the generative process in a compressed latent space rather than at full pixel resolution. A Variational Autoencoder [22] is trained to map high-resolution inputs into a lower-dimensional latent space; once trained, its encoder and decoder are frozen, and all generative modelling is performed within this latent space. This compression dramatically reduces the computational cost relative to pixel-space diffusion models such as DALL-E [23].

The diffusion process follows the DDPM framework [24], in which a forward process corrupts a clean latent by adding Gaussian noise over T timesteps until it approximates pure Gaussian noise. A learned reverse process, implemented as a U-Net [25], iteratively denoises a noisy latent to recover a clean sample. Text conditioning is introduced through cross-attention in the U-Net’s intermediate layers, where a pretrained CLIP text encoder [16] produces embeddings injected at multiple spatial resolutions

Classifier-Free Guidance

Classifier-Free Guidance (CFG) [26] is applied during inference to control the strength of text conditioning. During training, the model is stochastically shown the image without its text prompt, allowing the same network to learn both conditional generation and unconditional denoising. At inference, two forward passes are performed per denoising step and the final noise estimate is computed as:

$$\tilde{\epsilon}_{\theta}(z_t, c) = \epsilon_{\theta}(z_t, \emptyset) + s \cdot (\epsilon_{\theta}(z_t, c) - \epsilon_{\theta}(z_t, \emptyset)) \quad (2.1)$$

where z_t is the noisy latent at timestep t , c is the text conditioning, $\emptyset$ denotes the null unconditional token, and $s \geq 1$ is the guidance scale. Higher guidance scales produce outputs that adhere more strongly to the prompt, but at the cost of stochastic diversity.

Training Lineage

Stable Diffusion v1.5 was released by RunwayML as a continuation of training from v1.2, itself initialised from v1.1 weights produced by CompVis [18]. Training was conducted on LAION-Aesthetics v2 5+ using the AdamW optimiser with a batch size of 2048 at 512×512 pixel resolution. The training corpus is a direct derivative of LAION-2B-en and therefore inherits its web-scraped composition, demographic imbalances, and spurious statistical regularities documented in Section 2.1. The open-source availability of the model weights and the documented training data lineage makes Stable Diffusion v1.5 particularly well-suited for mechanistic bias analysis, as findings can be independently verified and directly linked to known properties of the training corpus. Later models in the Stable Diffusion family, including SDXL [27] and Stable Diffusion 3 [28], rely on proprietary training corpora whose composition has not been disclosed, substantially limiting independent bias auditing and reinforcing the methodological value of focusing on v1.5.

2.2.1 Low-Rank Adaptation of Diffusion Models

Low-Rank Adaptation (LoRA) [29] is a parameter-efficient fine-tuning technique in which the weight update ΔW for a matrix $W \in \mathbb{R}^{d \times k}$ is decomposed into two low-rank matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$, where $r \ll \min(d, k)$:

$$W' = W + \Delta W = W + AB$$

Because only A and B are trained while W remains frozen, the number of trainable parameters is reduced from $d \times k$ to $r(d + k)$. LoRA adapters can be fused directly into base weights at inference time, eliminating runtime overhead.

Latent Consistency Models (LCMs) [30] introduce a distillation-based objective that enables high-quality generation in two to four denoising steps, rather than the twenty to fifty required by standard DDPM samplers. LCM-LoRA [31] applies this distillation as a LoRA adapter rather than requiring full model retraining, implementing a consistency function that maps predictions at different noise levels to the same clean image, allowing the sampling trajectory to skip intermediate timesteps.

Crucially, LCM-LoRA’s training objective is purely geometric: the adapter is optimised to reproduce the output distribution of the base model more efficiently, with no training signal derived from demographic content, aesthetic labels, or semantic categories. The LCM distillation loss operates on the consistency of latent-space trajectories across noise levels, ensuring that predictions at different timesteps map to the same clean image. The adapter weights therefore encode information about the structure of the denoising process, rather than the content of the training images. This

distinguishes speed-based LoRAs from concept LoRAs or style LoRAs, which are explicitly trained to shift the model’s semantic prior toward a specific concept or appearance, and which consequently alter what the model generates. Consequently, applying LCM-LoRA to Stable Diffusion v1.5 does not introduce additional demographic signal and is therefore unlikely to confound measurements of the base model’s demographic prior, making it suitable for the generation pipeline used in this thesis.

2.2.2 Aesthetic Quality Assessment in Large-Scale Image Datasets

Aesthetic quality varies substantially in web-scraped datasets. To address this, Schuhmann [17] trained a linear predictor on CLIP ViT-L/14 embeddings using human preference ratings from the Simulacra Aesthetic Captions (SAC) dataset [32], producing a lightweight scalar aesthetic score. This predictor was used to construct the LAION-Aesthetics v2 subsets, with the 5+ threshold retaining images above a score of 5.0 out of 10.

In the context of evaluating generative outputs, aesthetic scoring serves a different purpose. Generated images may exhibit quality failures such as posterisation or large undifferentiated regions arising from incomplete denoising, none of which are captured by detection-based validation. A conservative threshold of 3.0 is adopted in this work as a minimum viability threshold, positioned well below the training cutoff to avoid introducing systematic upward quality bias into the generated subset while still rejecting genuinely corrupted outputs.

2.2.3 Face Detection Models and Evaluation Benchmarks

Face detection produces axis-aligned bounding boxes with confidence scores for human faces within an image. Modern detectors are predominantly single-stage architectures that perform classification and bounding-box regression jointly, offering substantially higher throughput than two-stage approaches [33]. The WIDER Face benchmark [34] is the standard evaluation dataset, comprising 32,203 images with 393,703 annotated faces across three difficulty tiers based on face size and occlusion. For applications requiring detection of clearly visible, unoccluded faces in generative or retrieval settings, Easy and Medium Average Precision are the more relevant indicators, as the Hard partition targets conditions unlikely to arise in such contexts.

RetinaFace [33] established a strong baseline through joint detection and five-point landmark regression within a Feature Pyramid Network. SCRFD [35] introduced sample and computation redistribution to produce a family of efficient detectors at reduced FLOPs. YOLOv8-face adapts the YOLOv8 architecture [36] for

face detection, offering competitive accuracy on Easy and Medium partitions with substantially higher throughput, making it well-suited for large-scale filtering pipelines where speed is a primary constraint.

For the filtering pipeline used in this work, the most relevant detector properties are single-face precision and inference throughput, since candidate images are processed at large scale. Small-face recall is a secondary concern because the downstream structural validation criteria require face regions large enough to support reliable FaceMesh landmark extraction.

2.2.4 Facial Landmark Estimation: MediaPipe FaceMesh

Facial landmark estimation recovers the positions of anatomically defined keypoints within a detected face region. MediaPipe FaceMesh [37] predicts 468 three-dimensional face landmarks from a single RGB image using a lightweight convolutional network, producing dense coverage across the full facial surface including eye contours, eyebrows, lips, nose, and outer face boundary.

FaceMesh serves a functionally distinct role from face detection within the structural validation pipeline. Detection confirms the presence and approximate location of a face; landmark estimation confirms that the detected region contains a geometrically coherent face with sufficient surface detail for downstream processing. A detected bounding box may enclose a face-like region that is too blurred, occluded, or poorly rendered to yield reliable landmarks, and FaceMesh failure on such regions provides a principled rejection criterion. The convex hull of the predicted landmarks also provides a precise polygonal mask for the facial skin region, enabling tightly cropped face extractions that exclude surrounding hair, clothing, or background, a property that is directly relevant when the extracted region is used for skin tone classification where non-skin pixels would introduce spurious colour signal.

This functional complementarity motivates the two-stage face processing pipeline adopted in this work: YOLO-based detection identifies candidate face regions efficiently at scale, and FaceMesh subsequently validates geometric coherence and produces the segmented face crop used in downstream demographic annotation.

2.3 Similarity Search in High-Dimensional Embedding Spaces

CLIP embeddings [16] are typically 512- or 768-dimensional floating-point vectors. Image retrieval is therefore framed as a nearest-neighbour search in this shared embedding space. When similarity is measured using cosine similarity, vectors are L2-normalised prior to indexing and inner product search is performed, since for unit vectors the two are equivalent.

Naive exact search scales as $O(Nd)$, where N is the number of indexed vectors and d is the embedding dimensionality, requiring $N \times d$ operations per query, which becomes computationally intractable at the scale of large web-scraped corpora such as Re-LAION-5B. Efficient similarity search therefore requires indexing structures that reduce the number of comparisons required per query.

The FAISS Library

The Facebook AI Similarity Search (FAISS) library [38] is a widely adopted open-source library providing indexing structures and algorithms for similarity search and clustering of dense vectors. It supports both CPU and GPU implementations and spans a spectrum from exact to approximate nearest-neighbour search.

Exact Search: IndexFlatIP

The simplest index, IndexFlatIP, performs exact inner product search with $O(Nd)$ complexity, exhaustively computing the inner product between a query and every indexed vector. Since no approximation is applied, the true nearest neighbours are guaranteed to be returned for every query. Despite being exhaustive, it substantially outperforms a naive linear scan through the use of hardware-optimised matrix operations.

Approximate Search Methods

For databases where exact search is too slow, FAISS provides approximate nearest-neighbour index types that trade retrieval exactness for substantially reduced query times. IndexIVFFlat partitions the embedding space into k Voronoi cells using k-means clustering, searching only the n_{probe} nearest cells at query time and reducing complexity to approximately $O(n_{\text{probe}} \cdot N/k)$ [39]. IndexHNSWFlat organises vectors into a Hierarchical Navigable Small World graph [40], yielding empirically logarithmic query complexity $O(\log N)$ with strong recall and low memory overhead. IndexIVFPQ combines inverted file partitioning with Product Quantisation [39], compressing vectors into quantised subvectors to enable billion-scale retrieval on commodity hardware at the cost of quantisation error.

Approximate search methods additionally introduce search error: the failure to visit index regions containing true nearest neighbours during traversal. For applications where retrieval correctness is critical, exact search with IndexFlatIP eliminates both error sources and is therefore adopted in this work to ensure retrieval-induced approximation error does not confound downstream bias measurements.

2.4 Spurious Features in Machine Learning

Defining Spurious Features and Spurious Correlations

A fundamental challenge in machine learning is the tendency of models to learn features that correlate with target labels in the training distribution but carry no causal or semantic relationship to the target concept. Such statistical associations are known as spurious correlations [41, 42], and the features that embody them are termed spurious features. Formally, a spurious correlation exists when a feature f co-occurs with a label y in the training distribution but this co-occurrence does not persist under distribution shift, such that $p_{\text{train}}(f | y) \neq p_{\text{test}}(f | y)$, or when f carries no causal information about y despite being statistically correlated with it. A model that exploits spurious features instead of causally relevant ones is said to learn a shortcut [43], a rational consequence of empirical risk minimisation since gradient-based optimisation will favour whichever feature reduces training loss most efficiently.

Taxonomies of Spurious Features in Computer Vision

Spurious features in visual data can be broadly categorised according to the level of the image hierarchy at which they reside and whether they reflect properties of the subject, the context, or the acquisition process.

At the pixel-statistical level, low-level image attributes including mean colour channel values, brightness distributions, contrast statistics, and jpeg compression artefacts can serve as reliable proxies for class labels when the dataset collection process introduces systematic correlations between acquisition conditions and class membership. Jo and Bengio [44] demonstrate that CNNs trained on benchmark datasets tend to learn spurious statistical regularities rather than high-level abstractions, achieving high accuracy even when these regularities bear no semantic relationship to the class concept. Geirhos et al. [45] further demonstrate that ImageNet-trained CNNs exhibit a strong texture bias, relying on texture for object classification rather than shape.

At the frequency-domain level, Wang et al. [46] demonstrate that CNNs can exploit high-frequency features imperceptible to the human eye as spurious shortcuts, and that adversarially introduced high-frequency perturbations can substantially alter model decisions without any visible change to image content. This sensitivity is particularly concerning in web-scraped datasets where different image origins carry systematically different frequency signatures arising from variation in camera hardware, processing pipelines, and compression settings.

At the scene-context level, spurious features can arise from statistical

regularities in scene composition rather than in the subject itself. Beery et al. [47] show that classifiers trained on images where animals consistently appear against certain background types fail severely when the same animal appears against an unfamiliar background. Xiao et al. [48] demonstrate that models achieve non-trivial classification accuracy when trained on background-only images with foreground objects masked, establishing that contextual scene statistics constitute a pervasive spurious signal. Torralba and Efros [49] show that models trained on one visual dataset and evaluated on another exhibit sharp performance drops, demonstrating that much of what models learn reflects dataset-specific regularities rather than generalisable visual concepts.

At the co-occurrence level, spurious correlations can be encoded in higher-level statistical regularities such as the tendency for certain object categories to co-occur with certain scenes, or for certain demographic attributes to be associated with particular activities. Zhao et al. [50] demonstrate that gender-stereotypical object co-occurrences in captioning datasets cause models to amplify those associations in their outputs. Singh et al. [51] further show that performance degrades substantially when objects appear without their typical co-occurring context, confirming that semantic co-occurrence regularities function as learned shortcuts.

At the structural level, classical edge detection algorithms such as the Canny edge detector [52] reduce an image to a binary boundary map capturing object contours while discarding colour and texture. Class-agnostic segmentation tools such as the Segment Anything Model [53] extend this to higher-quality contour representations. Monocular depth estimation encodes the relative spatial geometry of a scene without reference to object semantics. Because these structural properties are shaped by the contexts in which images are captured, they can carry dataset-specific or demographic-specific spurious signals that are not visible in pixel-level statistics alone. Brendel and Bethge [54] demonstrate that models operating solely on small independent image patches achieve surprisingly high accuracy on ImageNet, confirming that local structural features within small receptive fields are sufficient to support classification without any holistic understanding of image content.

At the person level, appearance, shape, spatial location, and pose may each independently correlate with demographic group membership when the collection conditions of a dataset introduce systematic differences in how different groups are photographed. This has direct consequences for fairness-through-blindness approaches, which attempt to remove protected attribute information by occluding the person in an image [55, 56]. Because person-level spurious features are intertwined with background signals and encoded across multiple representational layers simultaneously, such occlusion-based interventions may leave substantial demographic information intact, a limitation examined in detail in Section 2.4.

At the semantic-representation level, richer image representations that preserve

object identity while discarding pixel-level detail can carry dataset-specific spurious signals. Semantic segmentation maps retain the spatial distribution of scene categories while removing texture and colour. Object detection representations preserve object co-occurrence statistics while discarding internal pixel content. Image captions discard pixel-level visual information in favour of natural language, while still encoding semantic regularities reflecting the statistical composition of the underlying image distribution. VAE reconstructions, as implemented in latent diffusion architectures [1], provide a further diagnostic: if reconstructed images remain discriminable after compression, this suggests that the discriminative signal resides primarily in semantic or structural content rather than low-level acquisition artefacts. Taken together, these representation types demonstrate that spurious features are not confined to the pixel domain, but can permeate the full hierarchy of visual abstraction.

Spurious Features and Demographic Bias

When a spurious feature correlates with a sensitive attribute such as gender, race, or skin tone, a model that exploits this shortcut will systematically produce disparate outcomes across demographic groups. Buolamwini and Gebru [6] provide a foundational empirical demonstration: commercial gender classifiers trained on datasets overrepresenting lighter-skinned individuals exhibit substantially higher error rates for darker-skinned individuals, attributable in part to the model’s reliance on skin tone correlates as classification shortcuts. Mehrabi et al. [5] catalogue the mechanisms by which demographic shortcuts enter models, including representation bias (underrepresentation of certain groups in training data), measurement bias (features measured differently across groups), and aggregation bias (heterogeneous sub-populations pooled under a single model).

Critically, features that appear neutral, including colour statistics, pose distributions, and background scenes, can function as proxies for demographic attributes when dataset composition creates systematic correlations between low-level properties and group membership. Meister et al. [9] term these as gender artefacts and demonstrate that they are effectively impossible to fully remove from standard visual datasets without also removing information necessary for downstream tasks. This proxy mechanism means that debiasing interventions that remove explicit demographic signals may leave intact the spurious features that encode equivalent information, producing only superficial corrections.

Spurious Features in Text-to-Image Generative Systems

Spurious features manifest differently in text-to-image generative models. Whereas discriminative models learn to exploit spurious correlations for prediction, generative models trained on biased data learn to reproduce them: statistical regularities in the training corpus become embedded in the model’s learned image prior, causing it to synthesise spurious patterns even when the conditioning prompt contains no information that should elicit those patterns.

The training pipeline of latent diffusion models such as Stable Diffusion [1] illustrates this process. The model learns a joint distribution over image latents conditioned on text embeddings derived from CLIP [16]. Because CLIP is itself trained on web-scale image-text data, it can encode the same compositional and societal biases present in internet imagery [2, 3], and these biases are subsequently inherited by diffusion models conditioned on its embeddings. Spurious correlations between text tokens and low-level image features in the training data are absorbed into the diffusion model’s score function, causing those features to be hallucinated at generation time even in the absence of explicit demographic prompting.

Bianchi et al. [8] demonstrate this empirically, showing that text-to-image models not only reflect demographic under-representation present in their training data but actively amplify stereotypical associations: occupation prompts produce generated images with demographic distributions more extreme than those in real-world labour statistics. Luccioni et al. [7] extend this finding across a broader set of models, confirming that the amplification effect is systemic. However, neither study addresses which specific spurious features encode these biases, treating the model as a black box and measuring only demographic outcomes.

Understanding Bias in Large-Scale Visual Datasets

The taxonomy above provides a conceptual decomposition of spurious features across different levels of image representation, but it does not quantify their relative contribution to dataset-level bias. Zeng et al. [10] operationalise this problem through a transformation-based framework, aiming to identify the concrete forms of bias that make large-scale pre-training datasets distinguishable by neural networks. Their work builds on the dataset-classification task introduced by Torralba and Efros [49] and revisited by Liu and He [57], who found that three widely-used datasets, YFCC100M [58], CC12M [59], and DataComp-1B [60], collectively referred to as YCD, can be classified with 84.7% accuracy. This confirms dataset distinguishability but not which image properties are responsible, a gap Zeng et al. address through a transformation-based framework.

Transformation-Based Bias Quantification

Their central contribution is a transformation-based framework: apply a transformation that preserves one type of visual information while removing all others, train a dataset classifier on the transformed images, and use the resulting accuracy to measure how much bias that type of information carries. For each transformation, 1M images per dataset are used for training and 10K for validation using a ConvNeXt-Tiny model trained for 30 epochs. On unmodified images, a reference accuracy of 82.0% is achieved.

At the semantic level, segmentation using ViT-Adapter-Large with the ADE20K vocabulary achieves 69.8%; object detection using ViTDet-Huge with LVIS achieves 61.9%; and image captioning with LLaVA 1.5 achieves 63.8% for short captions and 66.1% for long captions. These results show that semantic content, specifically the categories of objects appearing in each dataset, is a primary source of dataset bias.

The structural transformations produce a striking result: Canny edge detection achieves 71.0%, SAM-based contour extraction reaches 73.2%, and monocular depth estimation with Depth-Anything-V2 yields 73.1%. These representations contain no object labels, yet they outperform most semantic transformations, showing that object shapes and spatial arrangements differ significantly across datasets and carry as much discriminative information as object category.

Pixel and patch shuffling experiments separate local structure from global spatial layout. Randomly shuffling individual pixels removes all spatial structure, reducing the classifier to colour statistics alone; accuracy falls to 52.2%, just above the 33.3% chance level. Shuffling 16×16 patches instead recovers nearly all discriminability at 81.2%. This indicates that local spatial structure within small image regions is sufficient to distinguish datasets, and that global composition matters much less than local texture and shape. A mean-RGB experiment compressing each image to a single average colour per channel achieves 48.5%, approximately 15 percentage points above chance, suggesting that even the global colour palette of images differs systematically across datasets. Frequency domain analysis shows high-pass filters yield 79.2% and low-pass filters yield 70.4%, indicating that spurious signals exist across the full frequency spectrum and cannot be eliminated by targeting a single frequency range.

A VAE experiment further contextualises these findings: passing images through the bottleneck of a pretrained Stable Diffusion VAE reduces accuracy from 82.0% to 77.4%, showing that low-level properties such as jpeg compression artefacts contribute relatively little to overall discriminability compared with semantic and structural content. Finally, combining two transformations together increases accuracy beyond what either achieves alone, for example combining pixel shuffling with SAM contours reaches 74.2% compared to 73.1% for SAM contours alone, demonstrating

that different types of spurious signal carry partly independent information. This finding is directly relevant to the multi-transformation approach used in this thesis.

Propagation of Bias into Generative Models

A central question for this thesis is whether biases in training data are preserved in generative model outputs. Zeng et al. test this by training separate unconditional Diffusion Transformer (DiT) models on each of the YCD datasets and classifying the resulting synthetic images. These generated images can be classified with 77.6% accuracy, nearly matching the 82.0% reference on real images, demonstrating that generative models reproduce the distributional biases of their training data to a substantial degree. A second experiment using a single pretrained SDXL-Turbo model conditioned on LLaVA-generated captions achieves 58.1%, confirming that semantic bias can be partly transmitted even when the generative model is not dataset-specific. Together, these results establish a clear link between training data composition and generative output, directly supporting the central hypothesis of this thesis.

This distinction motivates a two-stage bias pipeline central to the present work: biases originate in the training dataset and are subsequently encoded into the generative model, where they may be preserved or amplified during synthesis. Re-LAION-5B represents the source stage of this pipeline, and Stable Diffusion v1.5 represents the generative stage, making the relationship between their respective spurious feature distributions a primary object of analysis.

Semantic Explanation of Dataset Differences

To examine why the datasets are distinguishable, Zeng et al. apply Grad-CAM to the reference classifier, showing that the model focuses on specific objects to identify dataset origin. A logistic regression model trained on binary object-presence vectors achieves 52.0–52.4% accuracy depending on vocabulary, well above chance but well below the 82.0% of the neural classifier, showing that object identity alone does not fully explain dataset discriminability and that lower-level structural signals also play a role. Open-ended language analysis using LDA topic modelling characterises YFCC as outdoor and human-centred, DataComp as product-focused with clean backgrounds, and CC as a mix of the two.

These findings motivate the probe design adopted in this thesis, where transformations are selected to isolate semantic, structural, and low-level statistical components of the image. By evaluating classification performance under each transformation, the contribution of each feature type to dataset-specific discriminability can be quantified independently. When these spurious features correlate with demographic attributes, they provide a mechanism through which

demographic information can be inferred. The probe-based framework therefore enables not only the detection of dataset bias, but also the identification of feature types that act as proxies for demographic attributes.

Evaluation of Image Captioning Quality

The transformation-based framework of Zeng et al. requires generated captions to faithfully reflect image content, because a captioning model that produces degenerate descriptions would corrupt the semantic signal the transformation is designed to preserve. Caption-based representations discard pixel-level information, thereby isolating semantic content. If dataset classification remains above chance under this transformation, this strongly suggests that the discriminative signal resides in systematic differences in object co-occurrence, scene composition, or semantic structure rather than in low-level image statistics. Selecting an appropriate captioning model therefore requires a principled evaluation protocol. Reference-based n-gram metrics such as BLEU [61], METEOR [62], and CIDEr [63] assess surface-level lexical overlap with human-authored references, while diversity metrics [64] detect template collapse invisible to precision measures. At the semantic level, Sentence-BERT [65] provides a paraphrase-robust measure of meaning preservation, and CLIPScore [66] offers reference-free visual grounding by computing cosine similarity between CLIP image and text embeddings. RefCLIP-S extends this as the harmonic mean of CLIPScore and maximum reference similarity, achieving the highest correlation with human judgements of any single metric. No single metric is sufficient, and the failure modes of each motivate the multi-metric evaluation protocol used in this thesis.

Gender Artefacts in Visual Datasets

Meister et al. [9] systematically investigate the nature and prevalence of gender artefacts in large-scale computer vision datasets. A gender artefact is defined as a visual cue that is both learnable by a modern image classifier and interpretable, meaning that it corresponds to a visually interpretable property such as colour, shape, or scene composition. The study is motivated by a gap in the fairness literature: fairness-through-blindness techniques assume that removing a person's visible appearance from an image is sufficient to eliminate gendered signal. Meister et al. challenge this by asking to what extent gendered information can truly be removed from a visual dataset. The gender artefact model is treated as a diagnostic tool for understanding dataset correlations, rather than as a system for predicting gender identity; the gender labels used are derived from external annotator perception and are used solely to reveal distributional differences in the data.

Datasets and Model

The analysis focuses on MS-COCO [67] and OpenImages [68]. Gender labels for COCO are inferred from captions using keyword matching [50]; for OpenImages they come from the MIAP subset [69]. Both datasets are skewed male at 69.2% and 62.9% respectively. The gender artefact model uses a ResNet-50 [70] pretrained on ImageNet, with performance reported as AUC where 50% represents chance-level discrimination.

Resolution and Colour Artefacts

Images downsampled to 7×7 pixels, at which almost all semantic content has been removed, yield an AUC of 59.2% on COCO and 63.9% on OpenImages in colour. Repeating the experiment in greyscale, the AUC falls to near chance at 28×28 , showing that colour is the primary carrier of gender-correlated signal at low resolutions. A logistic regression classifier trained on just the three mean RGB channel values achieves 58.0% and 59.1% on COCO and OpenImages respectively, likely because of a compositional confound: male-labelled images are disproportionately set in outdoor sports contexts dominated by green tones, while female-labelled images more often feature close-up portraits with lighter average pixel values.

Person and Background Artefacts

A systematic occlusion protocol separates gender artefacts arising from the person from those arising from the background context. With all person pixels visible, the model achieves 93.4% AUC on COCO. Removing the background reduces this only slightly to 92.7%, confirming that person appearance is the dominant signal under normal conditions.

When the person’s appearance is replaced by a white segmentation mask but the background remains visible, the model still achieves 79.6% AUC, showing that the silhouette alone carries substantial gender-correlated information. Removing the background as well reduces performance to 74.8%, confirming that silhouette shape and position are meaningful gender artefacts independent of scene context. Training on the 17 COCO keypoint locations alone yields 64.8%, and a normalised pose-only representation still achieves 58.0%, establishing body pose as an independent gender-correlated artefact.

Under the MaskRect condition, which matches the occlusion strategy used by several prominent mitigation methods [55, 56], the AUC remains at 70.8% on COCO and 63.1% on OpenImages, directly challenging the core assumption of fairness-through-blindness approaches. Class Activation Map analysis explains why:

when the person is occluded, the model shifts attention onto background objects, associating indoor objects with female labels and outdoor objects with male labels.

Contextual Object Artefacts

A logistic regression classifier trained on a binary object-presence vector achieves 75.4% AUC on COCO, with objects most predictive of female-labelled images including hair dryers, handbags, and teddy bears, and those most predictive of male-labelled images including skateboards, ties, and snowboards. Crucially, gendered information is broadly distributed across the object vocabulary rather than concentrated in a few obviously gendered objects: 30 randomly selected objects from the 80 annotated COCO categories still yield 66.8% AUC, confirming that removing a small subset of objects is unlikely to meaningfully eliminate object-level artefacts.

Analysis of Fairness-Through-Blindness Methods

Applying the adversarial debiasing approach of Wang et al. [56] to COCO images reveals that 76.8% of removed pixels come from the background rather than the person, with outdoor scene objects such as giraffes, trains, and buses most frequently affected. Because gendered signals are distributed throughout the image rather than localised to the person, the method suppresses scene-level information needed for downstream tasks rather than the person-centric gendered information it was designed to target.

Discussion and Broader Implications

The authors draw three main conclusions from these findings. First, gender artefacts are so widely distributed across image structure that removing them without also removing task-relevant information is not feasible. Second, if a model can infer gender labels from mean pixel colour or object presence, then its predictions reflect the compositional biases of the dataset rather than gender expression in any meaningful sense. Third, the authors recommend moving away from fairness-through-blindness approaches towards fairness-aware methods that explicitly account for distributional differences across demographic groups.

2.5 Demographic Annotation in Large-Scale Image Datasets

Studying demographic bias in large-scale visual datasets requires annotating images with sensitive attributes such as gender and skin tone. Unlike object category labels, for which clear ground truth can in principle be defined, demographic annotations are

contested, because they sit at the intersection of biological, social, perceptual, and cultural factors. This section surveys annotation practices, examines measurement bias and label uncertainty, reviews skin tone annotation frameworks, and outlines the core limitations of visual demographic inference.

2.5.1 Human and Automated Annotation Practices

Demographic attributes in visual datasets have traditionally been labelled by human annotators. Early benchmarks such as Labelled Faces in the Wild [71] exhibited variation in gender and race but were not designed for fairness analysis. MS-COCO [67] includes the category person without structured demographic annotations, while OpenImages [68] adds subclasses such as man and woman without a fairness-driven protocol. The More Inclusive Annotations for People (MIAP) dataset [69] introduces non-binary gender presentation labels under a formal multi-annotator framework. These efforts demonstrate that human demographic annotation is feasible at scale, but methodologically complex, with disagreement and label uncertainty being intrinsic to the task.

Inter-annotator agreement for demographic labels is typically lower than for object categories, reflecting the interpretive ambiguity of socially constructed attributes. Aroyo and Welty [72] argue that annotator disagreement often encodes genuine differences in perception and interpretation, and that collapsing annotations to a single majority label can obscure meaningful variation.

At the scale of web-scraped datasets, manual annotation is not practically feasible. Large-scale studies therefore rely on automated demographic inference using pretrained classifiers. FairFace [73] is commonly used due to its improved demographic coverage relative to earlier benchmarks. The Casual Conversations v2 (CCv2) dataset is a purpose-built benchmark for evaluating automated demographic classifiers [74], comprising over 26,000 video clips from more than 5,000 consenting participants recorded across diverse lighting and environmental conditions. Demographic labels in CCv2 were provided directly by participants through self-reporting, while trained annotators labelled apparent skin tone using both the Fitzpatrick Skin Type and Monk Skin Tone scales. This dual-annotation design separates self-identified attributes from perceived attributes, providing a principled basis for evaluating whether automated classifiers capture perceived appearance rather than identity.

A complementary benchmark is the Fairness in Computer Vision Evaluation (FACET) dataset [75], providing annotations for approximately 32,000 images drawn from the Segment Anything 1 Billion (SA-1B) dataset, covering perceived gender presentation, perceived skin tone using the Monk Skin Tone scale, perceived age group, and additional fairness-relevant attributes. Annotations were produced by trained

human raters under a structured multi-stage pipeline and support systematic evaluation across demographic attributes and their intersections.

However, automated annotation introduces a form of structural circularity. Demographic classifiers are themselves trained on human-annotated datasets that encode the same perceptual subjectivities and representational imbalances described above. As demonstrated by Buolamwini and Gebru [6], performance disparities in commercial gender classifiers reflect underlying dataset imbalances. When such models are used to annotate new datasets, they propagate the statistical structure and potential biases of their training data into downstream analyses, transforming annotation bias from observable human disagreement into a systematic model-mediated measurement artefact that is no longer directly observable or correctable.

2.5.2 Measurement Bias and Label Uncertainty

Measurement bias arises when the label assignment process systematically distorts the metric it intends to measure [5]. Unlike object categories, demographic attributes lack stable visual ground truth and must be inferred from perceptual cues, making measurement error structured rather than random. Buolamwini and Gebru [6] show that darker-skinned females experience significantly higher misclassification rates than lighter-skinned males. When automated classifiers trained on such data annotate large-scale web datasets, these subgroup-specific error patterns become embedded in the generated labels.

These issues are particularly consequential when demographic labels function as covariates against which feature distributions are measured. Structured annotation error can produce apparent correlations between visual artefacts and demographic attributes that reflect classifier error patterns rather than properties of the underlying data. Without ground truth labels at scale, disentangling genuine distributional structure from measurement artefact becomes statistically intractable. Careful selection and documentation of annotation models is therefore essential when demographic attributes are used to audit spurious correlations in large-scale visual datasets.

2.5.3 Skin Tone Annotation Frameworks

The most widely used skin tone framework in computer vision is the Fitzpatrick Skin Type (FST) scale, originally developed in dermatology to classify sun-reactive skin response to ultraviolet exposure [76]. Its categories are anchored in self-reported photoreactivity rather than directly observable colour properties, creating a construct

mismatch when applied to image-based annotation. Subsequent work has documented compression of variation at the darker end of the scale [77], reducing sensitivity precisely where subgroup disparities are often most pronounced.

The Monk Skin Tone (MST) scale was developed to address these limitations as a ten-point ordinal framework explicitly designed for technology and machine learning applications [78]. Unlike Fitzpatrick, the MST scale is anchored to visual reference swatches intended for photographic assessment rather than self-reported sun-reactive history. A large-scale survey comparing Fitzpatrick, MST, and a 40-shade commercial palette found that Fitzpatrick was perceived as significantly less representative, particularly among darker-skinned participants, while the ten-point MST scale was perceived as comparably inclusive to the much finer-grained commercial palette [79].

Empirical work evaluating MST estimation under diverse lighting demonstrates substantial degradation in fine-grained ten-class performance, particularly in intermediate tone ranges [80]. When tones are grouped into broader light, medium, and dark categories, estimation stability improves markedly under varying lighting conditions. For large-scale web-scraped imagery where illumination is uncontrolled, grouped treatment provides greater robustness and reduces boundary-level annotation noise. Grouping in this context is not merely a simplification of the construct, but a strategy to reduce measurement variance that could otherwise obscure or artificially induce spurious correlations.

A further methodological risk is shortcut learning during skin tone estimation. Experimental evidence shows that full-image CNN models exploit background, clothing, or contextual artefacts correlated with skin tone labels rather than pigmentation itself, as revealed by Grad-CAM analyses, whereas models trained on segmented facial skin regions exhibit activations concentrated on skin pixels [81]. For this reason, skin tone classification in this thesis is applied to detected facial skin regions rather than to full images.

Schumann et al. [78] introduced the Monk Skin Tone Examples (MST-E) dataset comprising 1,515 images and 31 videos of 19 subjects spanning the full ten-point MST scale, photographed across a range of poses, expressions, and lighting conditions. Final annotations were provided by the creator of the MST scale, Dr Ellis Monk. The dataset was designed to support annotator training and to investigate annotation consistency under varying environmental conditions, and also enables evaluation of automated skin tone estimation methods across specific capture conditions rather than relying solely on aggregate accuracy metrics.

2.5.4 Limitations of Visual Demographic Inference

A fundamental limitation of visual demographic annotation is the distinction between perceived and self-identified attributes. Systems that infer gender from images predict whether a subject’s appearance conforms to culturally dominant norms of gendered presentation, rather than gender identity. Scheuerman et al. [82] demonstrate that commercial gender classification systems operationalise gender as a binary perceptual category and exhibit systematic failures on individuals whose presentation diverges from normative expectations. Consequently, when generative outputs are compared to demographic statistics such as labour force participation rates, the annotation variable and the statistical benchmark may refer to different constructs.

A second limitation concerns the instability of photometric inference under varying acquisition conditions. Skin tone classifiers are sensitive to illumination colour temperature, exposure, and post-processing. In uncontrolled web-scale datasets such as Re-LAION-5B, images are captured under varied conditions, creating a confound between acquisition artefacts and demographic inference quality. This issue is further compounded in synthetic imagery, where lighting is a learned prior embedded in the generative model rather than an external environmental factor.

A related conceptual distinction concerns race and skin tone. Race is a socially constructed category shaped by historical, political, and cultural processes that cannot be reliably inferred from visual appearance alone. Skin tone, by contrast, is a photometric property of visible surface pigmentation. While correlated with racial categorisation in many societies, the two constructs are not equivalent, and fairness audits that operationalise race through perceived visual categorisation introduce additional layers of annotator subjectivity [83]. Skin tone, though still perceptually mediated, is more directly grounded in observable colour properties and more closely aligned with the spurious features examined in this analysis.

Taken together, these limitations show that demographic labels derived from visual inference function as measurement instruments rather than ground-truth variables. Recognising this constraint is essential when interpreting spurious feature distributions in both web-scraped datasets and generative model outputs.

2.6 Inference-Time Debiasing of Text-to-Image Diffusion Models

Approaches to mitigating demographic bias in text-to-image diffusion models can be categorised according to the stage of the pipeline at which they intervene.

Training-time methods address bias through modifications to the training data or objective function [11, 12]. Fine-tuning methods adjust pretrained weights on demographic reference corpora [13]. Both families require substantial computational resources and high-quality annotated data, placing them beyond reach for practitioners working with pretrained deployed systems [84].

Inference-time methods operate on a frozen model during generation, requiring no retraining and no access to weights beyond those already publicly available. This family is particularly relevant for widely deployed open-source systems such as Stable Diffusion v1.5, where the only available intervention points are the conditioning signal, the sampling trajectory, and the model’s internal attention mechanisms. The methods surveyed below address demographic bias through three broad mechanisms: editing internal model representations via closed-form weight modifications, manipulating the text-conditioning signal through learned inclusive embeddings, and guiding the denoising trajectory through external signals targeting spatial structure or colour distribution.

Weight Editing: Unified Concept Editing

Unified Concept Editing (UCE) [85] directly edits the cross-attention projections of a diffusion model using a closed-form solution, without gradient-based training. The value projection matrices W_v of the U-Net’s cross-attention layers encode associations between text-token embeddings and the visual features attended to during denoising. UCE frames debiasing as a constrained least-squares problem over W_v at each attention layer: given target concept embeddings representing biased associations to be corrected and anchor embeddings representing unrelated content to be preserved, UCE computes a closed-form update ΔW_v that minimises change to anchor representations while steering target representations toward a more balanced demographic prior. This closed-form solution is derived by linearising the attention layer’s behaviour around the target token embeddings, requiring no forward passes beyond computing a small number of text embeddings. Because the edit is applied layer-wise and accumulated across attention layers, each layer contributes independently to the overall concept redirection, and the closed-form solution at each layer does not account for the downstream effects of edits applied to other layers.

A key strength of UCE is its scalability to concurrent edits across multiple

concepts, demonstrated across up to 35 occupational prompts alongside style erasure and content moderation within a single edited model. However, because the edit applies uniformly across all spatial positions and all prompts that activate the targeted concept representation, UCE cannot selectively debias outputs in specific image regions or under specific contextual conditions. The linear approximation underlying the closed-form solution also constrains the method’s ability to correct strongly non-linear biases embedded simultaneously across multiple layers. Moreover, it provides no mechanism for specifying a precise target demographic distribution.

Inclusive Prompt Embeddings: ITI-GEN

ITI-GEN [86] operates at the level of the text conditioning signal rather than the model weights. It begins from the observation that demographic attributes such as skin tone can be difficult to express unambiguously through natural language, and addresses this by replacing linguistic specification with image-grounded specification. The method learns a set of inclusive tokens, which are short, learnable embedding vectors inserted into the CLIP text encoder’s input space, optimised to produce generated images uniformly distributed across the categories of a target demographic attribute.

For each attribute category, a set of reference images is selected, and the corresponding token embedding is trained via two losses by backpropagating through the frozen diffusion model. A direction alignment loss encourages the learned token to move the text embedding toward the CLIP image embedding of the reference images for that category. A semantic consistency loss penalises divergence from the original prompt’s semantic content, ensuring the inclusive token supplements rather than overwrites the user’s intended generation context. For A attributes each with K categories, the method requires $A \times K$ reference image sets rather than K^A joint combinations, and the resulting tokens are composable at inference time by concatenation. ITI-GEN does not modify model weights and is directly compatible with ControlNet [87] and other conditioning frameworks.

A reproducibility study by Gallo Fernandez et al. [88] identifies two important limitations. First, ITI-GEN is susceptible to learning spurious proxy features from its reference images: if reference images for a demographic category are systematically correlated with background, clothing, or lighting, the learned token may encode these spurious cues rather than the target attribute itself. Second, the exponential growth of the joint attribute combination space makes inclusive generation for all joint combinations increasingly difficult as the number of attributes increases.

Self-Discovering Latent Directions: FairGen

FairGen [84] addresses a structural limitation shared by UCE and ITI-GEN: their dependence on externally curated reference datasets or annotation models to define target demographic directions. FairGen instead discovers demographic latent directions directly from the model through a self-supervised procedure, by attaching lightweight adapters to the cross-attention layers of the frozen U-Net and optimising them to learn the direction in the model’s latent feature space along which a target attribute varies. This is achieved by manipulating the noise-prediction function through a noise-composition objective that exploits the structure of the DDPM training process, allowing the latent direction for a given demographic attribute to be extracted without requiring any labelled demographic data.

Orthogonal regularisation between adapters for different attributes is a central design choice: without it, adapters trained on correlated attributes such as gender and age, which co-vary in training data, tend to encode overlapping directions that interfere destructively when activated together, degrading both fairness and image quality. At inference time, a distribution indicator tracks the demographic composition of the images generated so far in a batch and steers subsequent generation to correct any accumulated imbalance, enforcing fairness globally across a batch rather than independently per image. The primary limitation is that the discovered directions are model-specific and non-transferable, requiring the self-discovering training process to be repeated for each base model architecture.

Spatial Conditioning: ControlNet

ControlNet [87] adds fine-grained spatial conditioning to a pretrained diffusion model, enabling the structure of generated images to be constrained by auxiliary signals such as pose skeletons, depth maps, Canny edge maps, or semantic segmentation masks. The central architectural challenge is to introduce spatial conditioning without destabilising the large pretrained backbone. ControlNet resolves this through a trainable copy of the frozen U-Net encoder and middle block that receives the additional conditioning signal and processes it through the same encoder structure. The outputs are injected back into the frozen backbone via residual connections at each encoder level, allowing the decoder to integrate spatial information during the upsampling pass where image structure is synthesised.

To prevent the randomly initialised conditioning branch from corrupting the frozen backbone during early training, all residual connections are implemented as zero-initialised convolution layers. At the start of training, these produce zero output, so the backbone behaves identically to the unmodified model. As training proceeds, the zero convolution weights grow from zero in proportion to the gradient signal,

introducing conditioning information progressively. Multiple ControlNet signals can be composed at inference time by summing their residual contributions, enabling simultaneous constraint of multiple spatial properties without retraining.

In the context of debiasing, ControlNet is relevant because many demographic spurious features operate at the structural level. As documented by Meister et al. [9], body pose distributions, person-to-frame scale, and spatial framing patterns constitute significant gender artefacts that are not removed by purely semantic or colour-based debiasing interventions. By providing an explicit structural conditioning signal such as a neutral canonical pose skeleton, ControlNet can constrain the spatial properties of generated images independently of the semantic and demographic properties governed by the text prompt and inclusive token embeddings, allowing both forms of control to be composed without mutual interference.

Colour Distribution Guidance: SW-Guidance and Chamfer-Based Alignment

A further class of inference-time methods directly targets the colour statistics of generated images, motivated by the observation that text-to-image models trained on aesthetically filtered web-scraped corpora learn strong colour priors that may correlate with demographic attributes. Consistent with the spurious feature analysis of Meister et al. [9], who demonstrate that mean RGB channel values are sufficient to discriminate gender-labelled images above chance, colour distributions in training data can function as spurious proxies for demographic group membership.

SW-Guidance [89] is a training-free method that introduces colour conditioning into the diffusion sampling process by modifying the noise prediction at each denoising step. At each step, the gradient of the Sliced 1-Wasserstein distance between the colour distribution of the current partially denoised image and a reference palette is computed and added to the denoising network’s noise prediction, functioning as an additional guidance term analogous to CFG [26]. A notable practical limitation is that computing the guidance gradient requires back propagation through the entire denoising U-Net at every timestep, introducing substantial overhead that makes SW-Guidance impractical for large-scale image generation.

Shum et al. [90] propose an alternative colour alignment approach that avoids backpropagation through the denoising network. Their method projects diffusion terms, either the partially denoised latent representation or the predicted clean image $\hat{x}_0$, directly into a conditional colour manifold defined by a reference palette, using the Chamfer distance to establish correspondences between the colour point clouds of the generated and reference images:

$$d_{\text{Chamfer}}(X, Y) = \sum_{x \in X} \min_{y \in Y} \|x - y\|_2^2 + \sum_{y \in Y} \min_{x \in X} \|x - y\|_2^2 \quad (2.2)$$

At each denoising step where colour alignment is active, an iterative matching procedure computes the Chamfer-optimal correspondence and takes a gradient step to project the predicted $\hat{x}_0$ toward the reference distribution, which is then used in place of the original predicted clean image in the standard DDPM recomposition formula. Alignment is applied only during early denoising timesteps, reflecting the empirical observation that global colour structure is established early in the denoising trajectory while fine structural and semantic details emerge during later steps. After a configurable point in the denoising process, colour alignment is disabled to avoid corrupting fine-grained content formed during later denoising steps.

2.7 Occupations as a Probe for Demographic Bias in Text-to-Image Generation

Having established how bias manifests in datasets and propagates through generative models, the next requirement is an appropriate probing framework for measuring these effects empirically. Occupations represent one of the most widely validated analytical lenses for this purpose. Job titles carry historically accumulated societal associations between professional roles and demographic attributes, extensively documented in real-world labour statistics and deeply reflected in the large-scale text and image corpora on which modern models are trained. Neutral occupation prompts therefore provide a transparent and interpretable window into a model’s learned demographic priors, making them a standard experimental instrument across both natural language processing and text-to-image generation research.

2.7.1 Occupational Stereotyping in Language and Vision Systems

Evidence that AI systems acquire and reproduce occupational stereotypes is well established across model families. Nadeem et al. [91] demonstrate through their large-scale StereoSet benchmark, covering 120 profession-related target terms, that all major pretrained language models including BERT, GPT-2, RoBERTa, and XLNet exhibit significant stereotypical preferences in profession-related contexts. Notably, they find a consistent positive relationship between a model’s language modelling ability and the strength of its stereotypical biases, suggesting that stereotype acquisition is a systematic consequence of learning from real-world text distributions. This is directly relevant to the present work because the CLIP text encoders that condition text-to-image models are pretrained on precisely such distributions, providing a theoretically grounded pathway through which occupational stereotypes may propagate into generated imagery.

Harrison et al. [92] demonstrate that human annotators produce sport-related names more frequently for male athletes than for female athletes performing similar activities, with an average of 46% sport-related names for male subjects versus 35% for female subjects. Since large vision-language models are trained on image-text datasets, biased naming patterns in those datasets can be learned by the models. These annotation asymmetries therefore provide a direct mechanism through which activity, role, and occupational concepts become demographically skewed in model representations.

2.7.2 Occupational Bias in Text-to-Image Models

Within text-to-image generation, Naik and Nushi [93] provide a comprehensive audit showing that occupation prompts reliably surface severe representational biases in both DALL-E v2 and Stable Diffusion v1. Generating images with neutral prompts such as *a photo of a doctor* and annotating outputs for perceived gender, race, and age, they find that model outputs deviate dramatically from real-world occupational demographics as reported by the US Bureau of Labor Statistics. Occupations such as chief executive officer and computer programmer yield images that are almost exclusively male and white, while nurse and housekeeper yield almost exclusively female images. Chuang et al. [94] further demonstrate that these biases are embedded in the structure of the CLIP text embedding space itself, with demographic directions geometrically separable from occupational content, confirming that occupation prompts are a particularly reliable and interpretable probe for the demographic biases arising from web-scale training data.

2.7.3 Occupation Classification: The ISCO-08 Framework

For occupation-based probing to produce generalisable results, the prompt set should be drawn from a principled taxonomy covering the full breadth of occupational space rather than over-representing stereotypically gendered or racialised roles selected by convenience. The International Labour Organization’s International Standard Classification of Occupations (ISCO-08) [95] provides this foundation, organising occupations into a four-level hierarchy of ten major groups, 43 sub-major groups, 130 minor groups, and 436 unit groups.

In this thesis, ISCO-08 serves as the organising framework for the occupation prompt set. Sampling across all ten major groups ensures bias measurements reflect the model’s behaviour across occupational space rather than a small subset of strongly stereotyped roles, and facilitates comparison with prior work that has likewise adopted ISCO-08 as a reference point [93].

3 Methodology

This chapter describes the experimental pipeline underlying the empirical analyses presented in this thesis. Figure 3.1 summarises the six principal stages and their interdependencies.

Section 3.1 describes Re-LAION-5B image retrieval and embedding computation. Section 3.2 details the construction and filtering of profession-conditioned image subsets across 200 occupations. Section 3.3 describes the Stable Diffusion v1.5 generation pipeline and validation procedure. Section 3.4 presents the gender and Monk Skin Tone annotation pipeline. Section 3.5 describes the transformation-based probing framework, and Section 3.6 presents the inference-time debiasing pipeline.

3.1 Image Retrieval

This section describes the data acquisition process applied to Re-LAION-2B-en-research-safe, covering shard subsetting, image validity criteria, URL decay, and CLIP embedding computation.

3.1.1 Data Acquisition and Subsetting

Re-LAION-2B-en-research-safe is distributed as 128 shards of approximately 3.62 GB each. Due to storage and computational constraints, processing was restricted to four randomly selected shards, yielding 65,552,809 URL–caption pairs. Because LAION shards are generated during distributed web crawling without semantic ordering, shard index does not correspond to content category, domain, or temporal structure; random selection therefore supports treating the subset as representative of the broader corpus. Under a binomial sampling model, this scale yields a worst-case 95% margin of error of approximately $\pm 0.04\%$ for proportion estimates. Nonetheless, unknown crawl-time biases such as domain saturation effects cannot be excluded without access to LAION crawl metadata, and all findings are reported as representative estimates rather than exact population parameters.

3.1.2 Image Source Distribution

Analysis of valid image origins showed that 18.79% of successfully retrieved images came from five domains: `cdn.shopify.com` (3,053,782 images), `i.pining.com` (2,585,519), `i.ebayimg.com` (966,820), `images-na.ssl-images-amazon.com` (886,054),

Methodology Overview

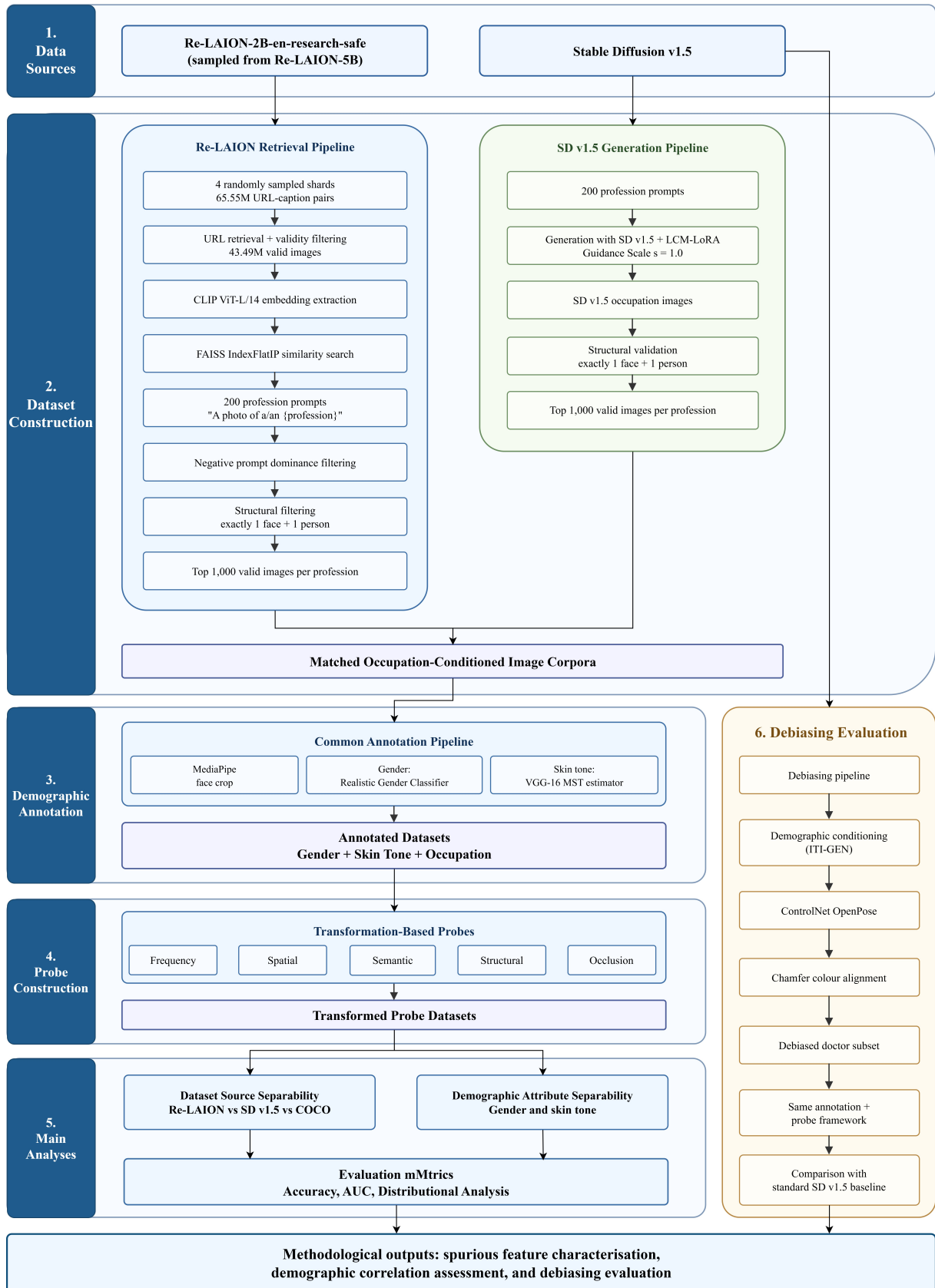


Figure 3.1 Overview of the methodology pipeline.

and `thumbs.dreamstime.com` (678,516). This concentration indicates strong domain centralisation, with large commercial platforms contributing disproportionately to the retrieved pool. Product imagery and stock photography may therefore be overrepresented relative to content from smaller independent domains.

3.1.3 Image Validity Criteria

Since Re-LAION-5B distributes URL-caption pairs rather than image binaries, each URL had to be resolved before downloading the image. An image was retained only if it satisfied all of the following conditions: a GET request completed successfully with an HTTP 2xx status code within a 10-second timeout; the response body was successfully decoded and converted to RGB; and embedding computation succeeded. Where embedding computation failed for a batch, recursive binary splitting was applied to isolate and exclude individual corrupted images without falling back to serial processing.

3.1.4 Decay Rate

From 65,552,809 attempted URLs, 43,491,938 images were retrieved and validated, yielding a decay rate of:

$$\text{Decay Rate} = 1 - \frac{43,491,938}{65,552,809} \approx 0.337$$

This level of link attrition is consistent with prior observations of web-scale dataset degradation [2, 19]. Link decay is not necessarily uniform: if domains or content types become unavailable at different rates, the surviving image pool may be skewed relative to the original dataset, introducing survivorship bias that cannot be quantified post-hoc.

3.1.5 Image Preprocessing and Embedding Computation

All images were preprocessed using the canonical `openai/clip-vit-large-patch14` pipeline: resizing to 224×224 pixels with aspect-ratio preservation, centre cropping, and CLIP-specific channel normalisation. CLIP image embeddings were extracted using the same model (embedding dimensionality: 768; patch resolution: 14×14) under `torch.no_grad()` in evaluation mode with GPU acceleration.

Each embedding vector $\mathbf{e}$ was L2-normalised:

$$\mathbf{e}_{\text{norm}} = \frac{\mathbf{e}}{\|\mathbf{e}\|_2}$$

This enforces unit-length representations and enables cosine similarity retrieval via inner-product equivalence, as described in Section 3.2.2. Results were incrementally flushed to Parquet files with checkpointing for fault tolerance and resumability.

While the original LAION-5B filtering pipeline employed CLIP ViT-B/32, this work uses the higher-capacity ViT-L/14 variant, which offers 768-dimensional embeddings, finer spatial granularity, and stronger semantic alignment in retrieval benchmarks. Since the objective is bias analysis rather than replication of LAION filtering thresholds, using a higher-capacity backbone is methodologically appropriate. The pipeline was executed on an NVIDIA RTX 3060 Ti GPU with 64 GB of system RAM over approximately two to three months.

Images were accessed exclusively via publicly available URLs furthermore, no authentication mechanisms were bypassed, no image binaries were redistributed, and no attempt was made to identify individuals. Analysis was conducted only at aggregate statistical levels to characterise bias, not to make individual-level demographic predictions or deploy demographic classification systems.

The retrieval process has five primary limitations: only 4 of 128 shards were processed; unknown crawl-time biases cannot be excluded without LAION crawl metadata; 33.7% of URLs were unavailable; five platforms account for 18.79% of valid images; and embeddings remain dependent on the CLIP encoder and its pretraining biases.

3.2 Profession-Conditioned Image Retrieval

This section describes how the 43 million retrieved embeddings were used to construct 200 profession-conditioned subsets, covering FAISS indexing, retrieval prompts, filtering, and the statistical justification for the 1,000-image-per-profession design.

3.2.1 Profession-Conditioned Subset Construction

Following embedding computation, profession-conditioned imagery was selected as the primary analytical lens because occupational prompts provide an interpretable probe for learned demographic priors, as established in Section 2.7. A set of 200 professions was aligned with ISCO-08 categories to ensure systematic coverage across skill levels and domains; the full list is provided in Appendix A.

An image is considered valid if it contains exactly one clearly detectable face associated with an identifiable person bounding box and satisfies the spatial and anatomical consistency criteria used in both pipelines. The retrieval-specific filtering

procedure is described in Section 3.2.8, and the corresponding generation validation procedure is described in Section 3.3.4.

3.2.2 FAISS Index Construction

To enable similarity-based retrieval over the 43 million L2-normalised embeddings, a FAISS IndexFlatIP index and position-to-path mapping were constructed per shard. Since embeddings were L2-normalised during extraction, inner product search over IndexFlatIP is equivalent to cosine similarity retrieval. Index construction processed each Parquet file row-group by row-group, with atomic checkpointing for crash recovery. At retrieval time, shard indexes were streamed sequentially, aggregated, and globally re-ranked by similarity score.

3.2.3 Prompt Design

The retrieval template used for each profession was:

"A photo of a/an {profession}"

The grammatical article was assigned programmatically based on the profession name. Gender-neutral phrasing was used to avoid reinforcing gender priors during retrieval.

3.2.4 Negative Prompt Dominance Filtering

A negative prompt dominance filter suppressed semantically irrelevant retrievals. For each candidate image, its CLIP embedding was compared against embeddings derived from the following negative prompts: Cartoon, NSFW, Sex, Naked, Clothing, Object, Sign, Logo, Document, Paper, Page, Animal. A candidate was rejected if the maximum cosine similarity to any negative concept satisfied:

$$\text{sim}_{\text{neg}} \geq \text{sim}_{\text{pos}} - \delta$$

where sim_{pos} is the cosine similarity to the target profession prompt and $\delta = 0.01$ is a dominance margin. This enforces strict positive dominance: an image is retained only when the target profession prompt exceeds every negative concept by more than the margin.

3.2.5 Embedding-Level Deduplication

To prevent duplicates within a profession’s result set, each candidate embedding was hashed by byte representation and duplicate hashes were discarded before accumulation. Cross-profession deduplication was not enforced as the same image could be relevant to multiple professions.

3.2.6 Retrieval Parameters

For each profession, the top 125,000 candidates were retrieved per shard and aggregated globally. A cosine similarity threshold of 0.15 removed near-random matches while preserving a large candidate pool for structural filtering. Following structural validation, the top 1,000 retrieval prompt aligned images were retained per profession. This two-stage design decouples semantic relevance from structural suitability.

3.2.7 Face Detection Model Selection

Face detection is a critical gating step in both retrieval and generation pipelines: images without a clearly detectable face are rejected before demographic annotation, as systematic failure on specific face scales or poses would introduce filtering bias. Four candidate models were evaluated on WIDER Face [34], reporting AP@0.5 overall, AP@0.5 by face size, and COCO-style mAP. Table 3.1 summarises all results and Figure 3.2 shows the corresponding precision-recall curves.

MediaPipe [96] achieved only 0.117 AP@0.5 with zero small-face recall, consistent with its design as a conservative single-face detector for mobile inference. RetinaFace [33], evaluated via the `batch_face` library, achieved 0.319 AP@0.5 with near-zero small-face recall; the precision-recall curve reveals systematic detection suppression rather than a precision-recall trade-off, and the library-level deviation from the original implementation limits interpretability. SCRFD [35] achieved 0.844 AP@0.5 overall but its small-face AP of 0.614 falls substantially below YOLOv12l-Face. YOLOv12l-Face achieved the highest AP@0.5 (0.905), highest mAP@0.50:0.95 (0.556), and best small-face AP (0.783). Its precision-recall curve sustains near-perfect precision to approximately 0.8 recall, wider than SCRFD, which begins degrading at around 0.6 recall. YOLOv12l-Face was therefore selected as the face detection model for all filtering and annotation pipelines.

Model	AP@0.5	mAP@0.5:0.95	Small	Medium	Large
YOLOv12l-Face	0.905	0.556	0.783	0.915	0.948
SCRFD	0.844	0.478	0.614	0.884	0.954
RetinaFace [†]	0.319	0.176	0.011	0.376	0.712
MediaPipe	0.117	0.040	0.000	0.015	0.497

Table 3.1 Face detection model evaluation on the WIDER Face validation set. AP@0.5 is reported overall and stratified by face size. mAP@0.50:0.95 is the COCO-style average across IoU thresholds. [†] Evaluated using the `batch_face` library, which uses a different model checkpoint and preprocessing pipeline from the original RetinaFace implementation; results reflect this deployment configuration rather than the model’s published capability.

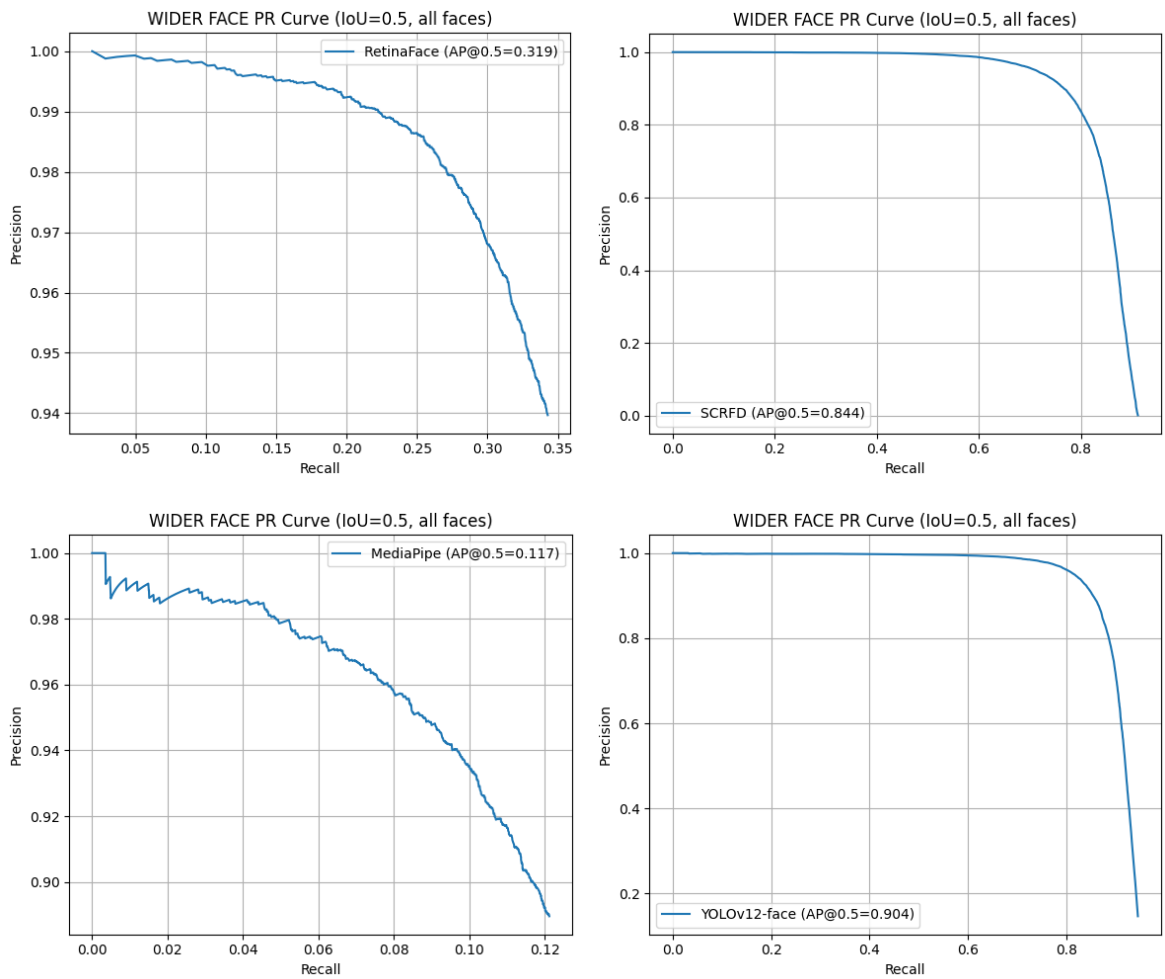


Figure 3.2 Precision-recall curves at IoU= 0.5 on the WIDER Face validation set for all four evaluated face detection models. RetinaFace and MediaPipe both exhibit a recall ceiling below 0.35 at near-perfect precision, indicating systematic detection suppression. SCRFD maintains high precision to approximately 0.6 recall before degrading. YOLOv12l-Face achieves the widest high-precision operating range, sustaining near-perfect precision to approximately 0.8 recall, consistent with its superior AP@0.5 and small-face detection performance.

3.2.8 Human-Centric Structural Filtering

CLIP-based retrieval does not guarantee clearly identifiable individuals; therefore, a multi-stage structural validation pipeline was applied to enforce human-centric suitability. Images were loaded in batches using TurboJPEG for JPEG inputs, with OpenCV as fallback. Face detection required exactly one detected face with confidence ≥ 0.6 ; person detection required at least one detected person with confidence ≥ 0.5 .

Following face detection, MediaPipe FaceMesh was applied to the detected face region. Images for which FaceMesh failed to return landmarks were discarded. For successful FaceMesh detections, landmarks defined a convex hull mask from which a tightly cropped face ROI was extracted with 10-pixel padding. Spatial consistency was then verified by requiring the face centroid to lie within a person bounding box, or IoU between the face and person bounding boxes to be ≥ 0.3 , with a face-to-person area ratio exceeding 0.02. Images with a valid face and successful FaceMesh segmentation were retained even when person detection was imperfect, avoiding unnecessary rejection where the face was clearly detectable but the person bounding box was slightly misaligned.

3.2.9 Filtering Outcomes and Acceptance Rate Distribution

Across all 200 professions, 25,125,679 candidate images were submitted for structural validation, of which 8,420,684 satisfied all criteria, yielding a mean acceptance rate of 33.51% (median 33.67%, standard deviation 10.71%).

Figure 3.3 illustrates the distribution of per-profession acceptance rates. The distribution is approximately unimodal with a visible left tail, consistent with the expectation that CLIP retrieval quality is profession-dependent: occupations associated with clear single-person visual conventions yield more structurally valid results than those whose online representation is dominated by equipment, environments, or group scenes. The 66% rejection rate confirms that CLIP retrieval alone is insufficient for demographic analysis.

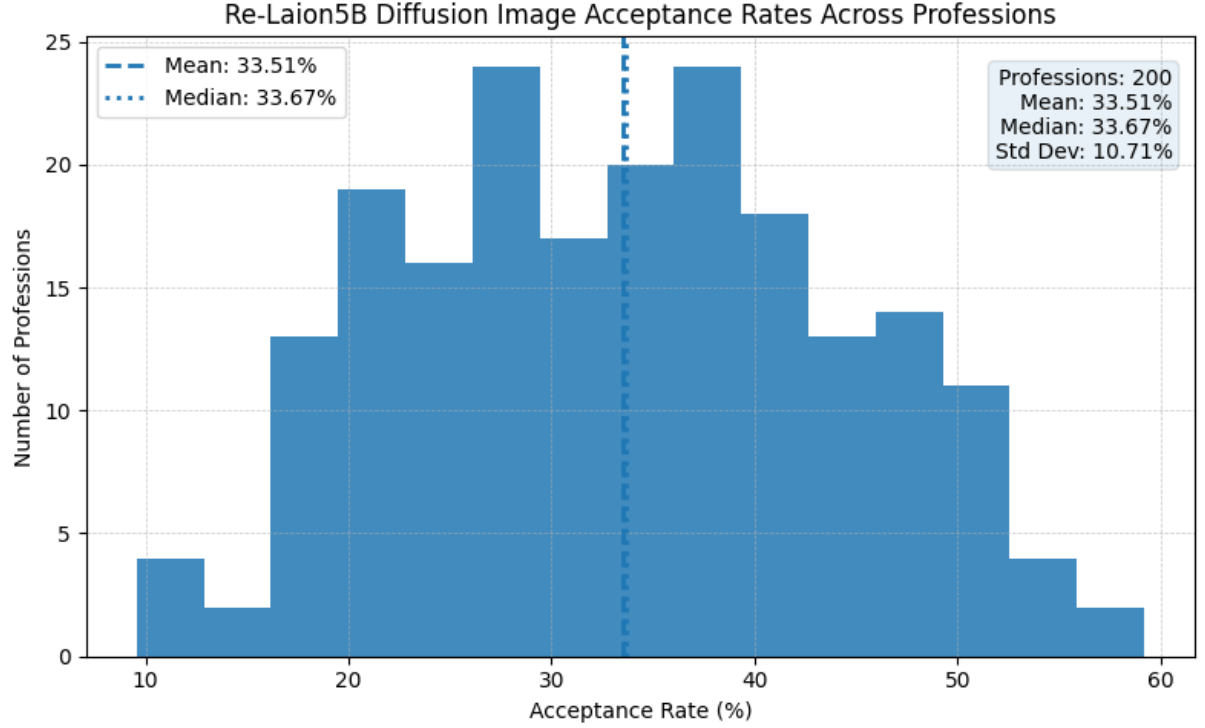


Figure 3.3 Distribution of structural acceptance rates across 200 professions following CLIP retrieval and YOLO-based structural filtering of Re-LAION-5B images. The distribution is approximately unimodal with a mean of 33.51% and median of 33.67%, with a visible left tail reflecting professions whose online visual representation is dominated by non-person content.

3.2.10 Subset Design and Statistical Justification

For each of the 200 professions, 1,000 images were retained following structural validation, yielding a balanced profession-conditioned subset of 200,000 images. Under a binomial model for binary feature presence, a per-profession sample of $n = 1,000$ yields a worst-case 95% margin of error of:

$$\text{MOE} = z_{0.975} \cdot \sqrt{\frac{p(1-p)}{n}} = 1.96 \cdot \sqrt{\frac{0.25}{1000}} \approx \pm 3.1\%$$

This margin is sufficient to detect meaningful cross-profession differences in spurious feature prevalence. Aggregating across all 200 professions yields a 200,000-image subset, for which binary proportion estimates have a worst-case 95% margin of error of approximately $\pm 0.22\%$. Equal sampling prevents high-frequency professions from dominating aggregate statistics, but means aggregate prevalence estimates reflect an equal-weighted occupational average rather than the natural profession frequency distribution within Re-LAION-5B.

3.3 Image Generation Pipeline

This section describes the synthetic corpus generated using Stable Diffusion v1.5 with LCM-LoRA acceleration, covering model selection, prompt design, guidance scale rationale, and structural validation.

3.3.1 Model Selection

A corresponding synthetic image set was required to compare training data with generative model outputs. Model selection was governed by two criteria: traceability to a documented LAION-derived corpus and throughput sufficient to generate 200,000 images within available constraints.

Applying the traceability criterion first, candidate models were restricted to those with a publicly documented training lineage derived from Re-LAION-5B. Three configurations satisfied this requirement: Stable Diffusion v1.5 [18], SD v1.5 with LCM-LoRA [18, 97], and Stable Diffusion v1.2 [98], all trained exclusively on LAION-Aesthetics v2 5+ [17]. Subsequent iterations were excluded: Stable Diffusion 2.0 and 2.1 employed a different text encoder architecture [99] introducing confounds; SDXL, Stable Diffusion 3, and later releases were trained on proprietary datasets of undisclosed composition [27, 28], severing the documentable link to Re-LAION-5B that this analysis requires.¹

Among the three qualifying configurations, final selection was based on inference throughput. Per-image inference times are reported in Table 3.2. SD v1.5 with LCM-LoRA achieved 1.5 seconds per image, compared to 5.6 seconds for standard SD v1.5 and 9.1 seconds for SD v1.2. At 200,000 images, this corresponds to approximately 83 hours versus 311 hours, making the LCM-accelerated configuration the only feasible option within the available compute window.

Model	Training Corpus	Time (s/image)
SD v1.5 LCM-LoRA [†] [18, 97]	LAION-Aesthetics v2 5+	1.5
SD v1.5 [18]	LAION-Aesthetics v2 5+	5.6
SD v1.2 [98]	LAION-Aesthetics v2 5+	9.1

Table 3.2 Per-image inference times for LAION-derived model configurations at 512×512 resolution. Times are the mean over four generation runs on an NVIDIA RTX 3060 Ti. [†]LCM-LoRA acceleration applied.

Sample outputs from each configuration are shown in Figure 3.4. SD v1.5 and SD v1.5 LCM-LoRA both produce consistently full-body, anatomically plausible

¹Additional architectures screened during preliminary evaluation — SSD-1B [100], PixArt-Alpha [101], and Segmind Vega [102] — were excluded solely on the basis of training data provenance.

compositions, with SD v1.5 LCM-LoRA retaining structural reliability at substantially reduced inference cost and only a marginal reduction in fine texture detail. SD v1.2 produces outputs of comparable quality but at 9.1 seconds per image, offering no throughput advantage.

SD v1.5 was selected as the base model, loaded with the latent-consistency/lcm-lora-sdv1-5 adapter [97] and fused for fast inference. As reviewed in Section 2.2.1, LCM-LoRA is used here as a throughput optimisation rather than as a demographic or semantic intervention. The default DDPM scheduler was replaced with LCMScheduler, enabling high-quality generation through consistency distillation [31]. A four-step inference schedule was used throughout; preliminary testing found that fewer steps introduced visible distortion, while additional steps yielded no measurable improvement. Generation was performed in half-precision (FP16) on GPU.

3.3.2 Prompt Design and Justification

The prompt template used for all profession-conditioned generation was:

```
full-body realistic photo of a {profession}, standing, centred,
in a professional setting appropriate for their occupation,
sharp focus, 35mm lens, natural professional lighting
```

Full-body framing was specified to override Stable Diffusion’s strong prior toward portrait-style close-ups [1] and to ensure compatibility with downstream structural validation, which requires a full detected person bounding box [103]. Standing and centred positioning reduces ambiguity from edge-cropping artefacts and multiple subjects [104]. An abstract contextual phrase was used instead of a specific environment, as explicit environment tokens can amplify demographic stereotypes [7, 13]. Sharp focus improves edge fidelity for face and person detection under low-step sampling [31], a 35mm focal length token promotes naturalistic full-body framing [105], and neutral professional lighting suppresses noise artefacts and supports aesthetic quality scores above the downstream threshold [13].

3.3.3 Guidance Scale and Its Methodological Implications

The classifier-free guidance scale was fixed at $s = 1.0$ for all validated generation runs. At $s = 1$, no guidance amplification is applied and the model produces outputs conditioned purely on its learned prior. Standard Stable Diffusion deployment often uses $s = 7-7.5$; this was avoided because, as described in Section 2.2, elevated guidance can amplify demographic or compositional bias encoded in the learned prior.



Figure 3.4 Sample outputs from the ten model configurations screened during preliminary evaluation, generated using a fixed Doctor prompt at 512×512 resolution. Four samples are shown per configuration. Of these, only SD v1.5, SD v1.5 LCM-LoRA, and SD v1.2 satisfy the LAION training lineage criterion and are formally evaluated for model selection.

By fixing $s = 1$, any detected demographic bias represents a conservative lower bound relative to higher deployment-scale guidance settings. If bias is detectable under minimal guidance, it is attributable to model weights and training data rather than guidance-scale amplification.

3.3.4 Structural Validation Criteria and Unfiltered Control Subset

Unlike Re-LAION-5B retrieval, generation provides no guarantee of compositional suitability. Common failure modes include multiple subjects, partially visible bodies, disconnected or misaligned faces, and low-frequency blurred renderings from low-step inference, thereby making structural validation necessary. Generation proceeded asynchronously: a ValidationWorker pool decoupled CPU-bound validation from GPU generation, allowing validation to run in parallel.

An image was accepted only if it satisfied the following conditions. The YOLO person detection model must detect at least one bounding box with confidence ≥ 0.4 . The YOLO face detection model must detect exactly one face with confidence ≥ 0.4 . MediaPipe FaceMesh must successfully return at least 10 landmarks for the face region, from which a convex hull mask and tightly cropped face ROI with 10-pixel padding were derived. For spatial and anatomical consistency, a composite score was computed per detected person bounding box:

$$\text{score} = 10 \cdot \mathbf{1}[\text{centroid_inside}] + 2 \cdot \mathbf{1}[\text{upper_half}] + \min\left(\frac{A_{\text{face}}}{A_{\text{person}}}, 0.05\right) \cdot 20$$

where $\mathbf{1}[\text{centroid_inside}]$ indicates that the face centroid lies within the person bounding box, $\mathbf{1}[\text{upper_half}]$ indicates the face centroid lies in the upper half of the person bounding box, and $A_{\text{face}}/A_{\text{person}}$ is the face-to-person area ratio. Images were rejected if no valid spatial association was established or if the face-to-person area ratio was below 0.002. Finally, each accepted image was scored using a CLIP-based aesthetic regressor (ViT-L-14 backbone, laion2b_s32b_b82k pretraining) and images with an aesthetic score below 3.0 were rejected, as described in Section 2.2.2.

3.3.5 Structural Acceptance Rates for Generated Images

Across all 200 professions, the mean structural acceptance rate for generated images was 66.38% (median 71.00%, standard deviation 16.60%), compared to 33.51% for Re-LAION-5B retrieval. The substantially higher acceptance rate is consistent with the effect of full-body prompt engineering, which biases the diffusion model toward compositions satisfying single-person detection and face-person spatial association

requirements. Nonetheless, per-profession variability remains substantial, indicating that certain occupations activate compositional priors that systematically resist full-body single-subject framing. The systematic difference in mean acceptance rate between retrieved and generated subsets must be considered when comparing demographic or spurious feature prevalence across the two data sources.

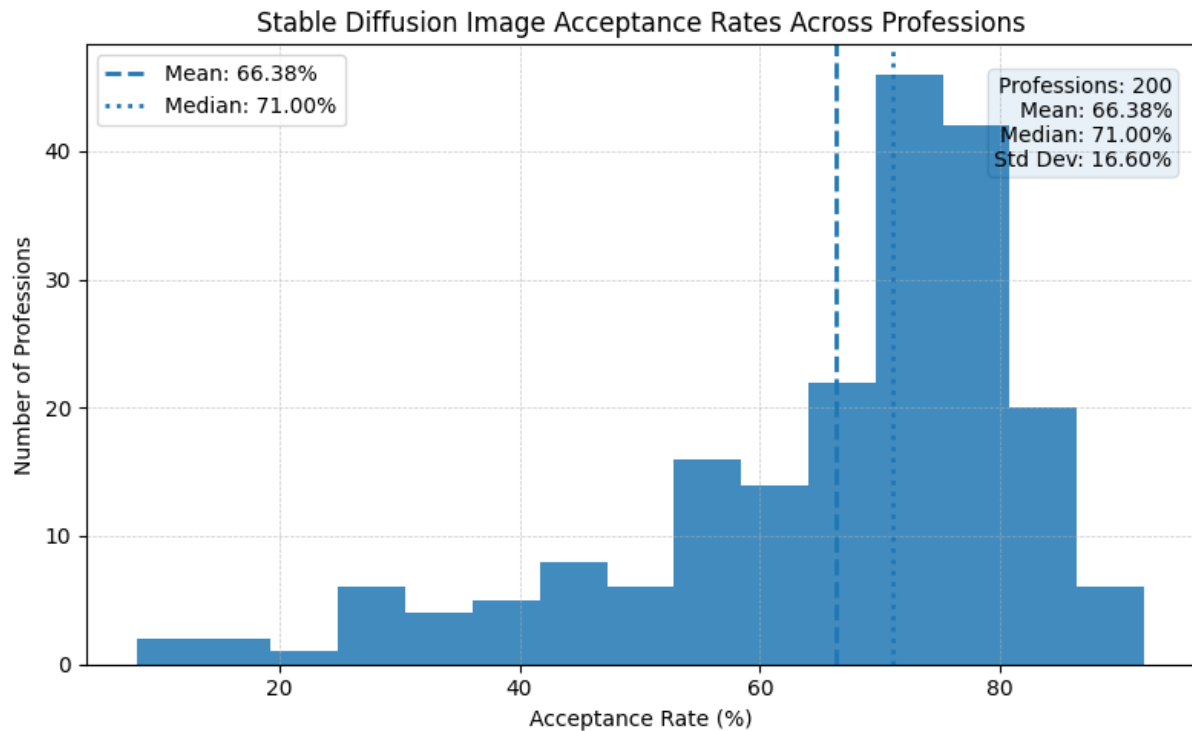


Figure 3.5 Distribution of structural acceptance rates across 200 professions for SD v1.5 generated images. The distribution is left-skewed with a mean of 66.38% and median of 71.00%, substantially higher than the Re-LAION-5B retrieval acceptance rate of 33.51%.

3.4 Demographic Annotation

This section describes the unified annotation pipeline applied to both the Re-LAION and Stable Diffusion corpora. It covers face pre-processing, gender classification, Monk Skin Tone regression, and the construction of real-world occupational demographic reference baselines from BLS and ILOSTAT data. All demographic labels are treated as model-derived proxies of perceived attributes rather than ground truth, and are used exclusively to analyse distributional correlations; the limitations of this approach are discussed in Section 2.5.4.

3.4.1 Pre-processing

All images were segmented using MediaPipe FaceMesh [37]. FaceMesh fits a dense 468-landmark mesh to the detected face region; a convex hull was computed from the landmarks, and pixels outside the hull were replaced with a uniform black fill, yielding a tightly cropped, background-removed face image. This black-background configuration was selected following the skin tone model ablation described in Section 3.4.3. The same FaceMesh configuration and face crop from structural validation were reused here. Applying segmented face crops rather than full bounding-box crops prevents models from exploiting background cues that may correlate incidentally with demographic attributes [81], and ensures that accuracy differences across candidate models reflect genuine differences in representational capacity rather than the image region each model receives.

Prior to FaceMesh pre-processing, an additional agreement-based filter was applied only to the FACET benchmark [75] for skin tone model evaluation. The agreement ratio for each sample was computed as the proportion of total annotator votes assigned to the majority skin tone label; only samples with an agreement ratio of at least 0.2 were retained. This threshold was selected as the minimum viable filter to suppress low-agreement samples that would otherwise inflate per-model error rates, while avoiding the substantial reduction in available sample count that a higher threshold would produce.

3.4.2 Gender Annotation

Candidate Models

Five off-the-shelf gender classifiers were evaluated on segmented face crops: HuggingFace Gender Classifier [106], a ViT-B/16 model fine-tuned for binary gender classification; the Realistic Gender Classifier [107], a SigLIP-backbone classifier trained on high-quality portrait imagery; DeepFace [108], a multi-attribute facial analysis library; InsightFace [109], a face analysis toolkit with an integrated demographic attribute head; and FairFace [73], a classifier trained on a demographically balanced dataset. Segmented face crops were used for all gender model evaluations to minimise background-derived demographic cues.

Evaluation Protocol

Models were evaluated on CCv2 [74] and FACET [75], both using the segmented face-only crops produced by the pre-processing pipeline described in Section 3.4.1. DeepFace was evaluated exclusively on FACET due to the absence of a batch inference

interface. Accuracy was computed as the proportion of correct binary predictions and decomposed by male and female subgroup.

Results

Table 3.3 summarises overall and per-class accuracy across models on segmented face-only crops from CCv2 and FACET. The Realistic Gender Classifier achieved an overall accuracy of 88.6% on CCv2, with a male–female gap of 6.2 percentage points. This gap is substantially smaller than those of the HuggingFace classifier (40.2 pp), FairFace (29.6 pp), and DeepFace on FACET (55.4 pp), all of which exhibit a systematic tendency to classify ambiguous or darker-toned faces as male. InsightFace achieved the smallest male–female gap (2.1 pp on CCv2) but at the cost of an 11.5 percentage point reduction in overall accuracy, indicating that its more balanced per-class rates arise from lower male recall rather than improved female recall.

Model	CCv2			FACET		
	Overall	M	F	Overall	M	F
HF Gender Classifier	0.715	0.963	0.561	0.789	0.877	0.569
Realistic Gender Clf.	0.886	0.924	0.862	0.823	0.875	0.695
InsightFace	0.771	0.784	0.763	0.772	0.821	0.643
FairFace	0.801	0.983	0.687	0.834	0.906	0.653
DeepFace	—	—	—	0.780	0.938	0.384

Table 3.3 Gender classification accuracy on segmented face-only crops from CCv2 and FACET. Overall, male (M), and female (F) accuracies are reported. DeepFace was not evaluated on CCv2 due to the absence of a batch inference interface. The Realistic Gender Classifier achieves the highest overall accuracy on CCv2 and the best balance between male and female subgroup accuracy.

Per-Skin Tone Performance of the Selected Model

Table 3.4 disaggregates the Realistic Gender Classifier’s CCv2 accuracy by MST level, quantifying the degree to which annotation quality varies across the demographic dimension the thesis analyses.

Overall accuracy peaks at MST 5 (90.6%) and declines toward both the lightest and darkest tones. Male accuracy remains consistently high across all MST levels while female accuracy degrades sharply at darker tones, falling to 48.8% at MST 9, consistent with the intersectional accuracy disparities documented in prior work [6]. These limitations must be accounted for when interpreting gender-disaggregated results at the darker end of the skin tone range. Despite these disparities, the Realistic Gender

Classifier offered the best combination of overall accuracy, male–female balance, and inference throughput and was therefore selected for final gender annotation.

MST	N	Overall	M Acc.	F Acc.
1	1,190	0.756	0.977	0.681
2	13,337	0.870	0.968	0.833
3	33,732	0.893	0.946	0.870
4	38,273	0.893	0.939	0.871
5	55,797	0.906	0.918	0.895
6	23,074	0.879	0.884	0.875
7	6,438	0.826	0.906	0.761
8	4,183	0.803	0.925	0.680
9	1,671	0.662	0.948	0.488
10	168	0.726	1.000	0.570

Table 3.4 Per-MST-level accuracy of the Realistic Gender Classifier on CCv2 segmented crops. Overall, male (M), and female (F) subgroup accuracies and sample counts (N) are reported. Accuracy is highest for mid-range tones (MST 4–6) and degrades at both extremes, with female accuracy declining more sharply at darker tones.

3.4.3 Skin Tone Annotation

Scale Selection

Building on the skin tone annotation frameworks reviewed in Section 2.5.3, the Monk Skin Tone (MST) scale [78] was adopted over the Fitzpatrick scale [76]. The Fitzpatrick scale was developed to predict ultraviolet radiation sensitivity and has been criticised for conflating clinical and perceptual dimensions and for providing insufficient resolution at dark skin tones [77, 78]. A large-scale survey by Heldreth et al. [79] found that Fitzpatrick was perceived as significantly less inclusive by darker-skinned respondents, while MST was rated comparably to a 40-shade commercial palette despite its coarser granularity.

Skin tone predictions were produced as ten-point MST scores and also collapsed into the three-bin grouping used by Matias and Neto [81]: Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10). The three-bin representation is used for aggregate bias analyses as it is more robust to annotation noise under variable imaging conditions, while full ten-point MST distributions are reported where finer inspection is warranted.

Candidate Models

Four skin tone estimators were evaluated: the SkinTone Classifier Library [110], an off-the-shelf tool deriving MST estimates from colour histograms; a RandomForest

histogram-based regressor trained on CCv2 following Matias and Neto [81]; DenseNet-121 [81] trained for MST classification in LAB colour space; and VGG-16 [80] fine-tuned for MST estimation and subsequently ablated across four design axes. All trained models were evaluated on a held-out CCv2 validation split, the Monk Skin Tone Estimation (MSTE) benchmark, which contains controlled skin tone examples under varying illumination conditions, and FACET, using exact accuracy, within-one accuracy, and three-bin accuracy.

Baseline Results

Table 3.5 reports baseline model accuracy. The SkinTone Library assigns near-zero probability to MST 1–3 across all datasets, making it unsuitable for reliable light-tone coverage. RandomForest generalises poorly beyond its training distribution, concentrating predictions in MST 4–6 and failing at the extremes. DenseNet-121 generalises more evenly across the tone range, but VGG-16 exceeds it on all CCv2 metrics and was therefore selected for the ablation study.

Model	MSTE			CCv2			FACET		
	E	± 1	B	E	± 1	B	E	± 1	B
SkinTone Library	0.112	0.281	0.425	0.084	0.298	0.603	0.073	0.239	0.421
RandomForest	0.132	0.335	0.410	0.346	0.756	0.700	0.176	0.550	0.484
DenseNet-121	0.210	0.517	0.564	0.303	0.692	0.697	0.210	0.529	0.576

Table 3.5 Global accuracy of baseline skin tone models. E = exact, ± 1 = within-one, B = three-bin. All trainable models were trained on CCv2.

VGG-16 Ablation Study

Ablation testing was conducted across four binary design dimensions, yielding sixteen VGG-16 variants evaluated under MST10 and MST3 label spaces. The four axes were output formulation (classification versus regression), colour space (RGB versus CIE $L^*a^*b^*$), label granularity (MST10 versus MST3), and background treatment (black background versus average skin tone background fill). Tables 3.6 and 3.7 report results for all variants.

Mode	Space / BG	MSTE			CCv2			FACET		
		E	± 1	B	E	± 1	B	E	± 1	B
Class.	RGB / BB	0.161	0.423	0.577	0.336	0.717	0.735	0.172	0.465	0.587
Class.	RGB / AB	0.217	0.445	0.529	0.341	0.636	0.717	0.119	0.384	0.547
Class.	LAB / BB	0.228	0.493	0.568	0.346	0.697	0.723	0.145	0.437	0.544
Class.	LAB / AB	0.162	0.445	0.492	0.283	0.669	0.674	0.148	0.442	0.583
Regr.	RGB / BB	0.159	0.424	0.566	0.375	0.804	0.778	0.165	0.471	0.593
Regr.	RGB / AB	0.132	0.367	0.528	0.374	0.784	0.783	0.173	0.499	0.572
Regr.	LAB / BB	0.146	0.423	0.549	0.350	0.776	0.734	0.185	0.532	0.587
Regr.	LAB / AB	0.163	0.494	0.556	0.370	0.778	0.775	0.207	0.567	0.596

Table 3.6 VGG-16 MST10 variant accuracy. E = exact, ± 1 = within-one, B = three-bin. BB = black background, AB = average skin tone background. Bold indicates the best value in each column.

Mode	Space / BG	MSTE		CCv2		FACET	
		Exact	± 1	Exact	± 1	Exact	± 1
Class.	RGB / BB	0.557	0.939	0.771	0.994	0.583	0.983
Class.	RGB / AB	0.545	0.945	0.765	0.993	0.581	0.972
Class.	LAB / BB	0.478	0.883	0.723	0.957	0.534	0.944
Class.	LAB / AB	0.540	0.901	0.727	0.990	0.502	0.908
Regr.	RGB / BB	0.627	0.992	0.801	1.000	0.640	0.996
Regr.	RGB / AB	0.584	0.978	0.799	1.000	0.619	0.997
Regr.	LAB / BB	0.578	0.996	0.749	1.000	0.604	0.997
Regr.	LAB / AB	0.526	0.995	0.777	0.999	0.598	0.998

Table 3.7 VGG-16 MST3 variant accuracy. Since the label space contains only three classes, exact accuracy and three-bin accuracy are identical. Bold indicates the best value in each column.

Model Selection

The ablation yields three conclusions. First, the regression formulation outperforms classification on the metrics most relevant to downstream analysis: in the MST10 setting, RGB regression achieves CCv2 within-one accuracy of 80.4% and three-bin accuracy of 77.8%, compared to 71.7% and 73.5% for the classification variant. Regression produces a continuous output rounded to the nearest integer at inference time, distributing prediction probability smoothly across adjacent MST classes rather than committing fully to a single discrete class, which is methodologically preferable when the downstream analysis relies on three-bin groupings. Second, RGB outperforms LAB as the input colour space, suggesting that the CCv2 training distribution already provides sufficient implicit lighting variation to support

luminance-robust representations. Third, black background consistently matches or outperforms the average skin tone background across regression variants, suggesting that the additional chrominance introduced by the mean face fill provides no representational benefit and may partially confound the skin tone signal.

The final selected model is the VGG-16 MST10 Regression RGB Black-Background variant, achieving the performance summarised in Table 3.8.

Dataset	Exact Acc.	Within-One Acc.	Three-Bin Acc.
CCv2 (held-out)	0.375	0.804	0.778
MSTE	0.159	0.424	0.566
FACET	0.165	0.471	0.593

Table 3.8 Final selected model performance: VGG-16 MST10 Regression RGB Black-Background.

The lower MSTE and FACET exact accuracies relative to CCv2 reflect distributional shift from the CCv2 training distribution and, in the case of MSTE, the benchmark’s assignment of identical ground-truth labels across varying illumination conditions despite the model being trained without explicit lighting augmentation. Three-bin accuracy is more robust to both effects, as broader grouping boundaries absorb within-class prediction variance.

Final Annotation Pipeline

The selected gender classifier and skin tone model were integrated into a unified annotation pipeline. For each input image the pipeline executes in sequence: FaceMesh segmentation produces a black-background face crop; the Realistic Gender Classifier yields a binary gender label and confidence score; the VGG-16 MST10 Regression model yields a continuous MST score rounded to the nearest integer and mapped to a Light/Mid/Dark bin. The resulting labels are written to a per-image JSONL manifest. The pipeline supports GPU batch inference and manifest-based resumability.

3.4.4 Real-World Occupational Demographic Reference Data

To contextualise observed demographic distributions against labour-force composition, two reference datasets were derived. The US-specific baseline was sourced from the Bureau of Labor Statistics (BLS) Current Population Survey Table 11 (CPSAAT11), reporting employed civilian workers by detailed occupation, sex, race, and Hispanic or Latino ethnicity for the 2024 reference year. The global baseline was sourced from ILOSTAT, providing employment data disaggregated by sex and two-digit ISCO-08 occupational group across reporting countries. The US baseline serves as the primary

reference due to the Western-web-dominated composition of large-scale web-scraped datasets such as Re-LAION-5B, as reflected in prior work on text-to-image models [7, 8].

The 200 professions were mapped to BLS labels using a hierarchical procedure. Ambiguous cases were first resolved through manual curation based on job role, task similarity, and occupational context, after which remaining professions were matched automatically using case-insensitive substring containment, keyword overlap scoring, and fuzzy string matching. Of the 200 professions, 197 were successfully matched and 3 were intentionally excluded. Professions mapped to BLS categories with fewer than 50,000 employed workers were retained for gender analysis but excluded from race-stratified comparisons.

For the global baseline, professions were mapped to ISCO-08 four-digit codes via the WageIndicator WISCO database [111], truncated to two-digit sub-major group codes for joining against ILOSTAT data. Because ILOSTAT reports at the ISCO-2 level, all professions sharing the same two-digit group receive identical global female percentages. This coarser granularity means the global baseline is a broad group-level reference rather than a per-profession ground truth.

3.4.5 Race to Skin Tone Distribution Mapping

Since the BLS reference data reports demographic composition in terms of racial categories while the annotation pipeline produces MST skin tone labels, an exploratory race-to-skin tone mapping was investigated. This mapping was derived empirically using FairFace [73] as a calibration corpus. The VGG-16 MST regression model was applied to FairFace face crops to predict a discrete MST label per image, and for each of the seven FairFace race categories a conditional distribution $P(\text{MST} \mid \text{race})$ was computed. The expected MST distribution for a given profession p was then computed as:

$$P(\text{MST} = m \mid p) = \sum_r P(\text{MST} = m \mid \text{race} = r) \cdot w_{r,p} \quad (3.1)$$

where $w_{r,p}$ is the BLS-reported racial composition fraction for race group r in profession p . This provides a profession-specific expected skin tone reference distribution against which observed MST distributions can be compared.

This approach carries several limitations and was ultimately treated only as an exploratory calibration attempt. The FairFace racial categories do not align perfectly with BLS categories, requiring approximate mappings for Asian and Hispanic subgroups. More importantly, inspection of the resulting conditional distributions showed that all racial groups mapped predominantly to the Mid bin, reflecting the tonal

range of the FairFace corpus and the behaviour of the VGG-16 estimator rather than a reliable population-level skin tone distribution. The resulting lack of variation across MST categories prevented the mapping from serving as a meaningful profession-level skin tone baseline. Consequently, skin tone results are interpreted as relative differences between Re-LAION-5B and Stable Diffusion v1.5 rather than against a real-world skin tone ground truth, as discussed further in Section 4.5.

3.5 Spurious Feature Probing Pipeline

Having established demographic annotation labels for both image sets, the analysis turns to characterising the spurious features present in Re-LAION-5B and Stable Diffusion v1.5 outputs. Unless otherwise stated, each sample corresponds to a single image-derived representation, such as a transformed image, caption, or segmented face region, with demographic and dataset labels inherited from the source image. This design pairs fixed labels with transformed inputs, ensuring that predictive performance reflects information retained by each representation rather than changes in annotation.

Within each task, the underlying image set, label assignments, data splits, and training protocols are held constant across transformations. Gender and skin tone use separate stratified splits, but each split remains fixed within its respective task. All probes are trained independently with fixed random seeds, allowing transformation-specific signal to be isolated without cross-representation leakage or training stochasticity. Comparisons between datasets are therefore interpreted within task, focusing on relative differences between transformations rather than absolute performance across demographic attributes.

The transformations serve two distinct probing objectives. The first, following Zeng et al. [10], applies image modifications to isolate specific visual feature categories as probes for dataset-discriminating signal, testing whether a classifier can distinguish Re-LAION-5B, Stable Diffusion v1.5 outputs, and COCO images from each transformation alone. The second, following Meister et al. [9], applies a subset of these representations as probes for demographic signal, testing whether gender and skin tone can be predicted from spurious features independently of semantic content. Where a transformation serves both objectives, it is described once and cross-referenced in the demographic probing context.

The dataset classification task is performed across three classes: Re-LAION-5B images, Stable Diffusion v1.5 generated images, and COCO images. The inclusion of COCO as a structurally independent reference enables per-class accuracy attribution from the confusion matrix. Under a binary classification over Re-LAION-5B and Stable Diffusion v1.5, a high overall accuracy cannot distinguish whether the discriminating

signal originates entirely from one dataset or from both simultaneously. By extending to a three-class problem, per-class accuracies (A_L, A_S, A_C) become individually interpretable: if $A_S \gg A_L \approx A_C \approx 1/3$, the signal is localised to the generative outputs; if $A_L \gg A_S \approx A_C \approx 1/3$, the signal is localised to Re-LAION-5B, pointing to properties of its web-scraping and filtering pipeline. The chance level $1/3$ serves as the lower bound for each class, replacing the binary chance level of $1/2$.

Not all transformations applied in the dataset classification setting are extended to the demographic probing setting. Transformations that present the full image to the probe model were deliberately excluded, as these representations retain both person and scene content without isolating either region. Any demographic signal recovered from such inputs cannot be attributed unambiguously to person appearance or to scene-level contextual correlates, undermining the interpretive goal of localising signal within specific, separable information channels. Where a full-image representation is nonetheless relevant to demographic probing, a modified representation is derived that discards scene-level content while retaining the target channel; this is described for each affected transformation below.

Where the same transformation appears in both settings, the probe model is matched to the structure and dimensionality of the representation and kept deliberately simple, following Meister et al. [9]. If a linear or shallow model recovers demographic signal from a compressed representation, that is strong evidence the signal is directly encoded there rather than being an artefact of a powerful classifier exploiting non-linear interactions. Accordingly, mean RGB is probed with logistic regression over the three-dimensional colour vector rather than a ConvNeXt-Tiny operating on a uniform-colour canvas, and object detection is probed via logistic regression over detection feature vectors rather than a convolutional classifier over the rendered white-canvas image, avoiding the reintroduction of spatial layout as a confound. Where a spatial input is unavoidable a ResNet-50 model is used, consistent with Meister et al. and providing a conservative upper bound relative to the ConvNeXt-Tiny used in dataset classification.

Monocular Depth Estimation

The depth transformation isolates the spatial geometry of each image by suppressing colour and texture cues while retaining only per-pixel relative depth values encoding scene structure, object positioning, and subject-to-background distance. Zeng et al. report that depth-transformed images achieve 73.1% classification accuracy across the YCD datasets, confirming that spatial geometry constitutes a meaningful axis of dataset-level bias. Depth maps were produced using Depth-Anything-V2 [112], resizing each image to 384×384 pixels prior to inference. The predicted depth tensor

was upsampled to original image dimensions via bicubic interpolation, normalised to $[0, 255]$, and saved as a three-channel PNG by replicating the greyscale channel across all colour channels for compatibility with the ConvNeXt-Tiny classifier. A final resize to 224×224 pixels was applied after depth computation. The ViT-S variant of Depth-Anything-V2 was used in place of the larger variant employed by Zeng et al.; their ablation demonstrates that pretrained model size has minimal impact on downstream classification accuracy for this transformation [10, Table 3]. Representative depth maps are shown in Figure 3.6.

For the demographic probing experiments, the dense depth map risks reintroducing person silhouette information that could correlate with gender independently of depth geometry. This work therefore samples the depth map only at the seventeen COCO keypoint locations identified by the pose estimator, producing a sparse seventeen-dimensional depth vector per image. Because only scalar depth values at discrete joint locations are retained, the representation exposes no silhouette geometry while still capturing the relative depth profile of the body. Depth values at visible keypoints were sampled at the nearest integer pixel coordinate, with joints below confidence 0.5 assigned a depth value of zero. The raw depth vector was per-person z-score normalised across visible joints to remove absolute subject-to-camera distance, with both raw and normalised variants saved to JSONL.

Edge Detection

Edge detection transforms an image into a sparse boundary map capturing object contours, silhouettes, and spatial intensity discontinuities while suppressing colour, texture, and semantic content. Zeng et al. report that Canny edge detection achieves 71.0% classification accuracy, while SAM-based contour extraction reaches 73.2% across the YCD datasets. The 2.2 percentage point difference is small relative to the computational overhead of SAM’s automatic mask generator across the full three-dataset corpus, making the additional cost disproportionate to the interpretive gain. Canny edge detection was therefore adopted as the sole edge representation.

Edge maps were derived using the classical Canny algorithm [52] as implemented in OpenCV. Each image was resized to 224×224 pixels, converted to greyscale, and smoothed with a 3×3 Gaussian kernel prior to gradient computation. Resizing before edge extraction avoids the edge thinning and thickening artefacts that arise when downsampling a precomputed edge map. Hysteresis thresholds of $t_1 \approx 85$ and $t_2 = 255$ were applied: pixels with gradient magnitude above t_2 were retained as strong edges, while pixels between t_1 and t_2 were retained only if connected to a strong edge. This matches the implementation used by Zeng et al. Example edge maps are shown alongside depth and pose outputs in Figure 3.6.

Frequency Domain Filtering

Frequency domain filtering decomposes an image into low-frequency and high-frequency content. Low-frequency components capture smooth colour gradients, global illumination, and coarse structure, while high-frequency components capture fine textures, sharp edges, and compression artefacts. Zeng et al. report classification accuracies of 70.4% for low-pass and 79.2% for high-pass filtered images, both close to the 82.0% reference. This confirms that spurious signals exist across the frequency spectrum and that mitigation strategies targeting only a single frequency range are insufficient.

Frequency filtering was applied using an ideal circular mask with a hard cutoff radius of $r = 40$ pixels, matching the primary configuration of Zeng et al. Each image was resized to 224×224 pixels, converted to greyscale, and the two-dimensional discrete Fourier transform computed via `numpy.fft.fft2` with frequency shift to centre the zero-frequency component. The low-pass mask retains all components within the circular region; the high-pass mask is its complement. Each masked spectrum was inverse-transformed, with the real component extracted and normalised to $[0, 255]$ via per-image min-max scaling before saving as a greyscale PNG. Both low-pass and high-pass representations were derived from a single FFT pass per image. Representative outputs are shown in Figure 3.7.

Pixel Shuffling, Patch Shuffling, and Mean RGB

The Pixel Shuffling, Patch Shuffling, and Mean RGB transformations progressively reduce the spatial information available to the classifier while preserving different aspects of the image signal. Pixel shuffling removes spatial organisation entirely, patch shuffling preserves local structure while disrupting global layout, and mean RGB compresses the image to its global colour statistics. Together, these transformations isolate the relative contribution of spatial layout, local texture, and colour distribution to dataset separability.

Pixel shuffling randomly permutes the spatial position of every pixel while preserving the overall colour distribution, reducing the available signal to global colour statistics alone. Zeng et al. report that this reduces classification accuracy from 82.0% to 52.2%, just above the 33.3% chance level, confirming that colour statistics carry only modest discriminative signal and that spatial structure is the dominant axis of dataset bias. Each image was resized to 224×224 pixels, and pixels were randomly permuted with a per-image deterministic seed derived from the sum of the RGB channel values of the top-left pixel.

Patch shuffling extends this procedure by permuting fixed-size non-overlapping patches rather than individual pixels. This preserves local structure within each patch

while randomising the global arrangement. Zeng et al. demonstrate that patch shuffling with 16×16 pixel patches recovers nearly all discriminability, achieving 81.2% classification accuracy, and establishes that local spatial structure within small image regions is sufficient to distinguish datasets. In this work, patch shuffling was applied at four scales: 2×2 , 4×4 , 8×8 , and 16×16 pixels. This allows the accuracy profile to be characterised across patch sizes rather than inferred from a single scale. Each patch size constitutes an independent classification experiment.

Mean RGB compresses each image to a single average colour per channel, filling the entire canvas with that uniform colour and leaving only the three-dimensional mean colour vector as the available signal. Zeng et al. report that even this extreme reduction achieves 48.5% accuracy, approximately 15 percentage points above chance, confirming that global colour distributions carry a non-trivial spurious signal. The raw mean RGB vector and a unit-range normalised equivalent were additionally recorded to a JSONL statistics file to support auxiliary linear probe experiments. Mean RGB also features in the demographic feature isolation experiments following Meister et al., where the raw three-dimensional vector is supplied directly to a logistic regression classifier rather than the filled uniform-colour image being passed to the ConvNeXt-Tiny network. Figure 3.8 illustrates the three spatial decomposition transformations.

Variational Autoencoder Reconstruction

The VAE reconstruction transformation passes each image through the encoder and decoder of a pretrained Variational Autoencoder, reducing fine-grained low-level properties such as JPEG compression artefacts and sensor noise while preserving semantically meaningful content. As a spurious feature probe, it isolates whether dataset-discriminating signal survives latent compression. Zeng et al. report that VAE-reconstructed images achieve 77.4% classification accuracy compared to the 82.0% reference, confirming that low-level acquisition properties contribute only marginally to the overall discriminative signal. Reconstructions were produced using the KL-regularised autoencoder (`autoencoder_kl_64x64x3`) from the Stable Diffusion v1.5 architecture [1], with images resized to 256×256 pixels prior to encoding and the posterior mode used for deterministic reconstructions. The decoded output was clamped to $[0, 1]$ and resized to 224×224 pixels for classifier compatibility.

A methodological consideration is that the VAE used for reconstruction is the same architecture and checkpoint embedded within Stable Diffusion v1.5. Generated images are therefore passed through the same decoder that produced them, meaning any latent-space artefacts introduced during generation are preserved with higher fidelity than for Re-LAION-5B or COCO images. This asymmetry should be considered

when interpreting per-class accuracy on VAE-reconstructed images: higher separability for Stable Diffusion outputs may partially reflect architectural proximity rather than solely the presence of spurious semantic features. Example VAE reconstructions are shown in Figure 3.9.

Semantic Segmentation

Semantic segmentation assigns a class label to every pixel using a fixed category vocabulary, retaining the spatial distribution of object categories while discarding all texture, colour, and appearance information. Zeng et al. report that segmentation using the ADE20K vocabulary achieves 69.8% classification accuracy, lower than purely structural transformations such as depth and edge detection, suggesting that object identity alone does not fully account for the discriminative signal captured by those representations.

Segmentation maps were produced using Mask2Former [113] with a Swin-Large backbone pretrained on ADE20K. Each image was resized to 512×512 pixels prior to inference. The predicted per-pixel class index map was resized to 224×224 pixels via nearest-neighbour interpolation before palette application, ensuring that interpolation operates directly on discrete class labels. This work uses the Swin-Large variant rather than the BEiT_{v2}-ViT-Adapter-Large backbone used by Zeng et al., which was excluded on computational grounds. Both models use the same ADE20K vocabulary and produce semantically equivalent output representations; since the downstream classification task operates on the category distribution rather than pixel-level precision, interpretive conclusions remain valid, though absolute accuracy may differ slightly from the 69.8% reported by Zeng et al. Representative segmentation maps are shown in Figure 3.10.

Object Detection

Object detection reduces each image to a set of labelled bounding boxes, retaining object identity, approximate spatial location, and co-occurrence statistics while discarding all pixel-level appearance information. Zeng et al. report that object detection achieves 61.9% classification accuracy across the YCD datasets, lower than segmentation and structural transformations, suggesting that object identity and co-occurrence alone account for a smaller share of dataset-level bias than spatial geometry.

Four detection vocabularies were considered. COCO, with 80 classes, was excluded because its vocabulary is too coarse to capture the range of occupation-relevant objects. LVIS, with 1,203 classes, was excluded because its long tail of rare categories adds detection noise without improving interpretability.

Object365, with 365 classes, was excluded because it lacks the breadth of occupation-relevant clothing and implement classes available in OpenImages V7. OpenImages V7, with 601 classes, was selected for its broad and balanced range of categories directly relevant to professional contexts.

Eight detection models were evaluated on a 10,000-image subset of Re-LAION-5B to assess throughput viability, with inference times summarised in Table 3.9. Throughput was measured on Re-LAION-5B rather than on OpenImages V7 because the objective was to estimate runtime under the image distribution and preprocessing conditions of the target corpus. Detection performance was evaluated separately on the OpenImages V7 validation set, as reported in Table 3.10. ViTDet-Huge [114] and GroundingDINO [115] both exceeded one hour of inference time and were excluded as impractical. RT-DETR-X [116] was excluded because its available pretrained weights target COCO and Object365 rather than OpenImages V7. The YOLOv8x-World variants achieved substantially degraded performance on OpenImages V7 when prompted with the class list rather than natively trained. YOLOv8x with OpenImages V7 pretrained weights was selected, achieving mAP@0.5:0.95 of 0.3614 with practical throughput of approximately 26 images per second.

Model	Time (10k images)	Architecture
YOLOv8x (OIV7)	~7 min 06 sec	CNN, closed-vocabulary
YOLOv8m (OIV7)	~5 min 47 sec	CNN, closed-vocabulary
YOLOv8x-World	~8 min 11 sec	CNN, open-vocabulary
YOLOv8x-Worldv2	~7 min 23 sec	CNN, open-vocabulary
YOLOv8x + World (Ensemble)	~12 min 44 sec	Ensemble
RT-DETR-X	~6 min 24 sec	Transformer, end-to-end
GroundingDINO	>60 min	Vision-language model
ViTDet-Huge	>60 min	Transformer backbone

Table 3.9 Object detection model inference time on a 10,000-image subset of Re-LAION-5B at 640×640 pixel input resolution.

Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLOv8x (OIV7)	0.4345	0.3614	0.4786	0.4374
YOLOv8m (OIV7)	0.4094	0.3336	0.4791	0.4007
YOLOv8x-World	0.1370	0.1140	0.2044	0.2568
YOLOv8x-Worldv2	0.1403	0.1160	0.1847	0.2600

Table 3.10 Detection performance on the OpenImages V7 validation set.

Detection was performed at 640×640 pixel resolution with a confidence threshold of 0.1 and an IoU threshold of 0.5. Twelve human body-part classes present in the OpenImages V7 vocabulary were excluded from all outputs, as these carry person-level appearance information rather than object co-occurrence signal. Detected bounding boxes were rendered as overlays on a uniform white canvas,

removing all pixel-level appearance information and retaining only object identity, spatial position, and bounding box geometry, with outputs resized to 224×224 pixels for classifier compatibility. Representative detection outputs are shown in Figure 3.9.

A label remapping strategy was applied to normalise class name variants and reduce gender-correlated label noise. The standard remapping consolidates gendered person labels into a single Person class. The restricted remapping additionally consolidates clothing, footwear, cosmetics, personal-care, and accessory labels carrying gender-correlated associations into generic counterparts. This enables a cleaner test of whether dataset-discriminating signal persists once gender-correlated object vocabulary is suppressed. Table 3.11 summarises both configurations.

Original Label	Standard Remap	Restricted Remap
Man, Woman, Boy, Girl	Person	Person
Dress, Skirt, Miniskirt, Brassiere	—	Clothing
Suit, Coat, Jacket, Jeans, Shirt	—	Clothing
Shorts, Trousers, Tie, Scarf, Belt	—	Clothing
Hat, Sun hat, Cowboy hat, Fedora, Sombrero	—	Clothing
High heels, Sandal, Boot, Sock	—	Footwear
Lipstick, Face powder	—	Cosmetics
Perfume, Hair spray	—	Personal care
Earrings, Necklace, Tiara, Handbag	—	Fashion accessory

Table 3.11 Label remapping configurations applied during object detection.

Image Captioning

Image captioning converts each image into a natural language description, reducing the visual representation to semantic content while discarding the underlying pixel representation. As a spurious feature probe, caption-based classification tests whether dataset-discriminating signal survives this extreme semantic compression: if a classifier operating on caption embeddings reliably distinguishes dataset origin, the bias must reside in systematic differences in scene content, object co-occurrence, or compositional conventions. Zeng et al. report classification accuracies of 63.8% for short captions and 66.1% for long captions using LLaVA 1.5 7B [117].

Ten models were assessed for throughput viability on a 25-image subset of Re-LAION-5B. A small subset was used because several autoregressive captioning models already required several minutes to process only 25 images, making larger-scale preliminary screening unnecessary for identifying models that were infeasible at full corpus scale. LLaVA 1.5 7B, TinyLLaVA, and LLaVA-OneVision were excluded due to prohibitive inference times from autoregressive generation requirements limiting effective batch sizes to 2. BLIP2-Flan-T5-XL was excluded because 8-bit quantisation

introduced an unnecessary constraint without a corresponding speed advantage. Moondream2 was excluded because it does not support batched inference. The five viable models, BLIP Large [118], Florence 2 Base [119], Kosmos 2 [120], ViT-GPT2, and Tiny, were advanced to quality evaluation. Table 3.12 summarises the throughput results.

Model	Batch Size	Time (25 images)	Viable
LLaVA 1.5 7B [117]	2	~27.0s	No
TinyLLaVA-Phi-2-SigLIP-3.1B	2	~180.0s	No
LLaVA-OneVision-Qwen2-0.5B	2	~180.0s	No
BLIP2-Flan-T5-XL [121]	25	~2.3s	No
Moondream2	1	N/A	No
BLIP Large [118]	25	~2.3s	Yes
Florence 2 Base [119]	25	~2.6s	Yes
Kosmos 2 [120]	25	~1.9s	Yes
ViT-GPT2	25	~2.1s	Yes
Tiny	25	~2.2s	Yes

Table 3.12 Image captioning model throughput assessment on a single GPU.

The five viable models were benchmarked on the COCO 2014 [67] and Flickr30k [122] Karpathy test splits using the caption quality metrics reviewed in Section 2.4: BLEU@1–4 [61], METEOR [62], CIDEr [63], 4-gram diversity, SBERT [65], CLIP-S, and RefCLIP-S [66]. Results are summarised in Tables 3.13 and 3.14.

Model	Dataset	BLEU@4	METEOR	CIDEr	4-gram Div.	SBERT
BLIP Large	COCO	0.1278	0.2096	0.0325	0.3406	0.1158
ViT-GPT2	COCO	0.1290	0.1918	0.0307	0.3246	0.0904
Tiny	COCO	0.1270	0.1964	0.0367	<u>0.0547</u>	0.0562
Kosmos 2	COCO	0.0947	0.2014	0.0050	0.3334	0.1449
Florence 2	COCO	0.0842	0.1754	0.0013	0.2978	0.1373
BLIP Large	Flickr30k	0.1254	0.1934	0.0268	0.4554	0.0866
ViT-GPT2	Flickr30k	0.1306	0.1890	0.0306	0.4935	0.0682
Tiny	Flickr30k	0.1386	0.1915	0.0375	<u>0.1499</u>	0.0479
Kosmos 2	Flickr30k	0.0992	0.2044	0.0051	0.4492	0.1079
Florence 2	Flickr30k	0.0855	0.1834	0.0011	0.4504	0.1194

Table 3.13 Classical caption quality metrics on the Karpathy test splits of COCO 2014 and Flickr30k. Low 4-gram diversity (underlined) indicates degenerate template-based generation.

BLIP Large achieved the highest scores across the most diagnostic metrics on both evaluation datasets. Its CLIP-S scores of 0.6646 on COCO and 0.6550 on Flickr30k indicate strong visual grounding, and its RefCLIP-S scores of 0.7279 and 0.6868 are the highest of all five candidates, confirming that its captions are simultaneously well-aligned with image content and semantically coherent relative to human references. A critical failure mode was identified in the Tiny model, which achieved a 4-gram diversity of only 0.0547 on COCO, indicating that fewer than 6% of all 4-grams across generated captions were unique. This is a hallmark of degenerate

Model	Dataset	CLIP-S	RefSim	RefCLIP-S
BLIP Large	COCO	0.6646	0.8148	0.7279
ViT-GPT2	COCO	0.6040	0.8289	0.6951
Tiny	COCO	0.4110	0.6087	0.4853
Kosmos 2	COCO	0.5568	0.6314	0.5861
Florence 2	COCO	0.5927	0.6536	0.6163
BLIP Large	Flickr30k	0.6550	0.7330	0.6868
ViT-GPT2	Flickr30k	0.5290	0.6713	0.5867
Tiny	Flickr30k	0.4777	0.6371	0.5409
Kosmos 2	Flickr30k	0.5600	0.5766	0.5619
Florence 2	Flickr30k	0.6059	0.5759	0.5853

Table 3.14 CLIP-based caption quality metrics on the Karpathy test splits of COCO 2014 and Flickr30k. CLIP-S measures image-caption alignment scaled to $[0, 2.5]$; RefCLIP-S is the harmonic mean of CLIP-S and maximum reference similarity.

template-based generation and demonstrates why traditional n-gram metrics alone are insufficient. BLIP Large was therefore selected as the captioning model.

A word-level gender-neutral remapping was applied to all generated captions prior to saving. Gendered person references such as woman and man were replaced with person, gendered child references such as boy and girl were replaced with child, and gendered pronouns such as he, she, his, and her were replaced with their gender-neutral equivalents. This ensures that downstream classifiers cannot exploit gendered language introduced by the captioning model as a proxy for dataset origin. Captions were generated using the prompt "An image of", following the captioning protocol of Peng et al. [120], with a maximum output length of 30 tokens and greedy decoding. Representative captions are shown alongside object detection and VAE outputs in Figure 3.9.

Pose Estimation

Pose estimation reduces each image to a set of COCO-17 body joint coordinates, retaining only skeletal geometry while discarding all pixel-level appearance information. Following Meister et al. [9], pose representations isolate whether demographic attributes can be inferred from body geometry alone. Unlike preceding transformations that produce rendered image outputs consumed by the ConvNeXt-Tiny classifier, the pose pipeline produces 34-dimensional normalised keypoint vectors as direct input to a low-capacity MLP probe. This transformation is applied exclusively to Re-LAION-5B and Stable Diffusion v1.5 image sets; COCO images are excluded as they are not guaranteed to contain people in the occupational contexts under analysis, and pose estimation on images without clearly visible subjects produces uninformative zero vectors.

Seventeen model configurations across five architectures were evaluated on all person-annotated images in the COCO 2017 validation set to identify the best

trade-off between accuracy and throughput, with results summarised in Table 3.15. For models requiring an external person detector, standard mode used their normal detection pipeline, while Oracle mode supplied ground-truth person bounding boxes to isolate keypoint estimation performance from detection failure. MoveNet and MediaPipe were excluded on accuracy grounds, achieving AP below 0.55 even in Oracle mode. HRNet variants achieved the highest AP values but operate at under 9 FPS end-to-end, rendering full-corpus extraction prohibitively slow. YOLOv8l-Pose was selected for its superior throughput of 39.7 FPS combined with an AP of 0.659, with the practical advantage of requiring no external person detector. The accuracy gap relative to YOLOv8x-Pose is 0.015 AP, which is small and does not affect the downstream representational objective, since the pipeline discards all appearance information and retains only normalised joint geometry.

Model	AP	AP ₅₀	AP ₇₅	AR	FPS
<i>YOLOv8 (integrated detection + keypoints)</i>					
YOLOv8n-Pose	0.479	0.744	0.520	0.538	27.0
YOLOv8s-Pose	0.581	0.830	0.642	0.638	37.1
YOLOv8m-Pose	0.633	0.860	0.704	0.691	33.0
YOLOv8l-Pose	0.659	0.871	0.735	0.717	39.7
YOLOv8x-Pose	0.674	0.871	0.747	0.730	27.9
<i>MoveNet (requires external person detector)</i>					
MoveNet-Lightning (Standard)	0.419	0.621	0.456	0.450	9.9
MoveNet-Thunder (Oracle)	0.548	0.779	0.594	0.584	4.1
<i>MediaPipe (requires external person detector)</i>					
MediaPipe-C2 (Standard)	0.243	0.405	0.255	0.333	3.6
<i>HRNet (requires external person detector)</i>					
HRNet-W32 (Standard)	0.714	0.879	0.789	0.769	8.7
HRNet-W32-384 (Oracle)	0.776	0.933	0.853	0.808	8.5
HRNet-W48 (Standard)	0.723	0.880	0.793	0.778	7.9

Table 3.15 Pose estimation model evaluation on COCO 2017 validation set. Standard mode uses the model’s integrated person detector. Oracle mode provides ground-truth bounding boxes. FPS is measured end-to-end on a single GPU.

For each image, YOLOv8l-Pose was applied at native image resolution with a person detection confidence threshold of 0.25. When multiple detections were returned, the detection with the highest box confidence score was retained. Joints with YOLO keypoint confidence at or above 0.5 were assigned visibility $v = 2$ and retained; joints below this threshold were zeroed and assigned $v = 0$. Pose geometry was normalised to remove absolute scale and image position: visible joints were centred by subtracting the mean joint position, and the bounding box area of visible joints was scaled to a fixed target area of 4,000 pixels, following the normalisation of Meister et al. The 34-dimensional normalised keypoint vector serves as direct input to the downstream MLP probe. Figure 3.6 shows skeleton overlays for visual reference only; no rendered pose image is used by the classifier.

Occlusion

Occlusion transformations selectively remove or isolate spatial regions of each image by replacing either the person or the background with a uniform fill, producing a controlled series of representations that expose different sources of demographic signal. Following Meister et al. [9], the occlusion family is applied exclusively to Re-LAION-5B and Stable Diffusion v1.5 images as a demographic probe. COCO images are excluded because they are not conditioned on occupational queries and are not guaranteed to depict people in the professional contexts required for gender and skin tone probing. This work replicates the occlusion experiments of Meister et al. in the context of profession-conditioned imagery and extends them to skin tone prediction in addition to gender.

The occlusion family comprises five variants: Full NoBg, which retains person pixels within the segmentation mask while suppressing the background; MaskSegm, which replaces all pixels within the person segmentation mask with white while retaining the original background; and MaskRect, which replaces the person bounding box with white while retaining the original background. The two NoBg masking variants, MaskSegm NoBg and MaskRect NoBg, apply the same segmentation-mask and bounding-box masks respectively, but remove the surrounding background.

A key methodological difference from Meister et al. is that their analysis operates on COCO images with ground-truth person segmentation masks. In this work, person masks must be predicted by a segmentation model. Five instance segmentation models were evaluated on single-person images from the COCO 2017 validation set for model selection, with results in Table 3.16. YOLACT was excluded because its detection rate of 79.5% means it misses approximately one in five persons. Lang-SAM achieved the highest IoU at 0.846, but its throughput of 0.35 iterations per second is approximately 3.7 times slower than Mask2Former COCO, rendering it impractical at full dataset scale. Mask2Former ADE20K performs semantic rather than instance segmentation and was excluded because it conflates all person pixels in multi-person images. Mask2Former COCO was selected because it achieved the highest detection rate of any evaluated model at 96.5%, combined with an IoU of 0.829 and precision of 0.899.

Model	Det. Rate	Success Rate	IoU	Prec.	Throughput
Mask2Former COCO	0.965	0.698	0.829	0.899	1.28 it/s
Mask R-CNN	0.941	0.707	0.813	—	1.63 it/s
Lang-SAM	0.891	0.637	0.846	0.931	0.35 it/s
Mask2Former ADE20K	0.946	0.676	0.785	—	1.86 it/s
YOLACT	0.795	0.613	0.789	—	3.00 it/s

Table 3.16 Segmentation model evaluation on single-person images from the COCO 2017 validation set, used for model selection only.

For each image, Mask2Former COCO was applied in half-precision under `torch.no_grad()` to produce a binary person mask, which was upsampled to the original image dimensions via nearest-neighbour interpolation. The five occlusion variants were derived as element-wise operations on the original RGB image. All occlusion fills use a uniform white value of $\text{RGB} = [255, 255, 255]$ for masked regions and a black value of $\text{RGB} = [0, 0, 0]$ for background suppression, consistent with Meister et al., who demonstrate that results are robust to the choice of fill colour. Final output images were resized to 224×224 pixels for classifier compatibility. Representative outputs are shown in Figures 3.10 and 3.11.

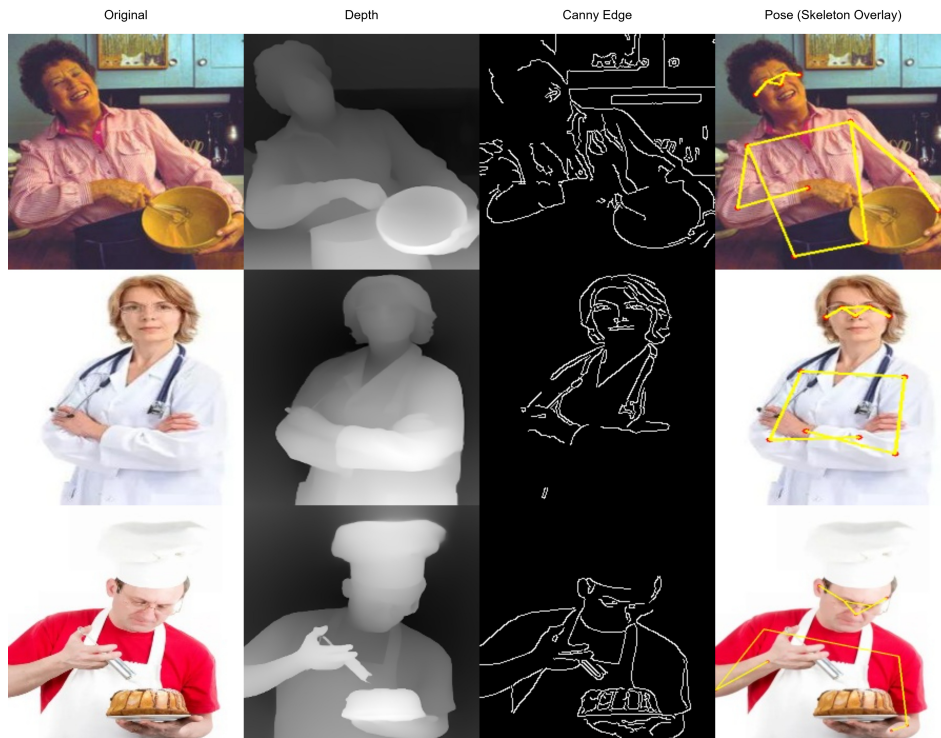


Figure 3.6 Structural transformations: depth map (Depth-Anything-V2), Canny edge map, and YOLOv8l-Pose skeleton overlay. The classifier input for pose is the 34-dimensional normalised keypoint vector, not the rendered overlay.



Figure 3.7 Frequency domain transformations: low-pass reconstruction retaining smooth luminance gradients (cutoff $r = 40$) and high-pass residual isolating fine textures and compression artefacts. Both derived from a single FFT pass per image.

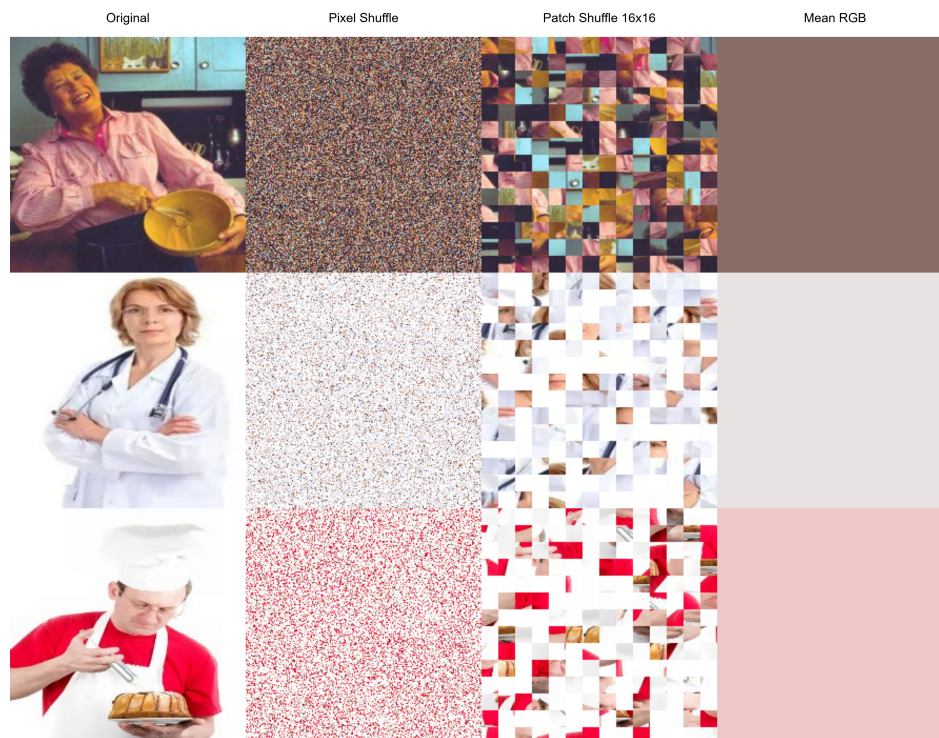


Figure 3.8 Spatial decomposition: pixel shuffle (global colour preserved, spatial structure destroyed), patch shuffle 16×16 (local texture preserved, global layout disrupted), and mean RGB (uniform per-channel colour fill).

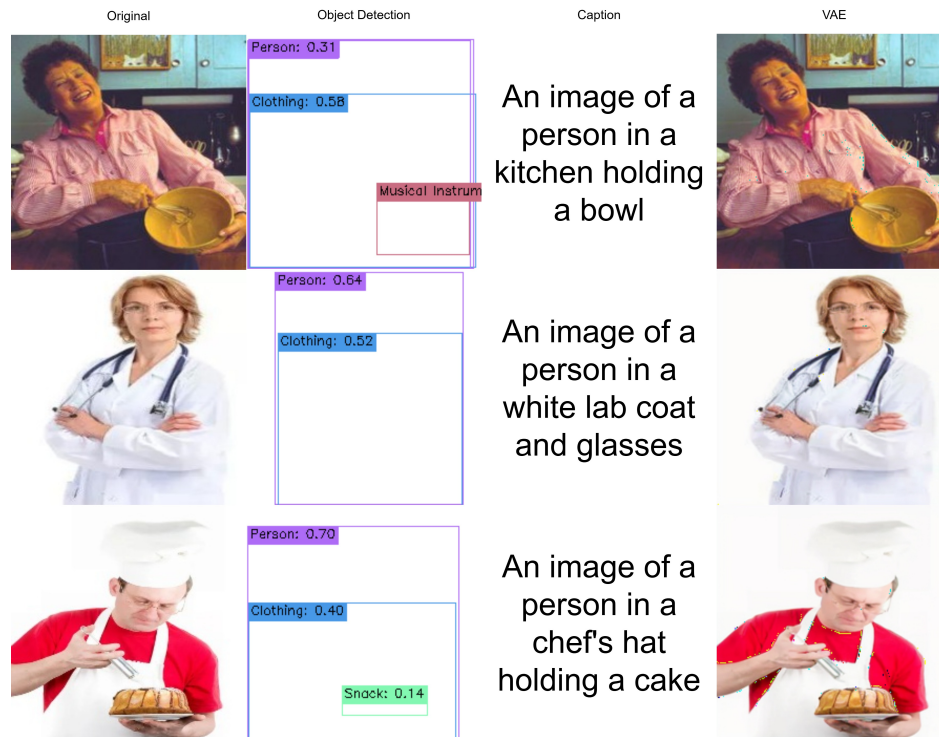


Figure 3.9 Semantic and latent-space transformations: YOLOv8x object detection bounding boxes on white canvas (OpenImages V7), BLIP Large captions with gender-neutral remapping, and SD v1.5 VAE reconstructions.

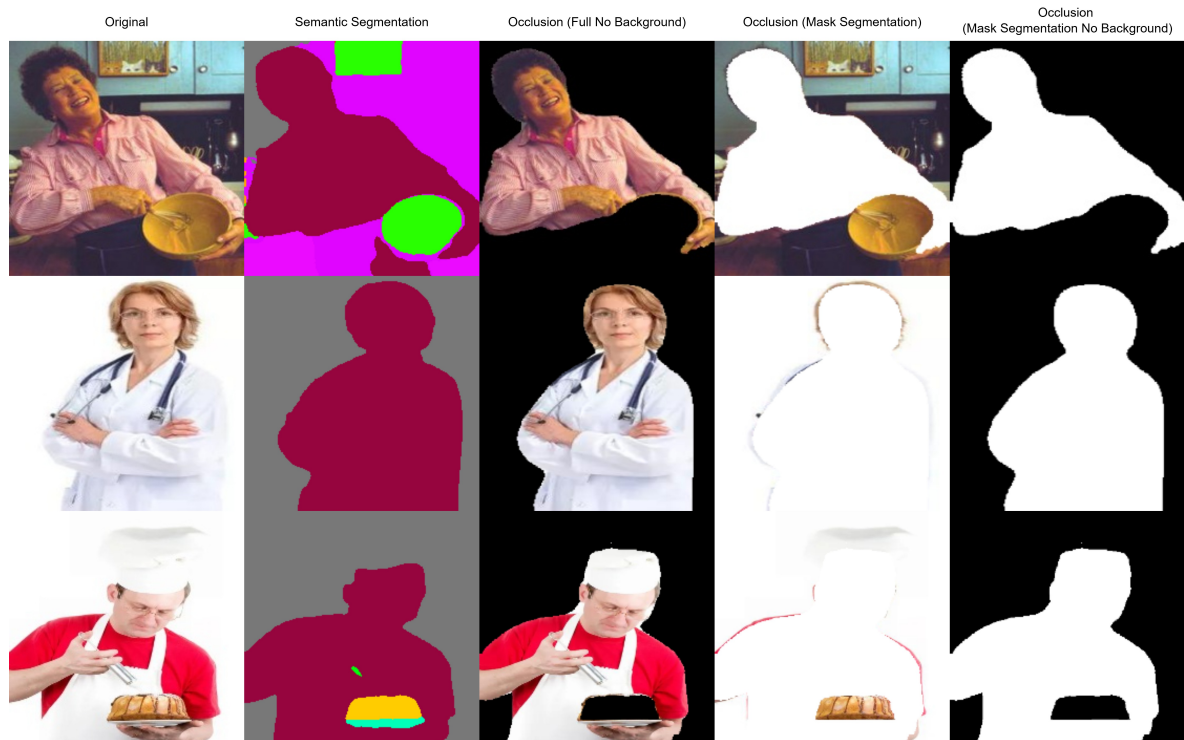


Figure 3.10 Segmentation and segmentation-derived occlusion: ADE20K pixel-label map (Mask2Former), full person isolated (no background), person masked with white fill (background retained), and white silhouette on black background.

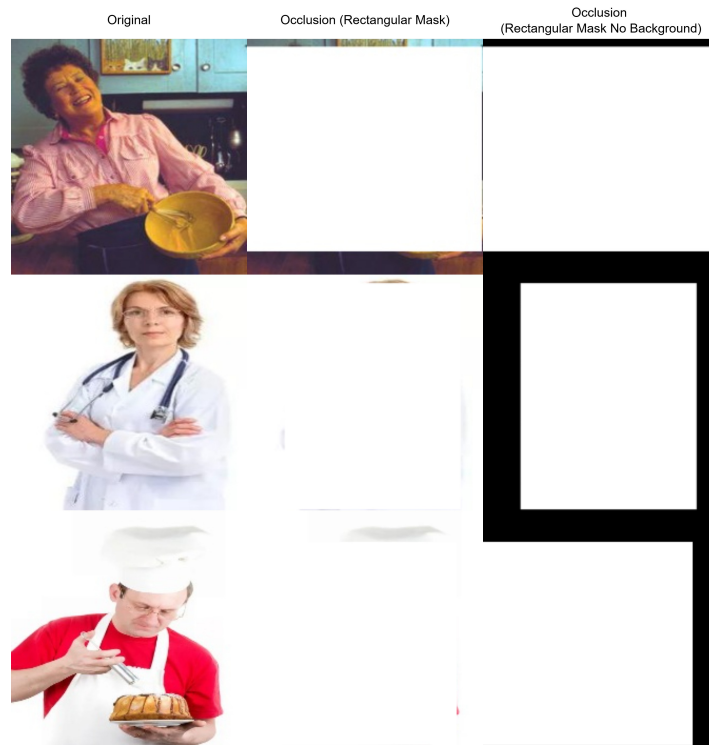


Figure 3.11 Rectangular occlusion: white rectangle replacing the person region (background retained) and inverse mask retaining only the occluded boundary region on a black background.

3.6 Inference-Time Debiasing of Stable Diffusion v1.5

The preceding analyses characterise the spurious features and demographic biases present in Re-LAION-5B and in the standard Stable Diffusion v1.5 generation pipeline. A complementary question is whether inference-time debiasing interventions applied to the same base model can produce measurable shifts in its demographic output distribution. This section describes the debiased generation pipeline used to produce a 10,000-image corpus against which the effectiveness of these interventions is evaluated in Section 4.4. Given the computational constraints of this work and the requirement that the base model remain Stable Diffusion v1.5, the analysis is restricted to inference-time approaches; retraining and weight editing methods are acknowledged as potentially more thorough interventions but are outside the scope of the present evaluation.

3.6.1 Experimental Design

The debiasing experiment was designed as a focused single-profession evaluation rather than a full 200-profession generation run. A total of 10,000 images were generated for the doctor profession, providing sufficient statistical power to detect

intervention-driven demographic shifts. Distributing the same generation budget across all 200 professions would yield only 50 images per profession per demographic condition, too few for reliable distributional comparison.

The doctor profession was selected for several converging reasons. First, it exhibits one of the most pronounced demographic skews in the standard Stable Diffusion v1.5 pipeline, making it a demanding test case against which any debiasing effect is clearly measurable. Second, it carries a well-documented real-world demographic baseline, enabling the debiased distribution to be assessed against established reference statistics. Third, computational constraints motivated restricting the debiasing evaluation to a single profession; the doctor prompt was preferred over alternatives because its strong learned prior constitutes a more stringent test of the pipeline’s effectiveness. Finally, the doctor prompt has been used as an evaluation case in prior work on occupational bias in text-to-image generation [7, 8], supporting direct comparability with the existing literature.

3.6.2 Demographic Conditioning

The demographic conditioning component was selected from inference-time approaches capable of steering Stable Diffusion v1.5 without modifying the base model weights, as reviewed in Sections 2.6 and 2.6. FairGen was evaluated but excluded because its adapter-based latent direction discovery did not produce sufficiently coherent Stable Diffusion v1.5 outputs for reliable large-scale evaluation. ITI-GEN was adopted instead because it produced coherent outputs while achieving the required demographic conditioning.

ITI-GEN [86] learns a set of inclusive token embeddings that steer the diffusion model toward a more balanced demographic distribution without modifying model weights. Token embeddings were trained on CCv2 [74] across two attribute axes: gender, represented by female and male categories, and skin tone, spanning the ten-point MST scale. This produced learned perturbation embeddings indexed over all gender–skin tone combinations. The training prompt was "an image of a person", following ITI-GEN’s neutral prompt convention. The trained embeddings were applied to the generation prompt "A picture of a doctor" via the `prompt_prepend` mechanism, concatenating inclusive embeddings with the CLIP text encoding before passing the combined representation to the UNet cross-attention layers.

3.6.3 Pose Conditioning via ControlNet

A fixed canonical pose was enforced across all debiased images using ControlNet [87] with OpenPose conditioning. The conditioning image was a single reference pose

skeleton depicting a neutral standing figure, applied at a ControlNet scale of 0.6. Without a fixed structural constraint, the model’s learned pose prior may itself be demographically skewed, allowing body-geometry variation to confound demographic annotation. A canonical pose therefore reduces pose-correlated compositional confounds in the debiased corpus.

3.6.4 Colour Alignment

ITI-GEN operates through semantic text conditioning and does not directly constrain image colour distributions. A complementary inference-time colour alignment strategy was therefore added to steer the denoising trajectory toward a neutral reference colour distribution derived from the standard SD v1.5 doctor generation corpus. Two colour-guidance approaches reviewed in Section 2.6 were implemented and compared: SW-Guidance and Chamfer colour alignment.

SW-Guidance [89] applies Sliced Wasserstein Distance guidance at each denoising step by decoding the current latent to pixel space and computing a gradient signal that minimises the SWD between the decoded image’s colour distribution and that of a reference. Despite producing favourable preliminary colour distributions, SW-Guidance was excluded on throughput grounds: the repeated decode–gradient–encode operations at each denoising step result in inference times incompatible with generating 10,000 images within the available computational window.

The Chamfer colour alignment approach of Shum et al. [90] performs colour steering directly in latent space without decoding at each step. A reference image is blurred and encoded to a latent representation at the start of generation. At each DDPM backward step, the predicted clean sample $\hat{x}_0$ is projected toward the reference latent distribution using an iterative Chamfer distance matching procedure: nearest-neighbour correspondences between spatial positions of the current and reference latents are established, a displacement field is computed from matched correspondences, and the projected latent is blended back into the denoising step via a learning-rate-scaled update. Because the Chamfer projection operates entirely in latent space, it avoids per-step decode–encode overhead and achieves throughput comparable to standard generation. Chamfer colour alignment was therefore selected for the full-scale debiased run.

3.6.5 Generation Configuration

The debiased corpus was generated using Stable Diffusion v1.5 (runwayml/stable-diffusion-v1-5) with a fine-tuned VAE

(stabilityai/sd-vae-ft-mse), producing 512×512 pixel images. The generation prompt was "A picture of a doctor", with classifier-free guidance at a scale of 7.0. Unlike the standard generation pipeline, which used $s = 1.0$ to minimise inference-time amplification of learned priors, the debiased pipeline uses the standard deployment scale to ensure that the model's demographic priors are fully active. This provides a more demanding test of whether the debiasing interventions can override those priors. The denoising schedule used 50 inference steps over the canonical 1,000-step scaled-linear noise schedule, with $\beta_{\text{start}} = 0.00085$ and $\beta_{\text{end}} = 0.0120$, following the standard Stable Diffusion v1.5 configuration. A fixed negative prompt was applied throughout: "longbody, lowres, bad anatomy, bad hands, missing fingers, extra digit, fewer digits, cropped, worst quality, low quality".

The full generation space comprised twenty demographic combinations: two gender categories, female and male, crossed with ten skin tone levels, MST 1–10, from the CCv2 attribute set. Five hundred images were generated per combination, yielding a total corpus of 10,000 images. A base random seed of 42 was used, with per-image seeds derived deterministically as $\text{seed} + n$ for iteration index n , ensuring full reproducibility across all combinations.

ITI-GEN inclusive token embeddings trained on CCv2 [74] under the training prompt "an image of a person" were used to condition the generation prompt via the `prompt_prepend` mechanism, as described in Section 3.6.2. ControlNet OpenPose conditioning (lllyasviel/control_v11p_sd15_openpose) was applied at a conditioning scale of 0.6 using a single fixed reference pose skeleton (`basic_pose.jpg`), enforcing a canonical standing posture across all generated images as described in Section 3.6.3. LCM-LoRA was not used because its reduced step count, combined with per-step Chamfer projection, produced images insufficiently coherent for annotation and probing.

The Chamfer colour alignment projection was applied to the predicted clean sample $\hat{x}_0$ at every denoising step where the DDPM timestep $t > 600$, corresponding to approximately the first twenty of the fifty inference steps. Restricting projection to early high-noise timesteps follows the placement strategy of Shum et al. [90], where colour steering applied at low-noise timesteps risks corrupting fine-grained semantic detail. The projection used a learning rate of 0.05 and a blur factor of 5. The blur operation downsamples the reference image to one-fifth of its spatial resolution before re-upsampling prior to VAE encoding, suppressing high-frequency texture in the reference signal and retaining only the broad colour distribution. The colour reference image was a uniform-fill image derived from the mean RGB of the standard SD v1.5 doctor generation corpus, providing a neutral colour prior without encoding any scene-level content.

4 Evaluation

This chapter presents the empirical results of the analyses described in the methodology. The evaluation proceeds in four stages. First, the demographic distributions of the annotated Re-LAION-5B and Stable Diffusion v1.5 corpora are characterised at both aggregate and profession level. Second, the dataset classification results from the spurious feature transformation pipeline are reported and interpreted, with per-class accuracy attribution used to localise the source of discriminative signal within each transformation. Third, the demographic probing results are presented, assessing whether gender and skin tone can be inferred from low-level image representations independently of semantic content. Fourth, the debiased generation corpus is evaluated against the aggregate Stable Diffusion v1.5 results to assess whether the inference-time interventions produce measurable demographic shifts. The chapter concludes with a unified discussion of findings, their implications, and the limitations of the evaluation.

4.1 Demographic Distribution Analysis

4.1.1 Aggregate Gender and Skin Tone Distributions

At the aggregate level, both corpora exhibit clear demographic imbalance, although the form and severity of this imbalance differ between gender and skin tone. For gender, both datasets show a pronounced male skew across the full 200-profession corpus, with the imbalance substantially more severe in Stable Diffusion v1.5 than in Re-LAION-5B. Re-LAION-5B contains 83,003 female-labelled images (41.5%) and 116,997 male-labelled images (58.5%), corresponding to a 17 percentage-point male–female gap. Stable Diffusion v1.5 widens this gap to 41.96 percentage points, producing only 58,037 female-labelled images (29.02%) compared with 141,963 male-labelled images (70.98%). The generative model therefore does not merely reproduce the gender distribution observed in the retrieved corpus, but amplifies it substantially, yielding outputs that are considerably more male-skewed than the LAION-derived web-scraped imagery underlying its training distribution. This gender comparison is visualised in Figure 4.1.

The skin tone results show a different form of imbalance. Rather than a binary skew, both datasets are overwhelmingly concentrated in the Mid bin (MST 4–7), with Re-LAION-5B at 86.74% and Stable Diffusion v1.5 at 76.99%. Within this Mid-bin concentration, however, the modal values shift toward lighter tones in the generated corpus: Re-LAION-5B peaks at MST 5 and MST 6, whereas Stable Diffusion v1.5 peaks

at MST 4 and MST 5. The more pronounced divergence appears at the extremes of the scale. Re-LAION-5B has a Light-bin proportion of 12.45% and a Dark-bin proportion of 0.81%, indicating that non-Mid predictions are distributed, albeit unevenly, across both ends of the tone scale. Stable Diffusion v1.5, by contrast, increases the Light-bin proportion to 22.87% while reducing the Dark-bin proportion to only 0.14%, an 83% reduction relative to Re-LAION-5B. MST values 9 and 10 are effectively absent in both datasets, with Stable Diffusion v1.5 generating near-zero proportions even at MST 8. This near-complete collapse of dark skin tone representation, combined with the shift toward lighter tones within the Mid bin, suggests that the model has absorbed a strong light skin tone prior from its training corpus and concentrates its outputs toward the lighter half of the tonal range. The three-bin skin tone distribution is visualised in Figure 4.2, while the full MST 1–10 distribution is shown in Figure 4.3.

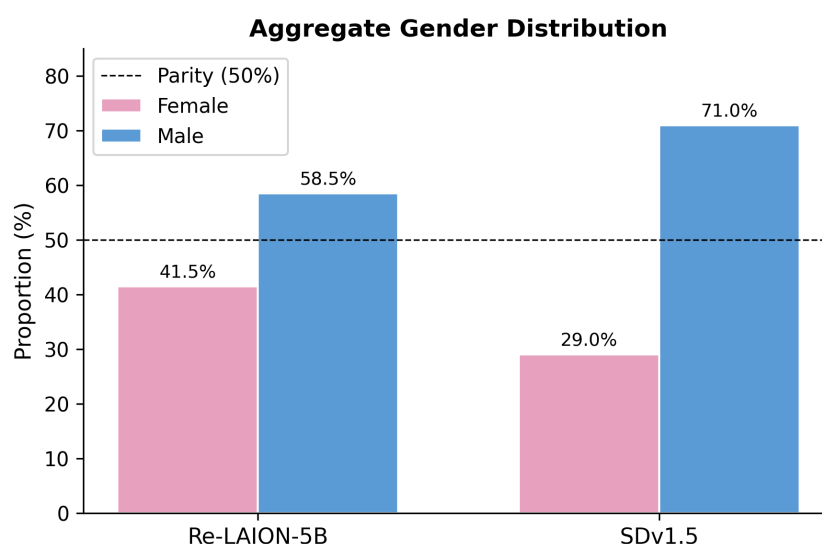


Figure 4.1 Aggregate gender distribution across all 200 professions for Re-LAION-5B and Stable Diffusion v1.5. The dashed line indicates demographic parity at 50%.

4.1.2 Per-Profession and Per-ISCO-Group Analysis

At the occupational-group level, gender skew is pervasive across both datasets, but its magnitude and pattern differ markedly. In Re-LAION-5B, Clerical Support is the only major ISCO-08 group to exceed demographic parity, at 58.26% female, consistent with the historically female-dominated composition of administrative and secretarial roles. All remaining groups fall below parity, with Plant & Machine Operators showing the strongest male skew at 77.85% male, followed by Armed Forces at 76.5% and Managers at 64.53%. In Stable Diffusion v1.5, the direction of skew is broadly preserved: Clerical Support remains the only group at or near parity, at 51.93% female. However, the magnitude of male skew increases substantially across every group

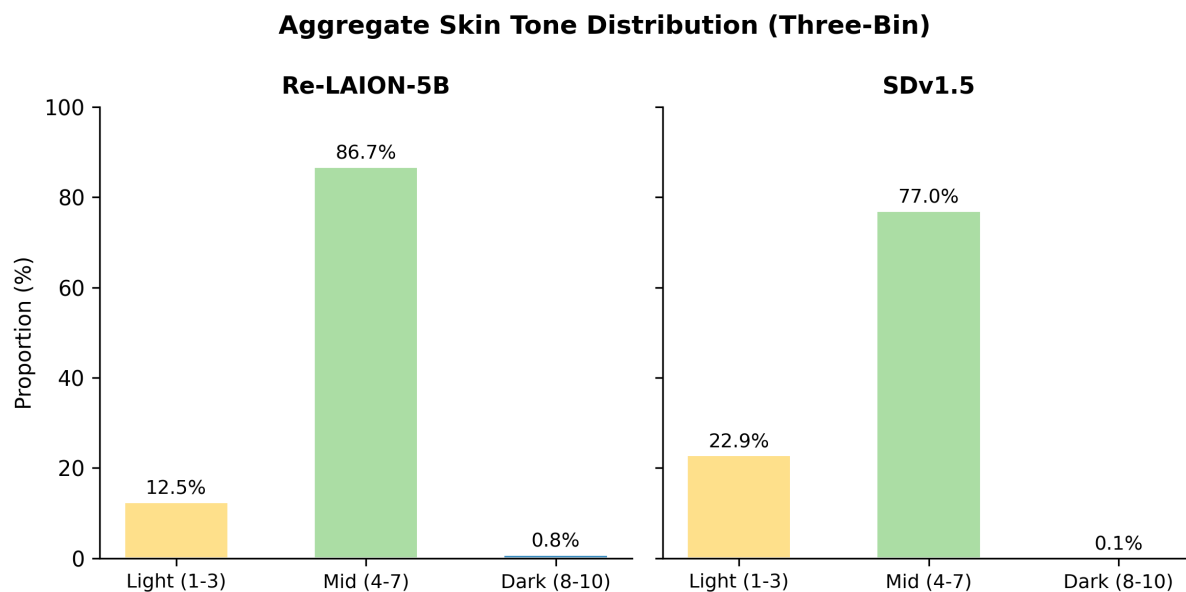


Figure 4.2 Aggregate skin tone distribution using the Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) three-bin grouping for Re-LAION-5B and Stable Diffusion v1.5.

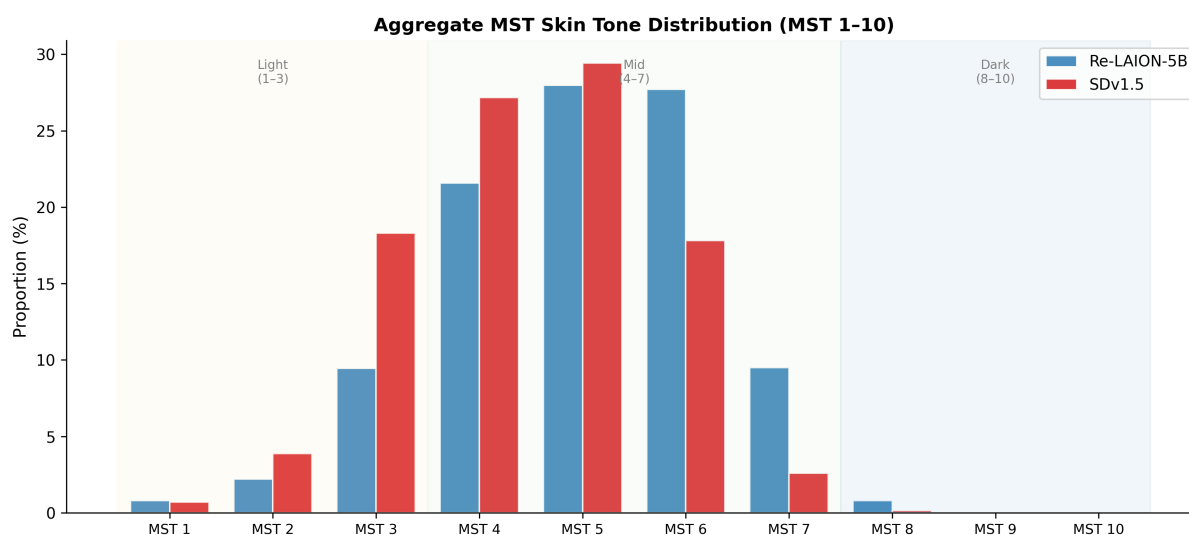


Figure 4.3 Aggregate MST 1–10 skin tone distribution for Re-LAION-5B and Stable Diffusion v1.5. Shaded regions indicate the Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) bin boundaries.

without exception. The most severe amplifications occur in Skilled Agricultural, where the female proportion falls from 43.3% to 7.3%, Armed Forces, where it falls from 23.5% to 7.7%, and Plant & Machine Operators, where it falls from 22.15% to 13.55%. This systematic reduction in female representation across occupational categories confirms that the amplification effect identified at the aggregate level is not driven by a small number of outlier professions, but reflects a broad structural tendency of the generative model. The ISCO-level gender distribution is visualised in Figure 4.4.

Comparing these occupational distributions against the US BLS 2024 and Global ILOSTAT reference markers further clarifies the direction of bias. Both reference baselines show broad agreement in their directional signal: groups identified as female-majority in US BLS data are also female-majority in the global statistics, while groups such as Skilled Agricultural, Craft & Trades, and Plant & Machine Operators are male-dominated in both. This convergence between the two independent baselines strengthens the validity of the comparison and indicates that the observed gaps are not artefacts of geographic or methodological differences between the two reference sources.

Against these baselines, Re-LAION-5B presents a mixed picture that depends on the reference used. Relative to US BLS 2024, Re-LAION-5B falls below the real-world female proportion in approximately six of ten ISCO groups, most notably in Professionals (approximately 43% versus 50%), Service & Sales (approximately 46% versus 51%), and Technicians & Associate Professionals (approximately 44% versus 49%). Against the Global ILOSTAT baseline, the picture is more balanced: Re-LAION-5B exceeds the global female proportion in several groups, including Skilled Agricultural, Craft & Trades, and Plant & Machine Operators, while falling below it in Professionals and Technicians & Associate Professionals. This divergence between the two baselines reflects genuine differences in occupational gender composition between the US and global labour markets rather than inconsistency in the dataset. It suggests that Re-LAION-5B's gender distributions are broadly consistent with global workforce composition in some occupational domains while remaining below US-specific norms in others. The BLS and ILOSTAT comparisons are included in Figure 4.4.

Taken together, these comparisons establish a two-stage bias pattern. Re-LAION-5B exhibits gender distributions that diverge from real-world labour force composition in a baseline-dependent manner, falling below US-specific norms in the majority of occupational groups while remaining broadly consistent with global workforce proportions in several others. Stable Diffusion v1.5 then amplifies the male skew further across nearly every group regardless of baseline, producing occupational gender distributions that are more male-skewed than the already imbalanced retrieved corpus. The generative model therefore does not merely inherit the distributional

properties of its training data, but exacerbates them, producing representations that diverge substantially from both demographic parity and empirically observed occupational gender composition.

At the individual profession level, the same amplification pattern is visible. A total of 167 of 200 professions (83.5%) show gender skew in the same direction in both datasets, indicating that Stable Diffusion v1.5 largely inherits the directional gender associations present in the retrieved corpus. However, in 178 of 200 professions (89.0%), the magnitude of skew is greater in Stable Diffusion v1.5 than in Re-LAION-5B. The most skewed professions illustrate this pattern clearly: Administrative Assistant, which already shows a strong female skew in Re-LAION-5B at 94.7%, reaches 96.4% in SD v1.5; Dietitian rises from 88.6% to 96.9%; and Receptionist rises from 92.0% to 97.3%. At the male-skewed end, the stereotyping is equally pronounced, with Logger (97.5% male), Farmer (97.0%), and Construction Worker (96.4%) among the most extreme SD v1.5 outputs. The 33 professions where the datasets disagree in skew direction most likely reflect cases where the retrieved-corpus signal is weak or near parity, making generation more sensitive to prompt phrasing or random seed variation and producing nominal directional changes that do not necessarily reflect meaningful stereotyping. These cases warrant further investigation but are beyond the scope of the current analysis. The professions with the strongest gender skew are listed in Table 4.1.

The ISCO-level skin tone analysis shows a similarly systematic pattern. In Re-LAION-5B, the Light-bin proportion varies narrowly between 6.9% for Armed Forces and 13.59% for Clerical Support, indicating that no occupational group is strongly associated with light skin tones in the retrieved corpus. In Stable Diffusion v1.5, by contrast, every ISCO group shows a substantially elevated Light-bin proportion, ranging from 16.85% for Plant & Machine Operators to 35.1% for Skilled Agricultural. The full MST distribution confirms that this increase is not simply a redistribution within the Light bin, but part of a broader leftward shift in the modal MST score. Whereas Re-LAION-5B concentrates mass at MST 5–6 across occupational groups, Stable Diffusion v1.5 systematically concentrates mass at MST 4–5, with correspondingly elevated MST 3 proportions. The Dark-bin collapse observed at the aggregate level is also consistent across occupational groups: Re-LAION-5B dark-bin proportions range from 0.5% to 1.8%, while in SD v1.5 they approach zero in most groups, with Skilled Agricultural recording no dark-bin images. The three-bin ISCO distribution is visualised in Figure 4.5, and the full MST group-level heatmap is shown in Figure 4.6.

At the individual profession level, the scale of the skin tone amplification in SD v1.5 is particularly stark. The most light-skewed profession in Re-LAION-5B is Makeup Artist, at 27.6% Light. In SD v1.5, the corresponding upper range is far higher, reaching

55.5% for Carer and 56.8% for Librarian. The SD v1.5 top ten light-skewed professions all exceed 40% Light, a threshold not reached by any profession in Re-LAION-5B. In 175 of 200 professions (87.5%), SD v1.5 generates a higher proportion of Light-bin images than Re-LAION-5B does for the same profession, establishing a near-universal tendency toward lighter skin tone generation that is not confined to a small subset of occupations. The professions with the highest Light-bin proportions are listed in Table 4.2.

Taken together, the Light-bin increase and Dark-bin collapse are best interpreted as linked outcomes of a single directional shift in the generated skin tone distribution, rather than as two unrelated effects. The aggregate three-bin results show that SD v1.5 nearly doubles the Light-bin proportion relative to Re-LAION-5B, from 12.45% to 22.87%, while reducing the already small Dark-bin proportion from 0.81% to 0.14%. The full MST distribution clarifies this pattern: the dominant mass in Re-LAION-5B lies around MST 5–6, whereas Stable Diffusion v1.5 concentrates around MST 4–5. The ISCO-level heatmap shows the same modal shift across occupational groups, indicating that the effect is not confined to a small number of professions. The Light-bin increase therefore reflects the positive expression of this leftward shift, while the Dark-bin collapse reflects its most severe consequence at the under-represented end of the scale. Because the Dark bin is already marginal in Re-LAION-5B, even a small absolute shift away from darker tones is sufficient to reduce representation to near-zero levels in the generated corpus.

This distinction matters for downstream use. The result is not simply that Stable Diffusion v1.5 generates slightly lighter depictions on average; rather, it produces an occupational image distribution in which darker skin tones are almost absent. Any downstream application that relies on such outputs for synthetic data augmentation, occupational image corpus construction, fairness auditing, or classifier training would therefore inherit a severe coverage gap for darker skin tones. In settings where dark skin tone representation matters, this creates two risks: first, the generated corpus cannot provide meaningful visual diversity for darker-toned individuals; second, models trained or evaluated on such generated data may underestimate failure modes affecting groups that are scarcely represented in the synthetic distribution. The observed Light-bin expansion and Dark-bin collapse should therefore be read as a representational compression of the occupational image space toward lighter skin tones, not merely as a minor shift in aggregate colour statistics.

Unlike the gender analysis, a direct comparison of skin tone distributions against real-world reference statistics was not conducted. The BLS CPS data reports labour force composition in terms of self-identified racial categories rather than skin tone measurements, and converting race proportions into expected MST distributions requires an intermediate race-to-skin-tone mapping. As described in Section 3.4.5, this

mapping was derived empirically from the FairFace dataset using the VGG-16 MST regression model. However, inspection of the resulting conditional distributions revealed that all racial groups map predominantly to the Mid bin (MST 4–7), reflecting the tonal range of the FairFace corpus rather than the full MST scale. The derived per-profession reference distributions therefore lack sufficient discriminative resolution to serve as meaningful baselines for the skin tone comparison, collapsing to near-identical Mid-dominant profiles regardless of occupational racial composition. As a result, skin tone analysis in this work is evaluated solely against the demographic parity baseline. The construction of a more granular and reliable race-to-skin-tone reference distribution, whether through a larger and more tonally diverse calibration dataset or through direct MST annotations on labour force samples, remains an open challenge and constitutes a limitation of the present analysis.

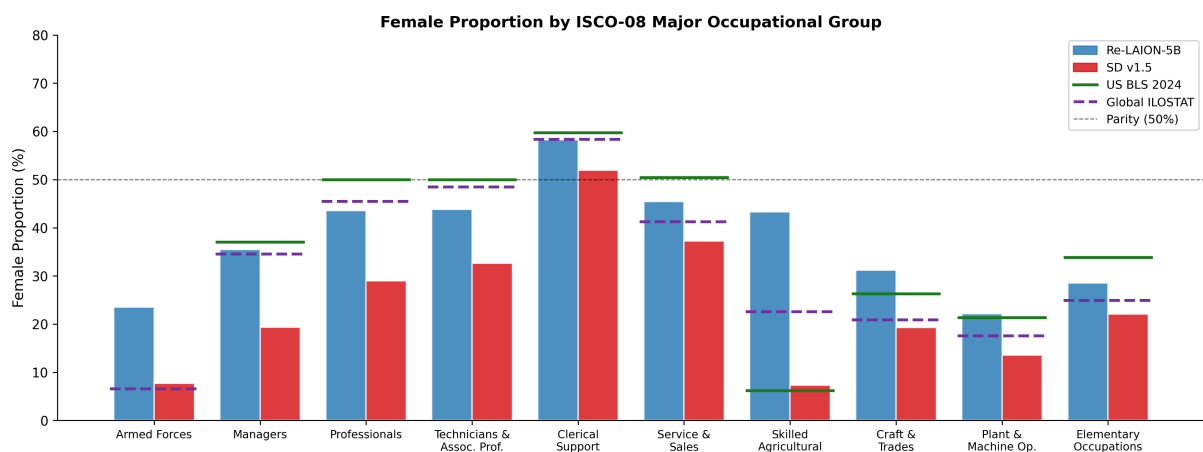


Figure 4.4 Female proportion by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5. The dashed line indicates demographic parity at 50%.

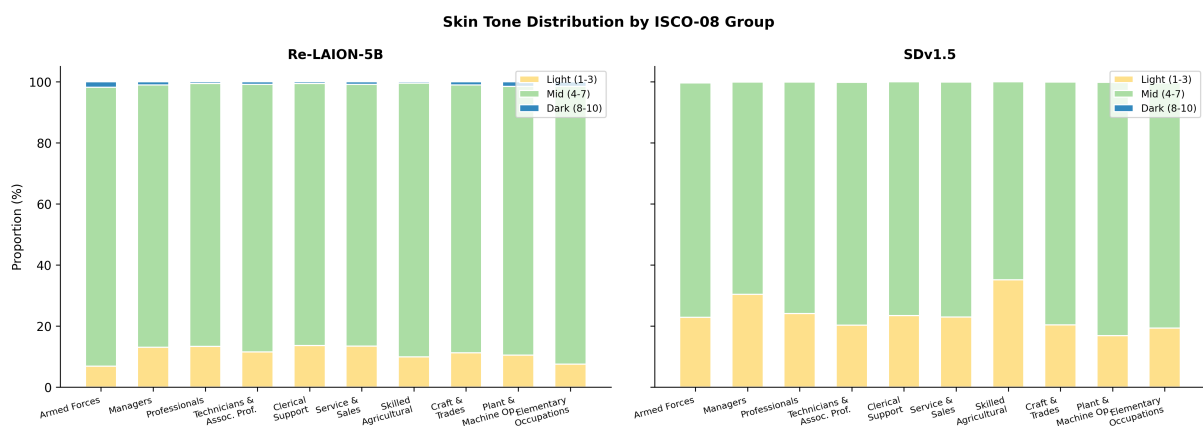


Figure 4.5 Skin tone distribution (Light, Mid, Dark bins) by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5.

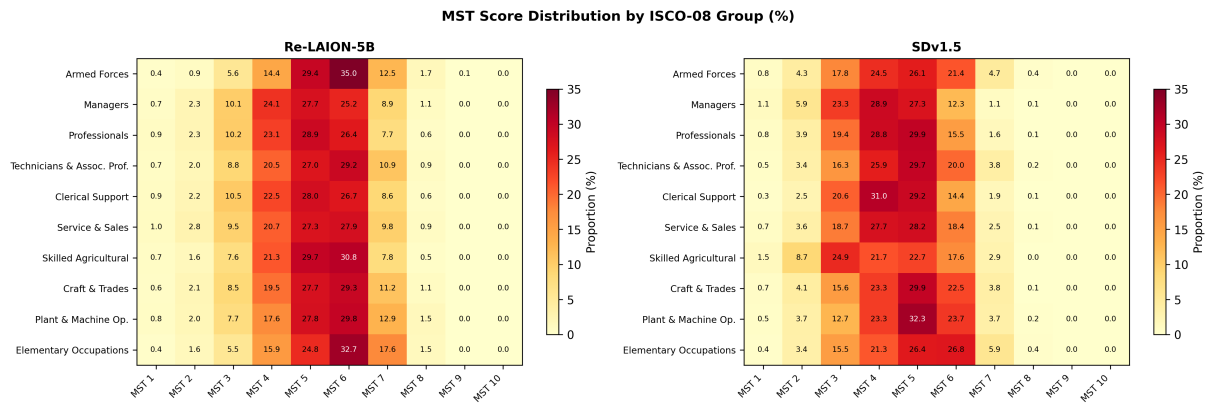


Figure 4.6 MST 1–10 score distribution (%) by ISCO-08 major occupational group for Re-LAION-5B and Stable Diffusion v1.5. Cell values indicate the proportion of images in each group assigned to each MST score.

Dataset	Profession	F%	M%	Skew
Re-LAION-5B	Administrative Assistant	94.7	5.3	0.447
Re-LAION-5B	Makeup Artist	92.7	7.3	0.427
Re-LAION-5B	Receptionist	92.0	8.0	0.420
Re-LAION-5B	Executive Assistant	91.5	8.5	0.415
Re-LAION-5B	Office Assistant	90.2	9.8	0.402
Re-LAION-5B	Dietitian	88.6	11.4	0.386
Re-LAION-5B	Paralegal	87.8	12.2	0.378
Re-LAION-5B	Bodyguard	13.0	87.0	0.370
Re-LAION-5B	Priest	13.3	86.7	0.367
Re-LAION-5B	Blacksmith	14.5	85.5	0.355
SD v1.5	Logger	2.5	97.5	0.475
SD v1.5	Nurse	97.5	2.5	0.475
SD v1.5	Receptionist	97.3	2.7	0.473
SD v1.5	Farmer	3.0	97.0	0.470
SD v1.5	Dietitian	96.9	3.1	0.469
SD v1.5	Makeup Artist	96.9	3.1	0.469
SD v1.5	Flight Attendant	96.6	3.4	0.466
SD v1.5	Administrative Assistant	96.4	3.6	0.464
SD v1.5	Construction Worker	3.6	96.4	0.464
SD v1.5	Miner	3.9	96.1	0.461

Table 4.1 Ten professions with the strongest gender skew in Re-LAION-5B and Stable Diffusion v1.5. Gender skew is the absolute deviation of the female proportion from 0.5; higher values indicate stronger skew regardless of direction. F% = female percentage; M% = male percentage.

Dataset	Profession	Light%	Mid%	Dark%
Re-LAION-5B	Makeup Artist	27.6	72.3	0.1
Re-LAION-5B	Hair Dresser	21.8	78.0	0.2
Re-LAION-5B	Dentist	20.4	79.3	0.3
Re-LAION-5B	Carer	19.9	79.8	0.3
Re-LAION-5B	Florist	18.7	81.1	0.2
Re-LAION-5B	Singer	18.6	81.1	0.3
Re-LAION-5B	Barber	18.5	80.2	1.3
Re-LAION-5B	Customer Support Specialist	18.5	81.2	0.3
Re-LAION-5B	Comedian	18.1	81.0	0.9
Re-LAION-5B	Actor	17.2	80.7	2.1
SD v1.5	Librarian	56.8	43.1	0.1
SD v1.5	Carer	55.5	44.4	0.1
SD v1.5	Politician	50.2	49.8	0.0
SD v1.5	Meteorologist	45.3	54.7	0.0
SD v1.5	Mayor	45.2	53.9	0.9
SD v1.5	News Reader	45.2	54.7	0.1
SD v1.5	Farmer	43.7	56.3	0.0
SD v1.5	Comedian	41.3	58.7	0.0
SD v1.5	Florist	40.5	59.5	0.0
SD v1.5	Conservation Officer	40.4	59.1	0.5

Table 4.2 Ten professions with the highest Light-bin (MST 1–3) skin tone proportion in Re-LAION-5B and Stable Diffusion v1.5.

4.2 Dataset Classification

Dataset classification experiments assess whether a ConvNeXt-Tiny classifier trained on transformed image representations can distinguish Re-LAION-5B, Stable Diffusion v1.5, and COCO images as a three-class problem. Each transformation is treated as an independent experiment: images are transformed, split into training, validation, and test sets, and the classifier is trained on the transformed training set with early stopping and evaluated on the held-out test set. Chance level for all experiments is $1/3 \approx 33.3\%$ under a balanced three-class prior. Above-chance accuracy on a given transformation indicates that the transformation retains dataset-discriminating signal; the magnitude of accuracy quantifies how strongly that signal is preserved. Classification accuracy is used as the primary metric throughout this section, consistent with the three-class evaluation protocol of Zeng et al. [10] from which the transformation framework is adopted.

The COCO dataset is included to extend the classification task beyond a binary comparison between Re-LAION-5B and Stable Diffusion v1.5, introducing an independent reference class and enabling class-specific diagnostic checks of whether separability is concentrated in one dataset or distributed across all three. As COCO

carries no profession-conditioned annotation for gender or skin tone within this pipeline, it does not appear in the demographic analyses that follow; the results in this section therefore reflect general distributional separability across datasets and should not be interpreted as direct evidence of demographic bias, which is examined independently in Section 4.3.

Table 4.3 summarises the results for all transformations evaluated. Per-class accuracy and AUC diagnostics were inspected during analysis but are not tabulated here, as they did not show substantial deviations from the aggregate pattern. The aggregate metrics are therefore sufficient for the purpose of this section, which is to characterise overall dataset separability across transformations rather than to provide a class-by-class error analysis.

Transformation	Val Acc (%)	Val AUC	Test Acc (%)	Test AUC
Base (unmodified)	99.56	0.9996	99.42	0.9996
VAE reconstruction	99.15	0.9992	99.15	0.9992
High-pass filter	98.65	0.9991	98.64	0.9988
Patch shuffle 8×8	98.41	0.9989	97.91	0.9983
Patch shuffle 16×16	98.60	0.9986	97.69	0.9983
Low-pass filter	97.30	0.9977	96.99	0.9969
Depth	96.41	0.9968	96.78	0.9965
Patch shuffle 4×4	97.20	0.9968	94.38	0.9949
Edge	95.17	0.9934	93.62	0.9905
Semantic segmentation	90.47	0.9814	91.36	0.9837
Captions [†]	—	—	90.99	0.9820
Patch shuffle 2×2	88.97	0.9839	89.91	0.9766
Object detection	85.88	0.9617	88.62	0.9728
Pixel shuffle	86.01	0.9574	80.67	0.9430
Mean RGB	70.80	0.8443	64.19	0.8207

[†] Logistic regression probe on BLIP Large captions encoded via SentenceT5-Base; all other rows use ConvNeXt-Tiny on rendered images.

Table 4.3 Dataset classification results per image transformation (three-class: Re-LAION-5B, Stable Diffusion v1.5, COCO). All models trained with ConvNeXt-Tiny, early stopping (patience = 2), seed = 0.

1. The three datasets are separable across every level of representation tested

Every transformation achieves well above the chance level of 33.3%, with even the most aggressive information compression, mean RGB at 64.19%, retaining substantial discriminating signal. This demonstrates that dataset-level differences between Re-LAION-5B, Stable Diffusion v1.5, and COCO persist even under extreme reduction of image content. Even when spatial structure, semantics, and most visual detail are removed, the classifier can still attribute images to their source dataset significantly better than chance.

The persistence of separability across such compressed representations suggests that dataset identity is encoded in multiple correlated visual cues rather than a single isolated artefact. These cues may include colour statistics, spatial patterns, semantic composition, or other structural regularities that differ systematically between datasets. Because these signals remain detectable even after large portions of the visual information are removed, they likely reflect broad statistical regularities rather than isolated features.

Importantly, these experiments demonstrate separability rather than its causal origins. The observed differences may arise from a combination of dataset curation practices, object distribution differences, scene composition biases, or visual statistics introduced by generative models. Consequently, the results indicate that the datasets possess distinct statistical signatures without directly identifying the specific factors responsible for those differences.

2. Dataset separability cannot be explained by acquisition artefacts alone The VAE result (99.15%) is virtually identical to the unmodified base (99.42%), a reduction of only 0.27 percentage points. Variational autoencoder reconstruction compresses each image through a latent bottleneck before reconstructing it, which typically suppresses certain low-level acquisition artefacts such as compression noise or encoding irregularities while preserving perceptually meaningful image structure.

Because VAE reconstruction largely preserves colour distributions, texture statistics, and spatial structure while reducing some superficial artefacts associated with file encoding or compression, the near-identical classification performance suggests that simple acquisition artefacts are unlikely to be the dominant source of discriminating signal. If dataset separability were primarily driven by compression differences, file-format characteristics, or other encoding artefacts, the VAE reconstruction process would be expected to reduce classification accuracy more substantially.

Instead, the results suggest that dataset differences are preserved in perceptual, structural, and statistical properties that survive latent reconstruction, rather than being explained primarily by superficial file-level acquisition artefacts. These may include differences in scene composition, object arrangement, photographic style, or statistical regularities introduced during dataset creation. However, VAE reconstruction does not eliminate all low-level visual statistics, such as colour distributions, texture patterns, sharpness, or local gradient structure. The result therefore indicates that acquisition artefacts are unlikely to dominate, rather than definitively ruling out all low-level contributions.

3. Local spatial statistics are highly dataset-discriminative Patch shuffling remains highly discriminative even when global layout is disrupted. With 2×2 patches, the classifier retains only minimal local spatial information, yet still achieves 89.91% test accuracy. Increasing the patch size further improves performance, with 8×8 patch shuffling reaching 97.91% and 16×16 achieving 97.69%.

These results indicate that dataset-specific signatures are embedded in local and patch-level statistics rather than solely in coherent global scene composition. Such statistics include short-range texture patterns, local gradient relationships, colour correlations, and small-scale structural regularities. Their strong discriminative power may arise from differences in photographic style, image processing pipelines, dataset curation practices, or generative model priors. However, the experiment does not isolate the causal origin of these statistics, and the observed separability may reflect a combination of these factors.

4. Both frequency bands carry strong discriminating signal, with a modest advantage for high-frequency components High-pass filtering achieves 98.64% test accuracy, outperforming low-pass filtering at 96.99% by 1.65 percentage points. This small gap indicates a modest advantage for high-frequency information, such as edges, fine textures, and small-scale spatial variation. However, the low-pass result remains very high, showing that coarse illumination patterns, global shape structure, and scene layout also retain substantial dataset-discriminating signal.

Taken together, the two frequency results indicate that dataset-level differences are distributed across multiple spatial scales. Dataset separability is therefore not concentrated exclusively in either high- or low-frequency components, but persists across the frequency spectrum.

5. A hierarchy of dataset-discriminating signal exists across representational levels Table 4.3 reveals a clear hierarchy of dataset-discriminating power across the transformation suite. Unmodified and VAE-reconstructed images form the strongest tier, both achieving near-perfect accuracy ($\geq 99\%$). Frequency-based and local spatial transformations follow closely: the high-pass filter and the 8×8 and 16×16 patch-shuffle variants all achieve 97–99% accuracy, showing that dataset-specific signatures remain highly recoverable even when global layout is disrupted.

A second tier is formed by transformations that retain broader structural information. Depth and Canny edge representations achieve 96.78% and 93.62% accuracy respectively, indicating that scene geometry and boundary structure alone are sufficient for reliable dataset classification without colour or texture. Patch shuffle 4×4 , at 94.38%, sits within this intermediate range, retaining more local structure than

2×2 shuffling but less coherent patch-level information than the larger 8×8 and 16×16 variants.

Semantic representations form the next tier. Semantic segmentation achieves 91.36%, caption embeddings achieve 90.99%, and object detection achieves 88.62%, indicating that object identity, scene semantics, and linguistic descriptions carry substantial dataset-discriminating signal, although generally less than low-level spatial and structural representations. Patch shuffle 2×2 , at 89.91%, falls close to this semantic tier, suggesting that even minimal local structure can preserve signal comparable to higher-level semantic summaries.

The weakest but still clearly above-chance tier consists of global colour statistics. Pixel shuffle achieves 80.67%, while mean RGB achieves 64.19%, approximately 31 percentage points above chance despite retaining only the three-dimensional average colour vector. Overall, this hierarchy indicates that dataset-discriminating information is distributed across colour, texture, spatial structure, geometry, semantics, and language. No single transformation eliminates separability, suggesting that dataset identity is not concentrated in one visual property but persists across multiple representational channels.

6. Semantic dataset differences survive compression to natural language

Caption-based classification using SentenceT5 embeddings achieves 90.99% test accuracy, despite the classifier receiving only textual descriptions rather than visual input. This indicates that dataset-discriminating signal survives compression from images into natural language.

The result suggests that differences in object co-occurrence, scene composition, and contextual relationships are reflected not only in visual statistics but also in the semantic content of generated captions. Dataset identity is therefore not confined to low-level visual patterns; it is also recoverable from linguistic descriptions of the images.

7. Absolute accuracies exceed those reported in the reference framework Across transformations shared with the Zeng et al. framework, the accuracies obtained here are substantially higher. However, these results are not directly comparable because the dataset configuration differs: Zeng et al. compare web-scraped corpora, whereas this work includes Stable Diffusion v1.5 outputs as one of the three classes.

This difference is nevertheless informative. Web-scraped corpora are internally heterogeneous because they aggregate images from many photographic sources, while Stable Diffusion v1.5 applies a consistent generative process across all outputs. This process can impose shared priors over colour, texture, spatial layout, and scene composition, producing stronger dataset-level regularities than those expected

between web-scraped datasets alone. The higher accuracies observed here, together with the persistence of separability under aggressive compression, are therefore consistent with the generative pipeline being a major contributor to dataset separability. This remains an inference rather than a direct causal finding, but the pervasiveness of separability across representational levels is difficult to explain without accounting for the systematic influence of the generative process.

4.3 Demographic Attribute Separability

To investigate whether demographic attributes are encoded in representation channels beyond direct person appearance, a series of probe experiments were conducted using transformations designed to isolate spatial structure, semantic content, body geometry, scene context, and global colour statistics. These probes reuse the transformation-based framework introduced in the dataset classification experiments, but replace dataset identity with gender or skin tone as the prediction target. The occlusion family is particularly important because it directly tests whether demographic signal remains recoverable from scene context or mask geometry after the visible person appearance has been removed. Results are reported as AUC, with chance level at 0.500 for binary gender classification and 0.500 macro-averaged AUC for multi-class skin tone classification.

To assess result stability, each probe was trained independently across four random seeds. The reported results correspond to the selected primary run, with the remaining runs used to verify that the qualitative conclusions are stable.

For the skin tone experiments, an important label-space caveat applies. MST categories 9 and 10 are absent from both datasets, as no images in either corpus received these labels under the VGG-16 MST10 Regression annotator. This is consistent with the distributional analysis in Section 4.1, which shows near-zero Dark-bin proportions in both corpora. Since the annotator assigned MST 9 and MST 10 labels during model evaluation, their absence in the profession-conditioned image sets is more likely to reflect the extreme rarity, or complete absence, of these tones than a systematic inability of the classifier to predict them. All skin tone probing experiments therefore operate over MST categories 1–8.

4.3.1 Gender Artefact Analysis

Table 4.4 reports the gender probing results across all representations for Re-LAION-5B and Stable Diffusion v1.5. The discussion first establishes the unmodified-image upper bound, then examines the occlusion probes as the most direct test of contextual gender signal. It then considers global colour statistics, geometric

representations, object-level scene composition, and caption-based semantic signal before synthesising the overall spurious-feature profiles of the two datasets.

Transformation	Probe	Re-LAION-5B AUC	SD v1.5 AUC
Base (unmodified)	ResNet-50	0.986	0.946
Occlusion (full, no background)	ResNet-50	0.499 [†]	0.948
Occlusion (masked rectangle)	ResNet-50	0.501 [†]	0.747
Occlusion (masked rectangle, no background)	ResNet-50	0.499 [†]	0.587 [‡]
Occlusion (masked segmentation)	ResNet-50	0.501 [†]	0.852
Occlusion (masked segmentation, no bg.)	ResNet-50	0.501 [†]	0.834
Depth at pose keypoints	Logistic Regression	0.559	0.592
Mean RGB	Logistic Regression	0.600	0.579
Object detection (restricted)	Logistic Regression	0.615	0.608
Object detection	Logistic Regression	0.692	0.608
Pose keypoints	MLP	0.697	0.680
Captions	SentenceT5	0.901	0.771

[†] AUC at chance level; model collapsed to majority-class prediction.

[‡] Near-chance AUC; recall ≈ 1.0 indicates near-complete majority-class prediction.

Table 4.4 Gender separability results per image transformation and dataset. Reported values correspond to the selected primary run, seed 0 unless otherwise noted. AUC is the primary metric; chance level is 0.500 for binary classification.

Unmodified images establish a strong separability upper bound The base ResNet-50 probe achieves AUC 0.986 on Re-LAION-5B and 0.946 on SD v1.5, confirming that gender-correlated visual signal is strongly encoded in both datasets before any transformation is applied. The slightly higher value on Re-LAION-5B likely reflects the greater photographic diversity of the web-scraped corpus, which provides a wider range of appearance cues correlated with gender labels, including facial structure, body morphology, hairstyle, and clothing. These results establish an upper bound against which the reduced representations are compared and confirm that the annotation pipeline produces labels sufficiently consistent for downstream probing.

Occlusion probes reveal a fundamental asymmetry between datasets The occlusion family is the most diagnostic set of gender probes, as it directly tests whether gender information is localised within the visible person appearance or distributed across the broader scene context.

On Re-LAION-5B, all person-hidden occlusion variants collapse to chance (AUC ≈ 0.500), regardless of whether masking uses a bounding rectangle or a tight segmentation contour, and regardless of whether the background is retained or removed. Variation across seeds was negligible for these near-chance results, confirming that the absence of recoverable signal is not due to random initialisation. This establishes that retained scene context carries no recoverable gender signal in

Re-LAION-5B. The full-person-no-background condition also yields chance performance (AUC 0.499), although this result is less straightforward to interpret: the probe collapses to majority-class prediction, which may reflect the difficulty of learning from isolated person crops under high photographic variability, annotation dependency on contextual features, or a genuine absence of learnable person-level signal in isolation. The unambiguous finding is that background context carries no independent demographic signal in Re-LAION-5B.

The pattern for SD v1.5 is qualitatively different. Both retained scene context and person-region geometry carry substantial gender signal. When the person is replaced by a white rectangle while the background is retained, the probe achieves AUC 0.747, indicating that the surrounding scene and coarse person-box geometry are strongly gender-correlated. When only the tight person-shaped silhouette is retained on a black background, performance rises to AUC 0.834, an 8.7 percentage-point increase over the rectangle-with-background condition. This indicates that the silhouette contour itself encodes gender signal beyond what is available from rectangular masking, likely through body shape and spatial proportions preserved by the tight segmentation boundary. When silhouette geometry and background context are available together under segmentation masking, performance reaches AUC 0.852, consistent with both sources contributing to the recovered gender signal. The central finding is therefore not which source is stronger in isolation, but that SD v1.5 encodes gender signal outside visible person appearance, whereas Re-LAION-5B does not. This is consistent with the contextual gender artefact identified by Meister et al. and shows that the effect is especially pronounced in the generated corpus.

Global colour statistics carry modest but consistent signal in both datasets Mean RGB classification achieves AUC 0.600 on Re-LAION-5B and 0.579 on SD v1.5. These values are comparable to the AUC of 0.580 reported by Meister et al. for mean RGB gender probing on COCO, suggesting that the magnitude of the photometric gender signal in both datasets is similar to that observed in a general-purpose image benchmark. The results indicate that gender-correlated photometric statistics remain detectable even when each image is compressed to three global colour values. These statistics may reflect systematic differences in clothing colour, lighting, background tone, or other photometric properties associated with gender labels. The similar performance across datasets suggests that this signal is not specific to either Re-LAION-5B or SD v1.5, but is a modest and recurring feature of occupational person imagery.

Geometric probes reveal body configuration as an independent signal source Pose keypoints achieve AUC 0.697 on Re-LAION-5B and 0.680 on SD v1.5, indicating that

normalised skeletal geometry correlates measurably with gender labels in both datasets. Because pose keypoints encode body proportions, posture, and limb arrangement without retaining colour or texture information, this result shows that body configuration contributes to gender separability independently of photometric appearance.

Depth values sampled at pose keypoint locations yield AUC 0.559 on Re-LAION-5B and 0.592 on SD v1.5, the lowest values among the above-chance probes. Although both exceed chance, the magnitude of separation is small, indicating that sparse 3D structure at skeletal locations carries only weak gender-correlated signal. Depth at keypoints is therefore substantially less informative than pose geometry, suggesting that the spatial configuration of pose keypoints carries more gender signal than the relative depth values sampled at those same keypoints. This suggests that sparse depth structure is not a dominant source of gender signal, although pose geometry itself remains informative.

Semantic probes reveal scene composition and language as additional signal carriers

Object detection probes show that scene composition and object co-occurrence carry gender-correlated information. On Re-LAION-5B, the unrestricted object detection probe achieves AUC 0.692, compared to 0.615 for the restricted variant. This 7.7 percentage-point gap indicates that objects appearing alongside people in Re-LAION-5B carry additional gender signal beyond the person-related labels. On SD v1.5, both object detection variants converge at AUC 0.608, suggesting that this form of object-level gender stereotyping is not amplified in the generated corpus to the same degree as the background signal observed under occlusion.

Caption-based probing shows that gender-correlated signal also survives compression to natural language. Captions achieve AUC 0.901 on Re-LAION-5B and 0.771 on SD v1.5, indicating that semantic descriptions of the images retain substantial gender signal in both datasets. Because the probe receives only textual input, these results show that gender associations are not confined to low-level image statistics: they are also present in the objects, activities, occupational contexts, and scene details captured by the captioning model. The stronger Re-LAION-5B caption result suggests that web-scraped images contain richer or more varied linguistic associations between gender, occupation, objects, and scene context. In SD v1.5, the caption signal remains clearly above chance, showing that gender stereotyping in the generated corpus is also reflected in semantic content, although less strongly than in the Re-LAION-5B captions.

The two datasets exhibit qualitatively distinct spurious gender feature profiles

Considered jointly, the results reveal a clear structural difference between the two

datasets. In Re-LAION-5B, retained background context carries no independent gender signal: all person-hidden occlusion variants collapse to chance regardless of whether the background is retained or removed. Other reduced representations, including mean RGB, object detection, and pose keypoints, carry modest above-chance signal, while captions retain strong gender signal. The full-person-no-background condition also collapses to chance, although this result is ambiguous and does not support a strong conclusion about person-level signal in isolation.

In SD v1.5, the defining feature is the presence of strong contextual gender signal. Gender is recoverable not only from visible person appearance, but also from retained scene context and person-region geometry. This pattern has no counterpart in Re-LAION-5B, where the same occlusion conditions remain at chance. Although some reduced representations, such as mean RGB, object detection, and pose keypoints, are comparable to or weaker than their Re-LAION-5B counterparts, SD v1.5 uniquely projects gender signal into the surrounding scene. This profile is consistent with the hypothesis that the generative model amplifies demographic priors from its LAION-derived training distribution, extending them beyond the generated person into background composition, spatial layout, and semantic content.

4.3.2 Skin Tone Artefact Analysis

Table 4.5 reports the macro-averaged skin tone probing results across all representations for Re-LAION-5B and Stable Diffusion v1.5, while Table 4.6 gives the corresponding per-class AUC values for MST 1–8. The discussion first establishes the unmodified-image upper bound, then examines the occlusion probes as the most direct test of contextual skin tone signal. It then considers global colour statistics, geometric representations, object-level scene composition, and caption-based semantic signal before synthesising the overall skin tone spurious-feature profiles of the two datasets.

Unmodified images establish a strong separability upper bound The base ResNet-50 probe achieves macro-AUC 0.968 on Re-LAION-5B and 0.900 on SD v1.5, confirming that skin tone information is strongly recoverable from unmodified images in both datasets. The per-class results are highly consistent across MST categories for Re-LAION-5B, ranging from 0.948 to 0.980. In SD v1.5, the per-class values vary more substantially, ranging from 0.779 to 0.996, with the lowest AUCs occurring for the mid-range tones MST4 and MST5. This indicates that skin tone is less uniformly separable across the tonal spectrum in the generated images than in the web-scraped corpus, a pattern that becomes more pronounced under reduced representations.

Transformation	Probe	Re-LAION-5B AUC	SD v1.5 AUC
Occlusion (full, no background)	ResNet-50	0.514 [†]	0.899
Occlusion (masked rectangle)	ResNet-50	0.514 [†]	0.728
Occlusion (masked rectangle, no background)	ResNet-50	0.515 [†]	0.660
Occlusion (masked segmentation)	ResNet-50	0.514 [†]	0.774
Occlusion (masked segmentation, no bg.)	ResNet-50	0.515 [†]	0.704
Mean RGB	Logistic Regression	0.563	0.543
Object detection (restricted)	Logistic Regression	0.591	0.626
Pose keypoints	MLP	0.596	0.666
Depth at pose keypoints	Logistic Regression	0.598	0.655
Object detection	Logistic Regression	0.611	0.663
Captions [‡]	SentenceT5	0.749	0.673
Base (unmodified)	ResNet-50	0.968	0.900

[†] Near-chance macro AUC; per-class AUCs at chance except MST8 (small-sample artefact).

[‡] Probe operates on BLIP Large captions encoded via SentenceT5-Base.

Table 4.5 Skin tone separability results per image transformation and dataset (MST 1–8). All probes trained with seed 0 unless otherwise noted. Macro-averaged AUC is the primary metric; chance level is 0.500.

Transf.	Dataset	MST1	MST2	MST3	MST4	MST5	MST6	MST7	MST8	MST9	MST10
Occ. (full., no bg)	Re-LAION-5B	0.499	0.501	0.501	0.499	0.501	0.499	0.499	0.614	—	—
Occ. (full., no bg)	SD v1.5	0.969	0.936	0.881	0.778	0.783	0.888	0.963	0.996	—	—
Occ. (rect.)	Re-LAION-5B	0.499	0.499	0.499	0.499	0.501	0.501	0.499	0.614	—	—
Occ. (rect.)	SD v1.5	0.837	0.777	0.715	0.594	0.625	0.685	0.741	0.853	—	—
Occ. (rect., no bg.)	Re-LAION-5B	0.499	0.501	0.501	0.501	0.501	0.499	0.501	0.614	—	—
Occ. (rect., no bg.)	SD v1.5	0.783	0.679	0.631	0.538	0.586	0.595	0.618	0.846	—	—
Occ. (seg.)	Re-LAION-5B	0.499	0.499	0.499	0.501	0.499	0.501	0.499	0.614	—	—
Occ. (seg.)	SD v1.5	0.876	0.821	0.751	0.625	0.645	0.738	0.815	0.921	—	—
Occ. (seg., no bg.)	Re-LAION-5B	0.501	0.501	0.499	0.501	0.501	0.499	0.501	0.614	—	—
Occ. (seg., no bg.)	SD v1.5	0.817	0.719	0.669	0.581	0.596	0.655	0.712	0.883	—	—
Mean RGB	Re-LAION-5B	0.546	0.586	0.570	0.576	0.518	0.563	0.583	0.558	—	—
Mean RGB	SD v1.5	0.585	0.594	0.525	0.520	0.529	0.538	0.551	0.499	—	—
Obj. det. (restr.)	Re-LAION-5B	0.682	0.596	0.578	0.550	0.540	0.548	0.577	0.653	—	—
Obj. det. (restr.)	SD v1.5	0.691	0.648	0.630	0.538	0.579	0.599	0.627	0.692	—	—
Pose keypoints	Re-LAION-5B	0.691	0.633	0.576	0.565	0.554	0.543	0.573	0.630	—	—
Pose keypoints	SD v1.5	0.770	0.668	0.640	0.539	0.588	0.610	0.638	0.875	—	—
Depth at pose	Re-LAION-5B	0.730	0.638	0.581	0.560	0.541	0.541	0.565	0.624	—	—
Depth at pose	SD v1.5	0.756	0.657	0.630	0.536	0.583	0.602	0.615	0.859	—	—
Obj. det.	Re-LAION-5B	0.705	0.639	0.586	0.564	0.550	0.564	0.588	0.691	—	—
Obj. det.	SD v1.5	0.735	0.683	0.649	0.566	0.589	0.637	0.672	0.775	—	—
Captions [‡]	Re-LAION-5B	0.784	0.732	0.719	0.760	0.734	0.751	0.745	0.769	—	—
Captions [‡]	SD v1.5	0.772	0.688	0.653	0.582	0.594	0.653	0.692	0.747	—	—
Base	Re-LAION-5B	0.976	0.976	0.969	0.961	0.948	0.959	0.973	0.980	—	—
Base	SD v1.5	0.966	0.935	0.883	0.779	0.783	0.895	0.964	0.996	—	—

[‡] Probe operates on BLIP Large captions encoded via SentenceT5-Base.

Table 4.6 Per-class skin tone AUC per image transformation and dataset (MST 1–8). MST 9 and MST 10 are absent from both datasets and omitted. All probes trained with seed 0 unless otherwise noted.

Occlusion probes reveal the same dataset asymmetry observed for gender, with per-class variation All occlusion variants on Re-LAION-5B produce macro-AUC values at chance (0.514–0.515). The per-class breakdown confirms that this is not caused by cancellation across classes: individual AUCs for MST1 through MST7 remain at chance (0.499–0.501) under every masking strategy. The exception is MST8, which reaches AUC 0.614 consistently across all variants, including those where the background is fully removed. Because this value is stable across masking strategies, it most plausibly reflects a small-sample artefact in this category rather than genuine contextual signal.

The person-hidden variants therefore show that retained scene context carries no recoverable skin tone signal in Re-LAION-5B. As with gender, the full-person-no-background condition also collapses to chance (0.514), although this result is less straightforward to interpret. It may reflect the photographic variability of web-scraped person crops, annotation dependency on contextual features, or a genuine absence of learnable person-level signal in isolation. The unambiguous finding is that background context carries no independent skin tone signal in Re-LAION-5B, matching the pattern observed for gender.

On SD v1.5, occlusion probes retain substantial above-chance performance across all masking strategies, with macro-AUC ranging from 0.660 to 0.899. The per-class results show that this signal is strongest at the extremes of the tonal distribution. MST1 and MST8 achieve the highest per-class AUCs across all variants, including 0.876 and 0.921 under segmentation masking, while mid-range tones such as MST4 and MST5 remain substantially lower. This indicates that the generative model encodes the strongest contextual skin tone priors at the tonal extremes, where scene associations may be most pronounced.

The same occlusion gradient observed for gender is present here. Segmentation-based masking consistently outperforms rectangular masking, and retaining the background alongside person occlusion consistently outperforms removing it. This indicates that skin tone signal in SD v1.5 is distributed across both person-region geometry and retained scene context, with the background providing additional information beyond mask geometry alone.

Global colour statistics carry weaker photometric signal than for gender Mean RGB achieves macro-AUC 0.563 on Re-LAION-5B and 0.543 on SD v1.5, lower than the corresponding gender values of 0.600 and 0.579. This is consistent with the expectation that global colour statistics carry weaker signal for skin tone than for gender. Skin tone is primarily a localised photometric property of face and body regions, so averaging colour across the full image dilutes this signal. By contrast, gender-correlated colour variation may also include broader cues such as clothing, hair,

lighting, and background tone.

The per-class mean RGB results for Re-LAION-5B are broadly distributed across MST categories, ranging from 0.518 to 0.586, with no clear concentration at the tonal extremes. This suggests that global colour statistics carry only weak skin tone signal, without a clear concentration in any specific MST category.

Geometric probes show stronger skin tone signal in SD v1.5 Pose keypoints and depth at pose keypoints produce similar macro-AUC values within each dataset. On Re-LAION-5B, pose keypoints achieve 0.596 and depth at pose keypoints achieves 0.598. On SD v1.5, the corresponding values are higher, at 0.666 and 0.655. This indicates that body configuration and sparse depth structure carry moderate skin tone signal, with stronger encoding in the generated corpus than in the web-scraped corpus.

The per-class pose results show that, as with the occlusion probes, the strongest SD v1.5 signal is concentrated at the tonal extremes. MST1 and MST8 achieve AUC 0.770 and 0.875 respectively, while MST4 remains close to chance at 0.539. This indicates that the skin tone signal recovered from geometric representations is not uniform across the tonal scale. Instead, SD v1.5 shows stronger separability for the lightest and darkest available MST categories, while mid-range tones remain harder to distinguish. This repeated pattern across occlusion and pose probes suggests that the skin tone signal in SD v1.5 is not limited to background context. It also appears in body-configuration cues, especially at the lightest and darkest available MST categories.

Object detection probes show that scene composition encodes skin tone in both datasets Both object detection variants achieve moderate macro-AUC values across datasets. The restricted probe reaches 0.591 on Re-LAION-5B and 0.626 on SD v1.5, while the unrestricted probe reaches 0.611 and 0.663 respectively. In contrast to the gender results, where the unrestricted Re-LAION-5B probe substantially outperforms the restricted variant (0.692 versus 0.615), the gap is modest for skin tone: approximately 2 percentage points for Re-LAION-5B and 3.7 percentage points for SD v1.5. This suggests that additional object co-occurrence contributes less skin tone signal than gender signal.

The consistently higher SD v1.5 values indicate that skin tone is more strongly encoded in object-level scene composition in the generated corpus than in the web-scraped corpus. This is consistent with the broader pattern observed across the reduced visual probes, where SD v1.5 tends to retain stronger demographic signal outside direct person appearance.

Caption probing replicates the cross-dataset reversal observed for gender

Caption-based probing replicates the cross-dataset reversal already observed in the gender caption results. Re-LAION-5B achieves macro-AUC 0.749, compared to 0.673 for SD v1.5. This differs from most non-caption visual probes, where SD v1.5 generally shows stronger skin tone signal than Re-LAION-5B. The same reversal appearing for both gender and skin tone suggests that demographic associations are encoded differently in the captions of real and generated images.

The Re-LAION-5B per-class caption AUCs are relatively consistent across tones, ranging from 0.719 to 0.784. This indicates that skin tone signal in captions is distributed broadly across the MST scale rather than concentrated only at the extremes. In SD v1.5, the per-class caption AUCs follow the pattern seen in several visual probes: MST1 and MST8 are more separable, while mid-range tones produce lower values. Together, these results suggest that SD v1.5 embeds skin tone priors more strongly in visual scene statistics, while Re-LAION-5B retains stronger and more evenly distributed skin tone associations in caption-level semantic content.

Per-class results reveal weaker separability for mid-range skin tones Across most reduced representations, especially in SD v1.5, MST1 and MST8 achieve higher per-class AUCs than the mid-range categories MST4 and MST5. This pattern suggests that the lightest and darkest available MST categories are linked to more distinctive non-skin cues, while mid-range tones are harder to separate from neighbouring categories. The result is important methodologically because macro-averaged AUC can obscure this per-class variation: a probe may appear moderately effective overall while still being much weaker for specific MST categories.

Skin tone and gender exhibit analogous but not identical spurious feature profiles

The skin tone results largely follow the structural pattern observed for gender. In Re-LAION-5B, background context carries no independent skin tone signal, while SD v1.5 distributes skin tone signal across scene context, mask geometry, object composition, body configuration, and captions. The main differences are quantitative rather than structural: skin tone AUCs under occlusion are generally lower than the corresponding gender AUCs, mean RGB is weaker for skin tone than for gender, and the caption results show a reversal relative to the dominant visual-probe pattern. The central finding nevertheless remains the same for both attributes: SD v1.5 projects demographic signal beyond visible person appearance in a way that is not observed in the web-scraped corpus.

4.3.3 Connections to Dataset Classification Results

Several parallels link the demographic probe experiments to the earlier dataset classification findings. Together, the two sets of results show that statistical signal is distributed across multiple representation levels rather than confined to a single image property.

The most direct parallel is the persistence of signal across transformations. In the dataset classification experiments, separability remained above chance for every transformation tested, from mean RGB at 64.19% to caption embeddings at 90.99%. This indicates that dataset identity is encoded across several visual and semantic channels at once. The demographic probing results show a related pattern: gender and skin tone are not recoverable from one representation alone, but appear across photometric probes, geometric probes, semantic probes, and occlusion-based contextual probes. In both analyses, the relevant signal is therefore distributed across the representational spectrum rather than being an artefact of a single cue.

The clearest connection between the two analyses concerns Stable Diffusion v1.5. In the dataset classification task, the high accuracies suggest that SD v1.5 outputs contain strong dataset-specific regularities. This is consistent with the fact that the generative pipeline applies a shared set of learned priors across its outputs. The demographic probing results provide a more targeted view of this regularity. SD v1.5 encodes gender at AUC 0.946 in the unmodified image and AUC 0.747 under background-retained rectangular occlusion; the corresponding skin tone values are 0.900 and 0.728. These results indicate that the generated corpus contains stable associations between person appearance, scene context, and person-region geometry. Re-LAION-5B shows no comparable contextual signal under person-hidden occlusion conditions, indicating that its scene context does not carry the same independent demographic signal. The full-person-no-background condition also collapses to chance in Re-LAION-5B, although this result remains ambiguous and does not support a definitive conclusion about whether isolated person appearance is sufficient in real photographs.

The caption results provide a second connection between the two analyses. Dataset identity survives compression to natural language at 90.99% classification accuracy, placing caption-based classification close to semantic segmentation and above object detection in the dataset classification hierarchy. Demographic signal also survives in captions: gender achieves AUC 0.901 on Re-LAION-5B and 0.771 on SD v1.5, while skin tone achieves 0.749 and 0.673 respectively. Unlike the strongest occlusion results, where SD v1.5 shows the clearest contextual demographic signal, captions show stronger demographic separability in Re-LAION-5B. This suggests that web-scraped images contain richer or more varied semantic associations between

demographic attributes, occupations, objects, and activities. The caption findings therefore show that stereotyping is not confined to pixel statistics or geometry, but is also present in the semantic content of the images.

Finally, the depth and edge classification results connect to the depth-at-pose-keypoints probe used in the demographic analysis. Full depth maps achieve 96.78% dataset classification accuracy, and Canny edges achieve 93.62%, confirming that structural representations carry strong dataset-discriminating signal. At the demographic level, however, depth sampled only at pose keypoints is much weaker: it reaches AUC 0.559 for gender and 0.598 for skin tone on Re-LAION-5B, and 0.592 and 0.655 on SD v1.5. This contrast suggests that the structural signal captured by full depth maps is driven largely by broader composition, such as camera distance, scene depth, and background structure, rather than by sparse body-depth information alone. The depth-at-keypoints probe therefore isolates a smaller demographic component of the broader structural signal while discarding much of the scene-level information that makes full depth maps highly discriminative for dataset classification.

4.4 Debiasing Evaluation

The debiasing evaluation compares the demographic distribution and spurious feature profile of the 10,000-image debiased doctor corpus against the full Stable Diffusion v1.5 generation results reported in Sections 4.1 and 4.3. The evaluation uses the aggregate SD v1.5 findings as the reference point rather than constructing a separate doctor-specific baseline. This allows the debiased corpus to be assessed against the overall demographic and spurious-feature profile characteristic of standard Stable Diffusion v1.5 generation.

This comparison directly tests whether the combined ITI-GEN, ControlNet, and Chamfer colour alignment pipeline produces a measurable departure from the standard generation profile. The rationale for selecting the doctor profession is discussed in Section 3.6.1. In summary, doctor was selected because it exhibits one of the most pronounced demographic skews in the standard pipeline, carries a well-documented real-world demographic baseline, and has been used as an evaluation case in prior work on occupational bias in text-to-image generation [7, 8].

An important methodological distinction applies throughout this section. Demographic labels for the debiased corpus are not derived from automated classifiers, but are assigned directly from the generation configuration. Each image is labelled with the ITI-GEN gender token and MST skin tone token used to produce it. This removes annotation uncertainty as a confound and ensures that the probing classifiers measure whether the demographic conditioning signal is encoded in

different representations, rather than whether an external classifier can recover predicted demographic attributes.

For gender, high probing accuracy on person-level representations is therefore an expected consequence of successful ITI-GEN conditioning rather than evidence of spurious signal. For skin tone, the 10-class task is substantially harder than the binary gender task, and person-level probe values should be interpreted relative to the macro-averaged AUC chance level of 0.500 rather than compared directly with the binary gender results.

4.4.1 Demographic Distribution Shift

The debiased generation pipeline was configured to produce a balanced corpus across both gender and skin tone. ITI-GEN embeddings were applied for female and male gender categories in equal proportions across the 10,000-image corpus, while MST conditioning was applied uniformly across all ten skin tone levels, MST 1–10. This produced a target distribution of approximately 500 images per gender–skin tone combination. After structural validation using the same acceptance criteria as the standard generation pipeline, the effective composition of the debiased corpus remained aligned with this intended uniform design, subject to profession-specific rejection rates.

By construction, the debiased corpus achieves an approximately balanced gender distribution, with 50% female and 50% male conditioning. This contrasts directly with the standard Stable Diffusion v1.5 pipeline, which produced only 29.02% female images across all 200 professions. As discussed in Section 4.1, that standard distribution reflects substantial amplification of gender imbalance relative to the retrieved corpus.

The skin tone distribution is also balanced by construction across MST 1–10. This contrasts with the standard pipeline, where 76.99% of generated images fell in the Mid bin whilst Dark-bin representation approached zero across most professions. The debiased generation configuration therefore produces a distribution that is sharply different from the demographic profile observed under standard SD v1.5 generation.

Figure 4.7 shows representative debiased outputs at a fixed seed. Rows correspond to female and male gender conditioning, while columns correspond to MST skin tone levels 1 through 10.

4.4.2 Probing Accuracy on the Debiased Corpus

To assess whether the debiasing interventions reduced not only the aggregate demographic imbalance but also the learnability of demographic signal from spurious



Figure 4.7 Representative debiased generation samples at a fixed seed. Rows correspond to female (top) and male (bottom) ITI-GEN gender conditioning; columns correspond to MST skin tone levels 1 through 10. Skin tone transitions progressively from lighter values at MST 1 to darker values at MST 10, demonstrating that the ten-point conditioning axis produces a visually coherent and continuous tonal gradient. Gender presentation is clearly differentiated between rows through facial features, body morphology, and clothing style. The ControlNet OpenPose conditioning enforces a consistent standing, full-body, centred pose across all twenty images, and the Chamfer colour alignment is reflected in the broadly consistent warm tonal palette maintained throughout the grid. The background exhibits some variation across columns, reflecting the residual scene-level variation that pose conditioning does not fully constrain, consistent with the background-level gender signal identified in Section 4.4.2.

features, the same probing classifiers applied to the standard SD v1.5 corpus were applied to the debiased corpus under identical experimental conditions. Tables 4.7 and 4.8 report gender and skin tone probing AUC for the standard SD v1.5 corpus and the debiased corpus across all evaluated representations.

Gender Probing Results

Person-level representations confirm successful demographic conditioning. The unmodified image probe achieves $AUC \approx 1.000$ on the debiased corpus, while person-centred occlusion variants are also very high: 0.998 for full person appearance without background (FullNoBg), 0.999 for the segmentation-masked person region with background retained (MaskSegm), and 0.985 for the silhouette alone without background (MaskSegmNoBg). These values exceed the corresponding standard SD v1.5 results by substantial margins.

This outcome is expected. ITI-GEN directly conditions the person’s visual appearance, and labels in the debiased corpus are assigned from the conditioning token rather than from an external classifier. High person-level separability therefore confirms that the conditioning pipeline produced visibly gendered images and encoded the intended demographic signal in the depicted person.

Across most non-person representations, gender signal is substantially reduced.

Representation	Probe	SD v1.5	Debiased
Base (unmodified)	ResNet-50	0.946	≈ 1.000
Occlusion (full, no background)	ResNet-50	0.948	0.998
Occlusion (masked segmentation)	ResNet-50	0.852	0.999
Occlusion (masked segmentation, no bg.)	ResNet-50	0.834	0.985
Occlusion (masked rectangle)	ResNet-50	0.747	0.995
Occlusion (masked rectangle, no bg.)	ResNet-50	0.587 [†]	0.701
Mean RGB	Logistic Regression	0.579	0.507
Object detection	Logistic Regression	0.608	0.449
Object detection (restricted)	Logistic Regression	0.608	0.457
Depth at pose keypoints	Logistic Regression	0.592	0.454
Pose keypoints	MLP	0.680	0.475
Captions	SentenceT5	0.771	0.530

Table 4.7 Gender probing AUC on the standard Stable Diffusion v1.5 corpus and the debiased corpus. Chance level is 0.500 for binary classification. Labels for the debiased corpus are derived from the generation configuration rather than automated annotation. [†] Standard SD v1.5 value reflects near-complete majority-class collapse; see Section 4.3.1 for detail.

Representation	Probe	SD v1.5	Debiased
Base (unmodified)	ResNet-50	0.900	0.861
Occlusion (full, no background)	ResNet-50	0.899	0.823
Occlusion (masked segmentation)	ResNet-50	0.774	0.729
Occlusion (masked segmentation, no bg.)	ResNet-50	0.704	0.663
Occlusion (masked rectangle)	ResNet-50	0.728	0.632
Occlusion (masked rectangle, no bg.)	ResNet-50	0.660	0.543
Mean RGB	Logistic Regression	0.543	0.482
Object detection	Logistic Regression	0.663	0.439
Object detection (restricted)	Logistic Regression	0.626	0.461
Depth at pose keypoints	Logistic Regression	0.655	0.443
Pose keypoints	MLP	0.666	0.483
Captions	SentenceT5	0.673	0.490

Table 4.8 Skin tone probing macro-averaged AUC on the standard Stable Diffusion v1.5 corpus and the debiased corpus. The SD v1.5 column reports 8-class macro-averaged AUC (MST 1–8, as MST 9 and MST 10 are absent from the standard corpus); the debiased column reports 10-class macro-averaged AUC (MST 1–10, from the generation configuration). Chance level is 0.500 in both cases. Labels for the debiased corpus are derived from the generation configuration rather than automated annotation.

Mean RGB falls from 0.579 on the standard SD v1.5 corpus to 0.507 on the debiased corpus, effectively at the binary chance level of 0.500. Object detection falls to 0.449 for the unrestricted vocabulary and 0.457 for the restricted vocabulary, indicating no systematic gender-correlated object co-occurrence in the debiased corpus. Depth at pose keypoints falls to 0.454, and pose keypoints fall to 0.475, both close to chance and consistent with little recoverable gender signal in geometric representations. These reductions contrast with the standard SD v1.5 corpus, where the same probes achieved AUC values between 0.579 and 0.680.

Caption-based probing falls from 0.771 on the standard SD v1.5 corpus to 0.530 on the debiased corpus, a reduction of 24.1 percentage points. This indicates that the debiasing pipeline substantially reduces gender-correlated semantic content in generated images, even though caption generation is not directly controlled by the interventions. The result suggests that the reduction in gender signal extends beyond low-level visual features and also affects the semantic content captured by captions.

The main exception is the background-retained rectangular occlusion probe. This probe achieves AUC 0.995 on the debiased corpus, indicating that strong background-level gender signal persists even when the person region is hidden. This is expected because none of the debiasing interventions directly targets background composition: ITI-GEN conditions person appearance, ControlNet constrains skeletal pose, and Chamfer alignment steers global colour statistics. Scene-level content therefore remains free to co-vary with the gender conditioning token. The corresponding standard SD v1.5 value is 0.747, making this increase particularly important. The persistence of near-person-level AUC under background-retained occlusion shows that background-level demographic conditioning remains substantially entangled with the generation process and represents the primary avenue for further improvement.

Skin Tone Probing Results

The skin tone probing results show reductions across all representations relative to the standard SD v1.5 corpus. Unlike the gender case, person-level probes decrease rather than increase in the debiased corpus: the unmodified base probe falls from 0.900 to 0.861, and the full-person-no-background probe falls from 0.899 to 0.823. This reduction should be interpreted in light of the harder label space used for the debiased corpus. The standard SD v1.5 comparison uses MST 1–8, while the debiased corpus is evaluated across the full MST 1–10 range under a balanced distribution. The person-level values of 0.823–0.861 remain substantially above the macro-AUC chance level of 0.500, confirming that ITI-GEN conditioning successfully encodes skin tone signal at the person level across the ten-point scale.

The clearest evidence for reduced global photometric signal comes from the Mean RGB probe, which falls from 0.543 on the standard SD v1.5 corpus to 0.482 on the debiased corpus. This is the skin tone probe closest to the chance level of 0.500. Since Mean RGB compresses each image to its global average colour per channel, the near-chance value indicates that the debiased corpus contains little systematic skin-tone-correlated signal in global colour statistics. This supports the conclusion that the colour alignment component disrupts the photometric correlations that previously allowed global colour statistics to carry demographic signal.

The remaining non-person probes also collapse to at or near chance. Object detection falls from 0.663 to 0.439 for the unrestricted vocabulary and from 0.626 to 0.461 for the restricted vocabulary, indicating no systematic skin-tone-correlated object co-occurrence in the debiased corpus. Depth at pose keypoints falls from 0.655 to 0.443, and pose keypoints fall from 0.666 to 0.483, both close to chance and consistent with little recoverable skin tone signal in geometric representations. Caption probing falls from 0.673 to 0.490, indicating that the semantic content captured by generated captions is no longer systematically correlated with the skin tone conditioning token.

The occlusion probes show consistent reductions across all variants, but some residual contextual signal remains. MaskRectNoBg falls from 0.660 to 0.543, close to chance, indicating that rectangular person-box geometry alone carries little residual skin tone signal after debiasing. MaskRect falls from 0.728 to 0.632, indicating a partial but incomplete reduction of background-retained skin tone signal. The remaining above-chance performance in MaskRect suggests that scene composition retains some skin-tone-correlated properties, consistent with the background entanglement observed in the gender analysis.

4.5 Limitations

Several limitations qualify the interpretation of the results presented in this chapter.

Scope of the Re-LAION-5B sample. The analysis is conducted on four of 128 shards of Re-LAION-5B, representing approximately 3.1% of the full dataset. Findings are therefore estimates of corpus-level properties rather than exact population parameters, and biases introduced during the original web-crawling and filtering process cannot be fully assessed from the released shard data alone.

Annotation as measurement instrument. Demographic labels are produced by automated classifiers rather than ground-truth human annotation. Gender predictions

are binary and do not capture non-binary gender expression. Skin tone estimates are sensitive to imaging conditions and lighting variation in web-scraped data. Observed demographic distributions should therefore be interpreted as classifier-derived labels rather than definitive ground truth.

Absence of a real-world skin tone baseline. Unlike gender, skin tone could not be compared against a reliable real-world occupational baseline. The available BLS reference data reports racial categories rather than skin tone, introducing a further limitation because race and skin tone are related but not equivalent constructs. An exploratory race-to-MST calibration was attempted using FairFace as an intermediate reference corpus, but the resulting conditional distributions mapped all racial groups predominantly to the Mid bin, providing insufficient variation across MST categories for meaningful profession-level comparison. No alternative baseline was adopted because deriving population-level skin tone distributions from racial labour-force statistics would require unsupported assumptions about within-category skin tone variation. Skin tone results are therefore interpreted as relative differences between Re-LAION-5B and Stable Diffusion v1.5 rather than as deviations from a real-world skin tone ground truth.

Dataset classification and causation. Above-chance classification accuracy confirms that a transformation retains dataset-discriminating signal, but it does not establish which specific properties of each dataset produce that signal. The demographic probing experiments provide a more targeted analysis by testing whether gender and skin tone are recoverable from specific reduced representations, such as background context, pose, object detections, captions, and global colour statistics. However, these probes still identify statistical associations rather than causal mechanisms. Further controlled experiments would be required to isolate the exact image properties responsible for each source of separability.

Debiasing evaluation scope. The debiasing evaluation is restricted to a single profession and a single combination of inference-time interventions. It does not evaluate weight editing or fine-tuning approaches, and results may not generalise to other professions, prompts, or model architectures. The probing results show that spurious correlations are substantially reduced at the representation level for the doctor profession, but whether this reduction generalises across the full 200 profession set or persists under different prompt formulations remains untested. The residual signal identified under background-retained occlusion in both the gender and skin tone analyses indicates that background-level demographic conditioning remains

partially entangled with the generation process and constitutes the primary open question for future work on this pipeline.

4.6 Synthesis of Evaluation Findings

The dataset classification experiments establish that Re-LAION-5B, Stable Diffusion v1.5 generated images, and COCO are distinguishable with high accuracy across every level of visual representation tested, from unmodified images at 99.42% down to mean RGB compression at 64.19%. The persistence of above-chance separability even under the most aggressive information reduction demonstrates that dataset identity is encoded in multiple correlated visual cues rather than in any single isolated artefact. The VAE experiment further shows that this separability is unlikely to be explained primarily by low-level acquisition artefacts, as reconstruction through the Stable Diffusion latent bottleneck reduces accuracy by only 0.27 percentage points.

The demographic distribution analysis shows that Re-LAION-5B exhibits gender distributions that diverge from real-world labour force composition in a baseline-dependent manner, while Stable Diffusion v1.5 amplifies the male skew across nearly every occupational group. Gender amplification is observed in 178 of 200 professions, with the female proportion falling from 41.5% in Re-LAION-5B to 29.02% in Stable Diffusion v1.5. The skin tone analysis reveals a near-complete collapse of dark-toned representation in generated outputs: the Dark-bin proportion falls from 0.81% in Re-LAION-5B to 0.14% in Stable Diffusion v1.5, an 83% reduction, while the Light-bin proportion almost doubles from 12.45% to 22.87%. Skin tone amplification toward lighter tones is observed in 175 of 200 professions.

The demographic probing experiments reveal distinct spurious feature profiles in the two datasets. In Re-LAION-5B, retained background context carries no independent demographic signal: every person-hidden occlusion variant collapses to chance regardless of whether background is retained, and residual signal from colour statistics, object co-occurrence, and skeletal geometry is modest. The full-person-no-background condition also collapses to chance, although this result is ambiguous and does not support a strong conclusion about person-level signal in isolation.

In Stable Diffusion v1.5, demographic signal is recoverable from representations that do not preserve visible person appearance. It appears under background-retained occlusion, from person-region geometry, and in natural language captions. This pattern is not observed in Re-LAION-5B, where person-hidden occlusion variants collapse to chance. The strongest difference is therefore not simply that SD v1.5 depicts gender and skin tone in the person region, but that demographic signal is also embedded in the

surrounding scene and in the spatial structure of the generated image.

The caption-based results add a further distinction between the datasets. Captions retain stronger demographic signal for Re-LAION-5B than for Stable Diffusion v1.5, even though the strongest visual contextual signals occur in the generated corpus. One plausible explanation is the gender-neutral remapping applied before caption classification: generated images produce more repetitive captions, removing explicit gendered terms may remove a larger share of the available caption signal. Captions of web-scraped images, by contrast, reflect a broader range of occupational settings, activities, objects, and scene contexts, allowing demographic associations to remain recoverable even after remapping. This cross-modal persistence shows that demographic stereotyping is not confined to low-level pixel statistics, but extends across visual, geometric, contextual, and semantic representations.

The debiasing evaluation demonstrates that the combined ITI-GEN, ControlNet, and Chamfer colour alignment pipeline produces a measurable departure from the demographic and spurious feature profile of the standard generation pipeline. Most non-person probes collapse to at or near chance for both gender and skin tone, including mean RGB, object detection, pose, depth-at-pose, and caption-based representations. This indicates that the interventions substantially reduce low-level photometric correlations, object-level stereotyping, geometric signal, and semantic stereotyping in the debiased doctor corpus. However, residual signal persists under background-retained occlusion, indicating that background-level demographic conditioning remains partially entangled with the generation process even under explicit pose and appearance control.

5 Conclusions and Future Work

5.1 Conclusions

This thesis applied and extended established spurious feature detection and demographic probing methodologies to Re-LAION-5B and Stable Diffusion v1.5, producing a systematic empirical characterisation of how demographic bias is encoded, distributed, and amplified across a web-scale training corpus and the generative model trained on it. The findings address each of the five research objectives and collectively advance understanding of bias mechanisms in large-scale vision systems.

Objective 1: Dataset Acquisition, Preparation, and Annotation.

A profession-conditioned subset of 200,000 images was retrieved from Re-LAION-5B across 200 occupations aligned with ISCO-08 categories, and a corresponding set of 200,000 images was generated using Stable Diffusion v1.5 with LCM-LoRA acceleration for the standard generation pipeline. CLIP embeddings were recomputed from scratch using `openai/clip-vit-large-patch14` in the absence of pre-computed embeddings, and a two-stage structural validation pipeline was developed to enforce human-centric image suitability across both corpora. Demographic annotation was performed using the Realistic Gender Classifier and a VGG-16 MST10 Regression model selected through systematic evaluation against CCv2, MSTE, and FACET benchmarks, with skin tone extended to the full ten-point Monk Skin Tone scale beyond the scales used in prior work.

Objective 2: Spurious Feature Detection. The transformation-based framework of Zeng et al. was applied and extended to a three-dataset configuration comprising Re-LAION-5B, Stable Diffusion v1.5, and COCO. Fifteen transformations spanning frequency, geometry, semantics, and colour were evaluated. Every transformation exceeded the 33.3% chance level, from unmodified images at 99.42% down to mean RGB at 64.19%. These results establish that dataset-specific spurious signatures persist even under extreme information compression and are distributed across multiple correlated visual attributes rather than concentrated in any single representation.

Objective 3: Demographic Correlation Assessment. Demographic probing following the occlusion framework of Meister et al. revealed a fundamental asymmetry between the two datasets. In Re-LAION-5B, person-hidden occlusion variants collapse to chance for both gender and skin tone, indicating that retained background context carries no independent demographic signal. The full-person-no-background condition

also collapses to chance, making person-level signal in isolation difficult to interpret under the applied masking protocol. In Stable Diffusion v1.5, demographic information is recoverable from representations that do not preserve visible person appearance, including background-retained rectangular occlusion (gender AUC 0.747; skin tone AUC 0.728), person-region geometry (gender AUC 0.834 under segmentation-mask no-background occlusion), object co-occurrence, skeletal geometry, and natural language descriptions. This cross-representational persistence demonstrates that the generative model does not merely inherit demographic priors from its training corpus but projects them onto scene context, spatial structure, and semantic content in generated images. Gender amplification was confirmed in 178 of 200 professions and skin tone amplification toward lighter tones in 175 of 200 professions, with the Dark-bin proportion collapsing by 83% from Re-LAION-5B to Stable Diffusion v1.5.

Objective 4: Inference-Time Debiasing. A combined pipeline of ITI-GEN demographic conditioning, ControlNet OpenPose structural conditioning, and Chamfer latent-space colour alignment was implemented and evaluated on a 10,000-image debiased doctor corpus. The pipeline successfully overrides the model’s strongly skewed learned prior at the distributional level, achieving a uniform gender and skin tone distribution by construction. At the representation level, most non-person probes collapse to at or near chance for both gender and skin tone, including mean RGB (0.507 for gender; 0.482 for skin tone), object detection, pose keypoints, depth at keypoints, and caption-based probing (gender 0.530; skin tone 0.490). The near-chance Mean RGB skin tone result provides strong evidence that the colour alignment component disrupts the photometric correlations that previously allowed global colour statistics to carry demographic signal. However, residual background-level demographic signal persists in the debiased corpus. The background-retained rectangular occlusion probe achieves AUC 0.995 for gender and 0.632 for skin tone, while the rectangular no-background condition achieves AUC 0.701 for gender and 0.543 for skin tone. This indicates that scene composition remains partially entangled with the demographic conditioning signal even under pose and appearance control.

Objective 5: Synthesis of Bias Mechanisms and Mitigation. The combined evidence characterises a two-stage bias propagation process. Re-LAION-5B exhibits gender distributions that diverge from real-world labour force composition in a baseline-dependent manner, falling below US-specific norms in the majority of occupational groups while remaining broadly consistent with global workforce proportions in several others. It also exhibits modest demographic signal across colour, geometry, and semantic representations, but no independent demographic signal in retained background context. Stable Diffusion v1.5 then amplifies these imbalances

and transforms their distribution: the generative process projects demographic priors beyond the depicted person onto background, scene composition, object layout, and textual description, producing a more entrenched and pervasive form of bias than is observed in the retrieved corpus. Inference-time debiasing can measurably reduce spurious correlations at the representation level without model retraining, but background-level entanglement persists and represents the primary remaining challenge for this class of interventions.

Impact on the field. These findings have several direct implications for practitioners and researchers working with large-scale visual datasets and generative systems. First, auditing a generative model’s demographic outputs is insufficient without also auditing the spurious feature distributions of the training corpus, as the generative process can amplify and transform biases rather than simply reproduce them. Second, occlusion-based probing provides a practical and interpretable diagnostic for distinguishing person-localised from scene-distributed demographic signal, a distinction that aggregate distribution metrics cannot detect. Third, the persistence of demographic signal in natural language descriptions of generated images demonstrates that stereotyping in generative systems is not confined to pixel statistics but extends into semantic content, with implications for downstream applications that consume generated text-image pairs. Fourth, the debiasing evaluation demonstrates that inference-time interventions can substantially reduce many spurious correlations, but background-level demographic conditioning requires interventions that target scene composition directly rather than person appearance alone.

5.2 Future Work

Several directions present themselves as natural extensions of the work presented in this thesis, spanning improvements to the debiasing methodology, broader evaluation scope, and deeper investigation of the mechanisms identified here.

The most immediate extension concerns the debiasing pipeline. Residual background-level demographic signal persists at AUC 0.995 for gender and 0.632 for skin tone under background-retained occlusion, indicating that scene composition remains partially entangled with the demographic conditioning signal even under pose and appearance control. Future work could investigate targeted background conditioning mechanisms that explicitly decorrelate scene composition from demographic attributes, either through adversarial background debiasing approaches or through extension of the ITI-GEN framework to condition on background properties independently of person appearance. The weight editing approach of Unified Concept

Editing [85] offers a complementary avenue, as its direct modification of cross-attention projections may suppress background-level demographic associations that inference-time conditioning does not address. A systematic comparison of conditioning-based approaches against weight editing methods on the same probing protocol would establish whether background entanglement is specific to the conditioning paradigm or is a general property of inference-time interventions.

The debiasing evaluation in this thesis is restricted to a single profession. Extending the evaluation to a stratified sample spanning all ten ISCO-08 major groups would determine whether the reduction of spurious correlations observed for the doctor profession generalises across occupational contexts with different baseline demographic distributions and different magnitudes of stereotyping. Professions at the extremes of the observed skew distributions represent particularly informative test cases, as the debiasing pipeline would be required to counteract the strongest learned priors in the model.

The skin tone analysis in this thesis uses the full ten-point MST scale for the debiased corpus, with labels derived from the generation configuration rather than from automated classifiers. A complementary evaluation using validated automated skin tone estimation on the debiased corpus would quantify the photometric fidelity of the conditioning, assessing whether images conditioned on darker MST tokens exhibit measurably different skin pigmentation statistics and whether the Chamfer colour alignment produces coherent skin tone variation rather than global colour shifts. This would also enable the debiasing evaluation to be compared directly with the eight-class standard corpus results.

The transformation-based probing framework applied in this work uses ConvNeXt-Tiny as the classification backbone. Future work could evaluate whether the separability ceiling identified here reflects genuine representational differences between datasets or the capacity limitations of the chosen probe, by applying larger classification architectures or self-supervised foundation model representations. This is particularly relevant for the full-person-no-background condition in the gender analysis, where chance-level performance may reflect either a genuine absence of independently learnable person-level signal under the applied masking protocol or a limitation of the probe under high photographic variability. Evaluating this condition with a higher-capacity backbone, or on a larger balanced crop set, would help disambiguate these explanations. The original Zeng et al. ablation demonstrates limited impact of model size on dataset classification accuracy for structural transformations, but this was conducted in a different dataset context and may not generalise to the Re-LAION-5B versus Stable Diffusion v1.5 comparison.

Finally, the analysis is grounded in a subset of four Re-LAION-5B shards, covering approximately 3.1% of the full corpus. Extending the analysis to a larger shard

sample would determine whether the spurious feature profiles and demographic distributions identified here are stable properties of the corpus as a whole or are influenced by domain concentration effects specific to the retrieved subset. The link attrition and domain concentration analyses conducted in this work establish that the retrieved subset is non-uniform in its domain composition, and access to LAION crawl metadata would enable a more principled characterisation of the representativeness of the surviving URL pool.

References

- [1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2022, pp. 10 684–10 695.
- [2] C. Schuhmann et al., *Laion-5b: An open large-scale dataset for training next generation image-text models*, 2022. arXiv: 2210.08402 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2210.08402>.
- [3] A. Birhane, V. U. Prabhu, and E. Kahembwe, *Multimodal datasets: Misogyny, pornography, and malignant stereotypes*, 2021. arXiv: 2110.01963 [cs.CY]. [Online]. Available: <https://arxiv.org/abs/2110.01963>.
- [4] T. Bolukbasi, K.-W. Chang, J. Zou, V. Saligrama, and A. Kalai, *Man is to computer programmer as woman is to homemaker? debiasing word embeddings*, 2016. arXiv: 1607.06520 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/1607.06520>.
- [5] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, *A survey on bias and fairness in machine learning*, 2022. arXiv: 1908.09635 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1908.09635>.
- [6] J. Buolamwini and T. Gebru, “Gender shades: Intersectional accuracy disparities in commercial gender classification,” in *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, S. A. Friedler and C. Wilson, Eds., ser. *Proceedings of Machine Learning Research*, vol. 81, PMLR, 23–24 Feb 2018, pp. 77–91. [Online]. Available: <https://proceedings.mlr.press/v81/buolamwini18a.html>.
- [7] A. S. Luccioni, C. Akiki, M. Mitchell, and Y. Jernite, *Stable bias: Analyzing societal representations in diffusion models*, 2023. arXiv: 2303.11408 [cs.CY]. [Online]. Available: <https://arxiv.org/abs/2303.11408>.
- [8] F. Bianchi et al., “Easily accessible text-to-image generation amplifies demographic stereotypes at large scale,” in *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, ser. *FAccT '23*, Chicago, IL, USA: Association for Computing Machinery, 2023, pp. 1493–1504, ISBN: 9798400701924. DOI: 10.1145/3593013.3594095. [Online]. Available: <https://doi.org/10.1145/3593013.3594095>.
- [9] N. Meister, D. Zhao, A. Wang, V. V. Ramaswamy, R. Fong, and O. Russakovsky, *Gender artifacts in visual datasets*, 2023. arXiv: 2206.09191 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2206.09191>.

- [10] B. Zeng, Y. Yin, and Z. Liu, *Understanding bias in large-scale visual datasets*, 2024. arXiv: 2412.01876 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2412.01876>.
- [11] K. Yang, K. Qinami, L. Fei-Fei, J. Deng, and O. Russakovsky, "Towards fairer datasets: Filtering and balancing the distribution of the people subtree in the imagenet hierarchy," in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, ser. FAT* '20, ACM, Jan. 2020, pp. 547–558. DOI: 10.1145/3351095.3375709. [Online]. Available: <http://dx.doi.org/10.1145/3351095.3375709>.
- [12] B. H. Zhang, B. Lemoine, and M. Mitchell, *Mitigating unwanted biases with adversarial learning*, 2018. arXiv: 1801.07593 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1801.07593>.
- [13] F. Friedrich et al., *Fair diffusion: Instructing text-to-image generation models on fairness*, 2023. arXiv: 2302.10893 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2302.10893>.
- [14] Common Crawl, *Common crawl – open repository of web crawl data*, <https://commoncrawl.org>, Non-profit organisation. Accessed: February 2026, 2007.
- [15] LAION, *Releasing re-laion-5b: Transparent iteration on laion-5b with additional safety fixes*, <https://laion.ai/blog/relaion-5b/>, 2024.
- [16] A. Radford et al., *Learning transferable visual models from natural language supervision*, 2021. arXiv: 2103.00020 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2103.00020>.
- [17] C. Schuhmann, *LAION-Aesthetics*, <https://laion.ai/blog/laion-aesthetics/>, Aug. 2022.
- [18] R. Rombach and P. Esser, *Stable diffusion v1-5 model card*, <https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5>, 2022.
- [19] M. Klein et al., "Scholarly context not found: One in five articles suffers from reference rot," *PloS one*, vol. 9, e115253, Dec. 2014. DOI: 10.1371/journal.pone.0115253.
- [20] I. J. Goodfellow et al., *Generative adversarial networks*, 2014. arXiv: 1406.2661 [stat.ML]. [Online]. Available: <https://arxiv.org/abs/1406.2661>.
- [21] P. Dhariwal and A. Nichol, *Diffusion models beat gans on image synthesis*, 2021. arXiv: 2105.05233 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2105.05233>.

- [22] D. P. Kingma and M. Welling, *Auto-encoding variational bayes*, 2022. arXiv: 1312.6114 [stat.ML]. [Online]. Available: <https://arxiv.org/abs/1312.6114>.
- [23] A. Ramesh et al., *Zero-shot text-to-image generation*, 2021. arXiv: 2102.12092 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2102.12092>.
- [24] J. Ho, A. Jain, and P. Abbeel, *Denoising diffusion probabilistic models*, 2020. arXiv: 2006.11239 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2006.11239>.
- [25] O. Ronneberger, P. Fischer, and T. Brox, *U-net: Convolutional networks for biomedical image segmentation*, 2015. arXiv: 1505.04597 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1505.04597>.
- [26] J. Ho and T. Salimans, *Classifier-free diffusion guidance*, 2022. arXiv: 2207.12598 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2207.12598>.
- [27] D. Podell et al., *Sdxl: Improving latent diffusion models for high-resolution image synthesis*, 2023. arXiv: 2307.01952 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2307.01952>.
- [28] P. Esser et al., *Scaling rectified flow transformers for high-resolution image synthesis*, 2024. arXiv: 2403.03206 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2403.03206>.
- [29] E. J. Hu et al., *Lora: Low-rank adaptation of large language models*, 2021. arXiv: 2106.09685 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2106.09685>.
- [30] S. Luo, Y. Tan, L. Huang, J. Li, and H. Zhao, *Latent consistency models: Synthesizing high-resolution images with few-step inference*, 2023. arXiv: 2310.04378 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2310.04378>.
- [31] S. Luo et al., *Lcm-lora: A universal stable-diffusion acceleration module*, 2023. arXiv: 2311.05556 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2311.05556>.
- [32] J. D. Pressman, K. Crowson, and S. C. Contributors, "Simulacra aesthetic captions," Stability AI, Version 1.0, 2022, url <https://github.com/JD-P/simulacra-aesthetic-captions>.
- [33] J. Deng, J. Guo, Y. Zhou, J. Yu, I. Kotsia, and S. Zafeiriou, *Retinaface: Single-stage dense face localisation in the wild*, 2019. arXiv: 1905.00641 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1905.00641>.
- [34] S. Yang, P. Luo, C. C. Loy, and X. Tang, *Wider face: A face detection benchmark*, 2015. arXiv: 1511.06523 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1511.06523>.

- [35] J. Guo, J. Deng, A. Lattas, and S. Zafeiriou, *Sample and computation redistribution for efficient face detection*, 2021. arXiv: 2105.04714 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2105.04714>.
- [36] G. Jocher, A. Chaurasia, and J. Qiu, *Ultralytics YOLO*, version 8.0.0, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>.
- [37] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, *Real-time facial surface geometry from monocular video on mobile gpus*, 2019. arXiv: 1907.06724 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1907.06724>.
- [38] M. Douze et al., *The faiss library*, 2025. arXiv: 2401.08281 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2401.08281>.
- [39] H. Jégou, M. Douze, and C. Schmid, “Product Quantization for Nearest Neighbor Search,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 33, no. 1, pp. 117–128, Jan. 2011. DOI: 10.1109/TPAMI.2010.57. [Online]. Available: <https://inria.hal.science/inria-00514462>.
- [40] Y. A. Malkov and D. A. Yashunin, *Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs*, 2018. arXiv: 1603.09320 [cs.DS]. [Online]. Available: <https://arxiv.org/abs/1603.09320>.
- [41] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, *Invariant risk minimization*, 2020. arXiv: 1907.02893 [stat.ML]. [Online]. Available: <https://arxiv.org/abs/1907.02893>.
- [42] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang, *Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization*, 2020. arXiv: 1911.08731 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1911.08731>.
- [43] R. Geirhos et al., “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, Nov. 2020, ISSN: 2522-5839. DOI: 10.1038/s42256-020-00257-z. [Online]. Available: <http://dx.doi.org/10.1038/s42256-020-00257-z>.
- [44] J. Jo and Y. Bengio, *Measuring the tendency of cnns to learn surface statistical regularities*, 2017. arXiv: 1711.11561 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1711.11561>.
- [45] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel, *Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness*, 2022. arXiv: 1811.12231 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1811.12231>.

- [46] H. Wang, X. Wu, Z. Huang, and E. P. Xing, *High frequency component helps explain the generalization of convolutional neural networks*, 2020. arXiv: 1905.13545 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1905.13545>.
- [47] S. Beery, G. van Horn, and P. Perona, *Recognition in terra incognita*, 2018. arXiv: 1807.04975 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1807.04975>.
- [48] K. Xiao, L. Engstrom, A. Ilyas, and A. Madry, *Noise or signal: The role of image backgrounds in object recognition*, 2020. arXiv: 2006.09994 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2006.09994>.
- [49] A. Torralba and A. A. Efros, “Unbiased look at dataset bias,” in *CVPR 2011*, 2011, pp. 1521–1528. DOI: 10.1109/CVPR.2011.5995347.
- [50] J. Zhao, T. Wang, M. Yatskar, V. Ordonez, and K.-W. Chang, *Men also like shopping: Reducing gender bias amplification using corpus-level constraints*, 2017. arXiv: 1707.09457 [cs.AI]. [Online]. Available: <https://arxiv.org/abs/1707.09457>.
- [51] K. K. Singh, D. Mahajan, K. Grauman, Y. J. Lee, M. Feiszli, and D. Ghadiyaram, *Don’t judge an object by its context: Learning to overcome contextual bias*, 2020. arXiv: 2001.03152 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2001.03152>.
- [52] J. Canny, “A computational approach to edge detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. PAMI-8, no. 6, pp. 679–698, 1986. DOI: 10.1109/TPAMI.1986.4767851.
- [53] A. Kirillov et al., *Segment anything*, 2023. arXiv: 2304.02643 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2304.02643>.
- [54] W. Brendel and M. Bethge, *Approximating cnns with bag-of-local-features models works surprisingly well on imagenet*, 2019. arXiv: 1904.00760 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1904.00760>.
- [55] K. Burns, L. A. Hendricks, K. Saenko, T. Darrell, and A. Rohrbach, *Women also snowboard: Overcoming bias in captioning models*, 2019. arXiv: 1803.09797 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1803.09797>.
- [56] T. Wang, J. Zhao, M. Yatskar, K.-W. Chang, and V. Ordonez, *Balanced datasets are not enough: Estimating and mitigating gender bias in deep image representations*, 2019. arXiv: 1811.08489 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1811.08489>.
- [57] Z. Liu and K. He, *A decade’s battle on dataset bias: Are we there yet?* 2025. arXiv: 2403.08632 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2403.08632>.

- [58] B. Thomee et al., “Yfcc100m: The new data in multimedia research,” *Communications of the ACM*, vol. 59, no. 2, pp. 64–73, Jan. 2016, ISSN: 1557-7317. DOI: 10.1145/2812802. [Online]. Available: <http://dx.doi.org/10.1145/2812802>.
- [59] S. Changpinyo, P. Sharma, N. Ding, and R. Soricut, *Conceptual 12m: Pushing web-scale image-text pre-training to recognize long-tail visual concepts*, 2021. arXiv: 2102.08981 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2102.08981>.
- [60] S. Y. Gadre et al., *Datacomp: In search of the next generation of multimodal datasets*, 2023. arXiv: 2304.14108 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2304.14108>.
- [61] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “Bleu: A method for automatic evaluation of machine translation,” in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, P. Isabelle, E. Charniak, and D. Lin, Eds., Philadelphia, Pennsylvania, USA: Association for Computational Linguistics, Jul. 2002, pp. 311–318. DOI: 10.3115/1073083.1073135. [Online]. Available: <https://aclanthology.org/P02-1040/>.
- [62] S. Banerjee and A. Lavie, “METEOR: An automatic metric for MT evaluation with improved correlation with human judgments,” in *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, J. Goldstein, A. Lavie, C.-Y. Lin, and C. Voss, Eds., Ann Arbor, Michigan: Association for Computational Linguistics, Jun. 2005, pp. 65–72. [Online]. Available: <https://aclanthology.org/W05-0909/>.
- [63] R. Vedantam, C. L. Zitnick, and D. Parikh, *Cider: Consensus-based image description evaluation*, 2015. arXiv: 1411.5726 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1411.5726>.
- [64] J. Li, M. Galley, C. Brockett, J. Gao, and B. Dolan, *A diversity-promoting objective function for neural conversation models*, 2016. arXiv: 1510.03055 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/1510.03055>.
- [65] N. Reimers and I. Gurevych, *Sentence-bert: Sentence embeddings using siamese bert-networks*, 2019. arXiv: 1908.10084 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/1908.10084>.
- [66] J. Hessel, A. Holtzman, M. Forbes, R. L. Bras, and Y. Choi, *Clipscore: A reference-free evaluation metric for image captioning*, 2022. arXiv: 2104.08718 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2104.08718>.
- [67] T.-Y. Lin et al., *Microsoft coco: Common objects in context*, 2015. arXiv: 1405.0312 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1405.0312>.

- [68] A. Kuznetsova et al., "The open images dataset v4: Unified image classification, object detection, and visual relationship detection at scale," *International Journal of Computer Vision*, vol. 128, no. 7, pp. 1956–1981, Mar. 2020, ISSN: 1573-1405. DOI: 10.1007/s11263-020-01316-z. [Online]. Available: <http://dx.doi.org/10.1007/s11263-020-01316-z>.
- [69] C. Schumann, S. Ricco, U. Prabhu, V. Ferrari, and C. Pantofaru, "A step toward more inclusive people annotations for fairness," in *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, ser. AIES '21, ACM, Jul. 2021, pp. 916–925. DOI: 10.1145/3461702.3462594. [Online]. Available: <http://dx.doi.org/10.1145/3461702.3462594>.
- [70] K. He, X. Zhang, S. Ren, and J. Sun, *Deep residual learning for image recognition*, 2015. arXiv: 1512.03385 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1512.03385>.
- [71] G. B. Huang, M. Mattar, T. Berg, and E. Learned-Miller, "Labeled Faces in the Wild: A Database for Studying Face Recognition in Unconstrained Environments," in *Workshop on Faces in 'Real-Life' Images: Detection, Alignment, and Recognition*, Erik Learned-Miller and Andras Ferencz and Frédéric Jurie, Marseille, France, Oct. 2008. [Online]. Available: <https://inria.hal.science/inria-00321923>.
- [72] L. Aroyo and C. Welty, "Truth is a lie: Crowd truth and the seven myths of human annotation," *AI Magazine*, vol. 36, no. 1, pp. 15–24, Mar. 2015. DOI: 10.1609/aimag.v36i1.2564. [Online]. Available: <https://ojs.aaai.org/aimagazine/index.php/aimagazine/article/view/2564>.
- [73] K. Kärkkäinen and J. Joo, *Fairface: Face attribute dataset for balanced race, gender, and age*, 2019. arXiv: 1908.04913 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1908.04913>.
- [74] B. Porgali, V. Albiero, J. Ryda, C. C. Ferrer, and C. Hazirbas, *The casual conversations v2 dataset*, 2023. arXiv: 2303.04838 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2303.04838>.
- [75] L. Gustafson et al., *Facet: Fairness in computer vision evaluation benchmark*, 2023. arXiv: 2309.00035 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2309.00035>.
- [76] T. B. Fitzpatrick, "The validity and practicality of sun-reactive skin types i through vi," *Archives of Dermatology*, vol. 124, no. 6, pp. 869–871, Jun. 1988, ISSN: 0003-987X. DOI: 10.1001/archderm.1988.01670060015008. eprint: <https://jamanetwork.com/journals/jamadermatology/articlepdf/549509/>

- archderm_124_6_008.pdf. [Online]. Available: <https://doi.org/10.1001/archderm.1988.01670060015008>.
- [77] M. Groh et al., *Evaluating deep neural networks trained on clinical images in dermatology with the fitzpatrick 17k dataset*, 2021. arXiv: 2104.09957 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2104.09957>.
 - [78] C. Schumann, G. O. Olanubi, A. Wright, E. M. Jr., C. Heldreth, and S. Ricco, *Consensus and subjectivity of skin tone annotation for ml fairness*, 2024. arXiv: 2305.09073 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2305.09073>.
 - [79] C. M. Heldreth, E. P. Monk, A. T. Clark, C. Schumann, X. Eyee, and S. Ricco, "Which skin tone measures are the most inclusive? an investigation of skin tone measures for artificial intelligence," *ACM J. Responsib. Comput.*, vol. 1, no. 1, Mar. 2024. DOI: 10.1145/3632120. [Online]. Available: <https://doi.org/10.1145/3632120>.
 - [80] S. K. Mbatha, M. J. Booysen, and R. P. Theart, "Skin tone estimation under diverse lighting conditions," *Journal of Imaging*, vol. 10, no. 5, 2024, ISSN: 2313-433X. DOI: 10.3390/jimaging10050109. [Online]. Available: <https://www.mdpi.com/2313-433X/10/5/109>.
 - [81] V. Matias and J. Neto, "Enhancing fairness in machine learning: Skin tone classification using the monk skin tone scale," Sep. 2024, pp. 76–81. DOI: 10.5753/sibgrapi.est.2024.31648.
 - [82] M. K. Scheuerman, J. M. Paul, and J. R. Brubaker, "How computers see gender," *Proceedings of the ACM on Human-Computer Interaction*, vol. 3, pp. 1–33, 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:207939928>.
 - [83] I. D. Raji and J. Buolamwini, "Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial ai products," *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*, 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:91168921>.
 - [84] Y. Jiang, W. Li, Y. Zhang, M. Cai, and X. Yue, *Fairgen: Enhancing fairness in text-to-image diffusion models via self-discovering latent directions*, 2025. arXiv: 2412.18810 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2412.18810>.
 - [85] R. Gandikota, H. Orgad, Y. Belinkov, J. Materzyńska, and D. Bau, *Unified concept editing in diffusion models*, 2024. arXiv: 2308.14761 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2308.14761>.
 - [86] C. Zhang et al., *Iti-gen: Inclusive text-to-image generation*, 2023. arXiv: 2309.05569 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2309.05569>.

- [87] L. Zhang, A. Rao, and M. Agrawala, *Adding conditional control to text-to-image diffusion models*, 2023. arXiv: 2302.05543 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2302.05543>.
- [88] D. G. Fernández, R.-A. Matisan, A. M. Muñoz, and J. Partyka, *Reproducibility study of "iti-gen: Inclusive text-to-image generation"*, 2024. arXiv: 2407.19996 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2407.19996>.
- [89] A. Lobashev, M. Larchenko, and D. Guskov, *Color conditional generation with sliced wasserstein guidance*, 2025. arXiv: 2503.19034 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2503.19034>.
- [90] K. C. Shum, B.-S. Hua, D. T. Nguyen, and S.-K. Yeung, *Color alignment in diffusion*, 2025. arXiv: 2503.06746 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2503.06746>.
- [91] M. Nadeem, A. Bethke, and S. Reddy, *Stereoset: Measuring stereotypical bias in pretrained language models*, 2020. arXiv: 2004.09456 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2004.09456>.
- [92] S. Harrison, E. Gualdoni, and G. Boleda, *Run like a girl! sports-related gender bias in language and vision*, 2023. arXiv: 2305.14468 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2305.14468>.
- [93] R. Naik and B. Nushi, *Social biases through the text-to-image generation lens*, 2023. arXiv: 2304.06034 [cs.CY]. [Online]. Available: <https://arxiv.org/abs/2304.06034>.
- [94] C.-Y. Chuang, V. Jampani, Y. Li, A. Torralba, and S. Jegelka, *Debiasing vision-language models via biased prompts*, 2023. arXiv: 2302.00070 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2302.00070>.
- [95] International Labour Organization, *International standard classification of occupations (ISCO-08)*, Accessed: 2025, 2012. [Online]. Available: <https://ilostat.ilo.org/methods/concepts-and-definitions/classification-occupation/>.
- [96] C. Lugaresi et al., *Mediapipe: A framework for building perception pipelines*, 2019. arXiv: 1906.08172 [cs.DC]. [Online]. Available: <https://arxiv.org/abs/1906.08172>.
- [97] Latent Consistency, *LCM-LoRA for stable diffusion v1-5*, <https://huggingface.co/latent-consistency/lcm-lora-sdv1-5>, 2023.
- [98] CompVis, *Stable diffusion v1-2 model card*, <https://huggingface.co/CompVis/stable-diffusion-v1-2>, 2022.

- [99] M. Cherti et al., “Reproducible scaling laws for contrastive language-image learning,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, Jun. 2023, pp. 2818–2829. DOI: 10.1109/cvpr52729.2023.00276. [Online]. Available: <http://dx.doi.org/10.1109/CVPR52729.2023.00276>.
- [100] Segmind, *SSD-1B model card*, <https://huggingface.co/segmind/SSD-1B>, 2023.
- [101] PixArt-alpha, *PixArt-XL-2 1024 model card*, <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>, 2023.
- [102] Segmind, *Segmind-vega model card*, <https://huggingface.co/segmind/Segmind-Vega>, 2023.
- [103] K. Huang, C. Duan, K. Sun, E. Xie, Z. Li, and X. Liu, *T2i-compbench++: An enhanced and comprehensive benchmark for compositional text-to-image generation*, 2025. arXiv: 2307.06350 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2307.06350>.
- [104] A. Nichol et al., *Glide: Towards photorealistic image generation and editing with text-guided diffusion models*, 2022. arXiv: 2112.10741 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2112.10741>.
- [105] I.-S. Fang, Y.-H. Han, and J.-C. Chen, “Camera settings as tokens: Modeling photography on latent diffusion models,” in *SIGGRAPH Asia 2024 Conference Papers*, ser. SA '24, Tokyo, Japan: Association for Computing Machinery, 2024, ISBN: 9798400711312. DOI: 10.1145/3680528.3687635. [Online]. Available: <https://doi.org/10.1145/3680528.3687635>.
- [106] rizvandwiki, *Gender-classification*, Hugging Face model repository, Accessed: 2026-03-06, 2023. [Online]. Available: <https://huggingface.co/rizvandwiki/gender-classification>.
- [107] M. Tschannen et al., *Siglip 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features*, 2025. arXiv: 2502.14786 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2502.14786>.
- [108] S. I. Serengil and A. Ozpinar, “Hyperextended lightface: A facial attribute analysis framework,” in *2021 International Conference on Engineering and Emerging Technologies (ICEET)*, 2021, pp. 1–4. DOI: 10.1109/ICEET53442.2021.9659697.

- [109] J. Deng, J. Guo, J. Yang, N. Xue, I. Kotsia, and S. Zafeiriou, "Arcface: Additive angular margin loss for deep face recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 10, pp. 5962–5979, Oct. 2022, ISSN: 1939-3539. DOI: 10.1109/tpami.2021.3087709. [Online]. Available: <http://dx.doi.org/10.1109/TPAMI.2021.3087709>.
- [110] R. A. Rejón Piña and C. Ma, "Classification algorithm for skin color (casco): A new tool to measure skin color in social science research," *Social Science Quarterly*, vol. 104, no. 2, pp. 168–179, 2023. DOI: <https://doi.org/10.1111/ssqu.13242>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/ssqu.13242>. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/ssqu.13242>.
- [111] K. G. Tijdens, "Measuring occupations in web-surveys: The WISCO database of occupations," *AIAS Working Paper*, no. 10-108, 2010. [Online]. Available: <https://www.researchgate.net/publication/241752024>.
- [112] L. Yang et al., *Depth anything v2*, 2024. arXiv: 2406.09414 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2406.09414>.
- [113] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, *Masked-attention mask transformer for universal image segmentation*, 2022. arXiv: 2112.01527 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2112.01527>.
- [114] Y. Li, H. Mao, R. Girshick, and K. He, *Exploring plain vision transformer backbones for object detection*, 2022. arXiv: 2203.16527 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2203.16527>.
- [115] S. Liu et al., *Grounding dino: Marrying dino with grounded pre-training for open-set object detection*, 2024. arXiv: 2303.05499 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2303.05499>.
- [116] Y. Zhao et al., *Detrs beat yolos on real-time object detection*, 2024. arXiv: 2304.08069 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2304.08069>.
- [117] H. Liu, C. Li, Q. Wu, and Y. J. Lee, *Visual instruction tuning*, 2023. arXiv: 2304.08485 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2304.08485>.
- [118] J. Li, D. Li, C. Xiong, and S. Hoi, *Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation*, 2022. arXiv: 2201.12086 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2201.12086>.
- [119] B. Xiao et al., *Florence-2: Advancing a unified representation for a variety of vision tasks*, 2023. arXiv: 2311.06242 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2311.06242>.

- [120] Z. Peng et al., *Kosmos-2: Grounding multimodal large language models to the world*, 2023. arXiv: 2306.14824 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2306.14824>.
- [121] J. Li, D. Li, S. Savarese, and S. Hoi, *Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models*, 2023. arXiv: 2301.12597 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2301.12597>.
- [122] P. Young, A. Lai, M. Hodosh, and J. Hockenmaier, "From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions," *TACL*, vol. 2, pp. 67–78, Dec. 2014. DOI: 10.1162/tac1_a_00166.

Appendix A Occupation List

The following 200 occupations, listed alphabetically, constitute the full profession set used for profession-conditioned image retrieval from Re-LAION-5B and for image generation using Stable Diffusion v1.5. The set was constructed by sampling job titles systematically across all ten ISCO-08 major groups, ensuring representative coverage of skill levels, professional domains, and social prestige ranges.

# Occupation	# Occupation	# Occupation	# Occupation
1 Accountant	51 Cleaner	101 Hair Dresser	151 Pilot
2 Actor	52 Clerk	102 Handyperson	152 Plumber
3 Actuary	53 Coach	103 Health Inspector	153 Police Officer
4 Administrative Assistant	54 Comedian	104 Historian	154 Politician
5 Administrator	55 Compliance Officer	105 Hotel Concierge	155 Postal Worker
6 Air Traffic Controller	56 Composer	106 Hotel Manager	156 Priest
7 Animal Trainer	57 Conservation Officer	107 Human Resources Specialist	157 Procurement Officer
8 Anthropologist	58 Construction Worker	108 IT Support Specialist	158 Professor
9 Appraiser	59 Copywriter	109 Illustrator	159 Property Manager
10 Archaeologist	60 Court Reporter	110 Industrial Designer	160 Psychologist
11 Architect	61 Crime Scene Investigator	111 Insurance Underwriter	161 Quality Assurance Inspector
12 Archivist	62 Customer Support Specialist	112 Janitor	162 Real Estate Agent
13 Art Director	63 DJ	113 Jeweller	163 Receptionist
14 Artist	64 Dancer	114 Journalist	164 Researcher
15 Astronaut	65 Data Scientist	115 Judge	165 Roofer
16 Astronomer	66 Database Administrator	116 Lawyer	166 Safety Inspector
17 Athlete	67 Debt Counselor	117 Librarian	167 Sailor

Continued on next page

Table A.1 – continued

# Occupation	# Occupation	# Occupation	# Occupation
18 Audio Technician	68 Dentist	118 Lifeguard	168 Salesperson
19 Auditor	69 Detective	119 Loan Officer	169 Scientist
20 Automotive Designer	70 Development Officer	120 Logger	170 Security Officer
21 Baker	71 Dietitian	121 Logistics Manager	171 Shopkeeper
22 Banker	72 Director	122 Magician	172 Singer
23 Bankruptcy Specialist	73 Doctor	123 Makeup Artist	173 Skier
24 Barber	74 Dog Walker	124 Marine Biologist	174 Social Worker
25 Barista	75 Draughtsperson	125 Marketing Manager	175 Software Engineer
26 Bartender	76 Driver	126 Masseur	176 Soldier
27 Basketball Player	77 Economist	127 Mathematician	177 Sound Engineer
28 Biologist	78 Editor	128 Mayor	178 Statistician
29 Biomedical Engineer	79 Electrician	129 Mechanic	179 Street Vendor
30 Blacksmith	80 Emergency Management Specialist	130 Meteorologist	180 Surfer
31 Bodyguard	81 Entrepreneur	131 Midwife	181 Surgeon
32 Bounty Hunter	82 Environmental Engineer	132 Miner	182 Swimmer
33 Boxer	83 Ergonomist	133 Model	183 Tailor
34 Brand Manager	84 Estate Planner	134 Musician	184 Tattoo Artist
35 Brewer	85 Event Coordinator	135 News Reader	185 Teacher
36 Bricklayer	86 Executive Assistant	136 Nurse	186 Technician
37 Broker	87 Exterminator	137 Nutritionist	187 Tennis Player
38 Builder	88 Facilities Manager	138 Oceanographer	188 Therapist
39 Butcher	89 Farmer	139 Office Assistant	189 Translator

Continued on next page

Table A.1 – continued

# Occupation	# Occupation	# Occupation	# Occupation
40 CEO	90 Fashion Designer	140 Operations Manager	190 Umpire
41 Carer	91 Firefighter	141 Optician	191 Urban Planner
42 Carpenter	92 Fishmonger	142 Painter	192 Usher
43 Cartographer	93 Flight Attendant	143 Paralegal	193 Veterinarian
44 Cashier	94 Florist	144 Paramedic	194 Videographer
45 Chef	95 Football Player	145 Park Ranger	195 Waiter
46 Chemical Engineer	96 Forklift Operator	146 Payroll Specialist	196 Waste Collection Worker
47 Chemist	97 Gardener	147 Personal Trainer	197 Welder
48 Chiropractor	98 Geologist	148 Pharmacist	198 Wholesaler
49 Civil Engineer	99 Graphic Designer	149 Photographer	199 Writer
50 Claims Adjuster	100 Grocer	150 Physicist	200 Zoologist

Appendix B Code Repository

The code used throughout this Thesis is accessible via GitHub using the following link:
<https://github.com/JerOme123/Masters-Thesis>.